independent_project

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2026-04-30

Setup

- Load the .xlsx file containing the dataset

```
file <- "Kehrberger_Scientific_Reports.xlsx"
```

Convert every sheet to a separate data frame - Clean colnames using janitor - Manually rename leftover names for clarity

```
T1 <- read_excel(file, sheet = "T1") %>%
  as_tibble() %>%
  janitor::clean_names() %>%
  rename(
    j_day = julian_date_of_year_2015,
    flower_count = number_of_p_vulgaris_flowers,
    visitation_rate = day_specific_flower_visitation_rate_individuals_per_hour,
    competitor_count = day_specific_number_of_other_than_p_vulgaris_flowering_plant_species,
    temperature = temperature_c
  )

T2 <- read_excel(file, sheet = "T2") %>%
  as_tibble() %>%
  janitor::clean_names() %>%
  rename(
    week = week_of_the_year_2015,
    flower_count = number_of_p_vulgaris_flowers,
    bee_abundance = mean_bee_abundance,
    visitation_rate = mean_bee_visitation_rate_on_p_vulgaris_flowers,
    competitor_count = total_number_of_other_flowering_plant_species
  )

T3 <- read_excel(file, sheet = "T3") %>%
  as_tibble() %>%
  janitor::clean_names() %>%
  rename(
    j_open = julian_date_of_bud_opening,
    longevitiy = floral_longevity_days,
    pollinator_hours = flower_specific_number_of_pollinator_suitable_hours,
    seed_set = seed_set_percent,
    mean_temp = flower_specific_mean_temperature_c,
    bee_visits = estimated_total_number_of_bee_visits_per_flower_of_p_vulgaris
```

```r
  )

T4 <- read_excel(file, sheet = "T4") %>%
  as_tibble() %>%
  janitor::clean_names() %>%
  rename(
    j_open = julian_date_of_bud_opening,
    seed_set = seed_set_percent
  )

T5 <- read_excel(file, sheet = "T5") %>%
  as_tibble() %>%
  janitor::clean_names() %>%
  rename(
    j_open = julian_date_of_bud_opening,
    seed_set = seed_set_percent
  )
```

Convert Julian dates to Date objects -if j__day = 1, date should = 2015-01-01 Factorize categorical variables Fix incorrectly assigned numerical values (NAs as characters)

```r
T1 <- T1 %>%
  mutate(date = as.Date(j_day - 1, origin = "2015-01-01"),
         site = as.factor(site),
         visitation_rate = as.numeric(visitation_rate),
         temperature = as.numeric(temperature)
         )
```

```
## Warning: There were 2 warnings in ‘mutate()‘.
## The first warning was:
## i In argument: ‘visitation_rate = as.numeric(visitation_rate)‘.
## Caused by warning:
## ! NAs introduced by coercion
## i Run ‘dplyr::last_dplyr_warnings()‘ to see the 1 remaining warning.
```

```r
T2 <- T2 %>%
  mutate(site = as.factor(site),
         bee_abundance = as.numeric(bee_abundance),
         visitation_rate = as.numeric(visitation_rate),
         competitor_count = as.numeric(competitor_count)
         )
```

```
## Warning: There were 3 warnings in ‘mutate()‘.
## The first warning was:
## i In argument: ‘bee_abundance = as.numeric(bee_abundance)‘.
## Caused by warning:
## ! NAs introduced by coercion
## i Run ‘dplyr::last_dplyr_warnings()‘ to see the 2 remaining warnings.
```

```r
T3 <- T3 %>%
  mutate(date_open = as.Date(j_open - 1 , origin = "2015-01-01"),
         site = as.factor(site),
```

```r
         flower_id = as.factor(flower_id))

T4 <- T4 %>%
  mutate(date_open = as.Date(j_open - 1, origin = "2015-01-01"),
         site = as.factor(site),
         pollen_limitation = as.numeric(pollen_limitation))
```

```
## Warning: There was 1 warning in ‘mutate()‘.
## i In argument: ‘pollen_limitation = as.numeric(pollen_limitation)‘.
## Caused by warning:
## ! NAs introduced by coercion
```

```r
T5 <- T5 %>%
  mutate(date_open = as.Date(j_open - 1, origin = "2015-01-01"),
         site = as.factor(site),
         treatment = as.factor(treatment)
         )
```

T4 has a really nasty outlier that'll interfere with our statistics, lets remove it!

```r
Q1 <- quantile(T4$pollen_limitation, 0.25, na.rm = T)
Q3 <- quantile(T4$pollen_limitation, 0.75, na.rm = T)
IQR <- Q3 - Q1

T4 <- T4 %>%
  filter(pollen_limitation >= (Q1 - 1.5 * IQR) & pollen_limitation <= (Q3 + 1.5 * IQR))
```

Verify data structure before proceeding

```r
glimpse(c(T1, T2, T3, T4, T5))
```

```
## List of 32
##  $ site             : Factor w/ 8 levels "1","2","3","4",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ j_day            : num [1:192] 72 77 79 82 84 86 89 91 93 96 ...
##  $ flower_count     : num [1:192] 0 22 50 102 220 ...
##  $ visitation_rate  : num [1:192] NA 0.3836 0 0 0.0182 ...
##  $ competitor_count : num [1:192] 0 0 0 0 0 0 0 0 1 1 ...
##  $ temperature      : num [1:192] NA 17.9 17.7 7.1 22 NA NA 6.8 7.9 11.7 ...
##  $ date             : Date[1:192], format: "2015-03-13" "2015-03-18" ...
##  $ site             : Factor w/ 8 levels "1","2","3","4",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ week             : num [1:80] 11 12 13 14 15 16 17 18 19 20 ...
##  $ flower_count     : num [1:80] 0 36 161.6 164.2 82.2 ...
##  $ bee_abundance    : num [1:80] NA 14.63 5 2.89 19.67 ...
##  $ visitation_rate  : num [1:80] NA 0.19179 0.00911 0.00893 0.03251 ...
##  $ competitor_count : num [1:80] NA 0 0 1 2 4 4 6 6 NA ...
##  $ site             : Factor w/ 8 levels "1","2","3","4",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ flower_id        : Factor w/ 64 levels "F1","F10","F11",..: 1 12 23 34 45 56 62 63 64 2 ...
##  $ j_open           : num [1:64] 84 84 87 91 91 94 96 98 105 108 ...
##  $ longevitiy       : num [1:64] 12 9 16 9 5 9 7 7 7 9 ...
##  $ pollinator_hours : num [1:64] 45 32 106 59 20 86 79 87 86 107 ...
##  $ seed_set         : num [1:64] 70.2 96.9 69.1 0 0 ...
```

```
##  $ mean_temp        : num [1:64] 7.04 7.72 9.21 8.03 5.46 ...
##  $ bee_visits       : num [1:64] 2.129 0.583 2.585 1.528 1.918 ...
##  $ date_open        : Date[1:64], format: "2015-03-25" "2015-03-25" ...
##  $ site             : Factor w/ 7 levels "1","2","3","5",..: 1 1 1 1 1 2 2 2 2 2 ...
##  $ j_open           : num [1:35] 84 87 94 96 98 74 79 81 90 97 ...
##  $ pollen_limitation: num [1:35] 0.0536 0.6261 2.5697 0 1.4876 ...
##  $ seed_set         : num [1:35] 83.5 69.1 26.4 48.6 38.6 ...
##  $ date_open        : Date[1:35], format: "2015-03-25" "2015-03-28" ...
##  $ site             : Factor w/ 8 levels "1","2","3","4",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ treatment        : Factor w/ 3 levels "hand","open",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ j_open           : num [1:174] 80 84 87 91 94 94 96 98 103 108 ...
##  $ seed_set         : num [1:174] 23.33 4.48 43.28 100 66.67 ...
##  $ date_open        : Date[1:174], format: "2015-03-21" "2015-03-25" ...
```

Data Transformation

Summarize statistics across sites

```r
T1_site_summary <- T1 %>%
  group_by(site) %>%
  summarize(
    mean_flowers = mean(flower_count, na.rm = TRUE),
    mean_visitation = mean(visitation_rate, na.rm = TRUE),
    mean_temp = mean(temperature, na.rm = TRUE),
    total_competitors = sum(competitor_count, na.rm = TRUE)
  ) %>%
  ungroup()

T2_site_summary <- T2 %>%
  group_by(site) %>%
  summarize(
    mean_flowers = mean(flower_count, na.rm = T),
    mean_bees = mean(bee_abundance, na.rm = T),
    mean_visitation = mean(visitation_rate, na.rm = T)
    ) %>%
  ungroup()

T3_site_summary <- T3 %>%
  group_by(site) %>%
  summarize(
    mean_longevity = mean(longevitiy, na.rm = T),
    mean_pollinator_hours = mean(pollinator_hours, na.rm = T),
    mean_seed_set = mean(seed_set, na.rm = T)
  ) %>%
  ungroup()

T4_site_summary <- T4 %>%
  group_by(site) %>%
  summarize(
    mean_pollen_limitation <- mean(pollen_limitation, na.rm = T),
    mean_seed_set = mean(seed_set, na.rm = T)
  ) %>%
  ungroup()
```

```r
T5_site_summary <- T5 %>%
  group_by(site,treatment) %>%
  summarize(
    mean_open_date = mean(j_open, na.rm = T),
    mean_seed_set = mean(seed_set, na.rm = T)
  ) %>%
  ungroup()
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by site and treatment.
## i Output is grouped by site.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(site, treatment))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

Summarize stats across dates

```r
T1_date_summary <- T1 %>%
  group_by(date) %>%
  summarize(
    mean_flowers = mean(flower_count, na.rm = TRUE),
    mean_visitation = mean(visitation_rate, na.rm = TRUE),
    mean_temp = mean(temperature, na.rm = TRUE),
    total_competitors = sum(competitor_count, na.rm = TRUE)
  ) %>%
  ungroup()

T2_date_summary <- T2 %>%
  group_by(week) %>%
  summarize(
    mean_flowers = mean(flower_count, na.rm = T),
    mean_bees = mean(bee_abundance, na.rm = T),
    mean_visitation = mean(visitation_rate, na.rm = T)
    ) %>%
  ungroup()
```

Visualization

Visualize flowering abundance over time using gganimate

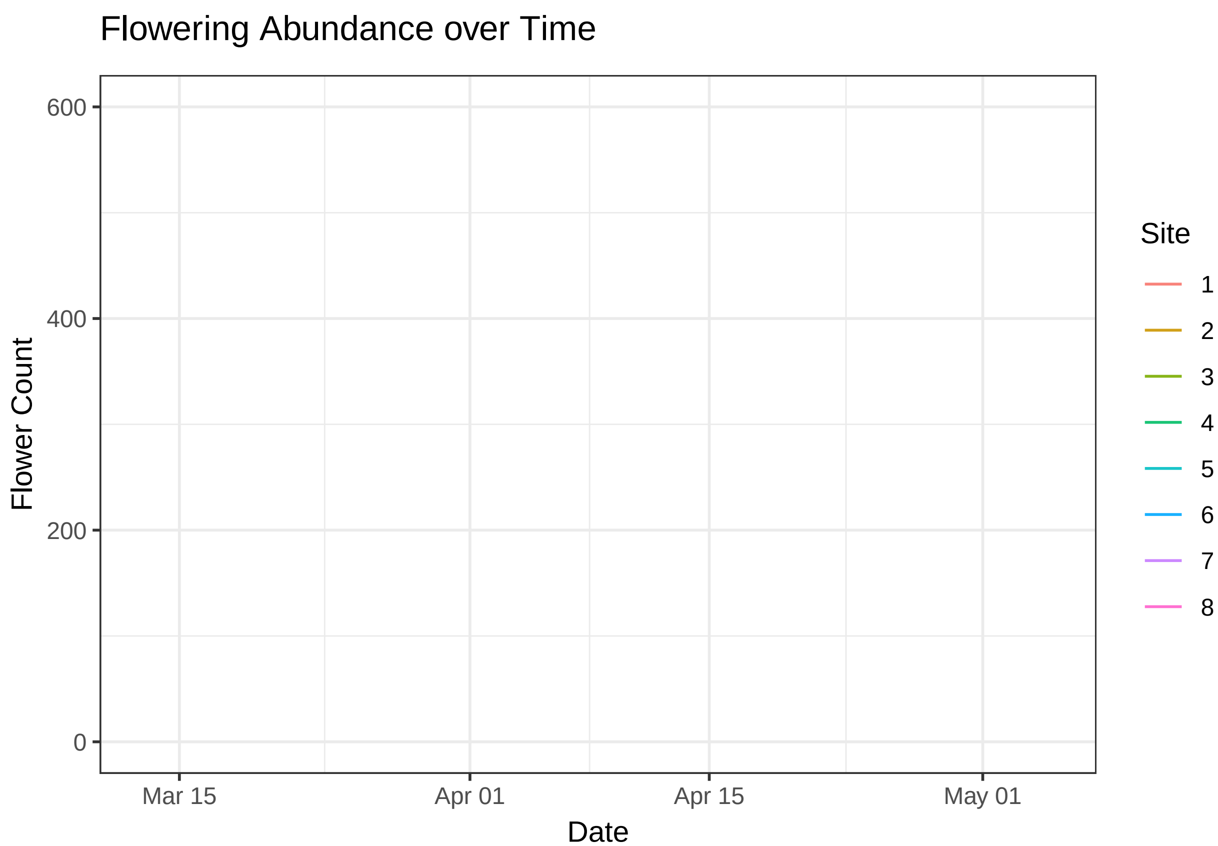
```r
(plot_T1_time <- T1 %>%
   ggplot(aes(x = date,
              y = flower_count,
              color = site,
              group = site)) +
   geom_line(alpha = 0.9) +
   theme_bw() +
   labs(title = "Flowering Abundance over Time",
        x = "Date",
        y = "Flower Count",
        color = "Site") +
```

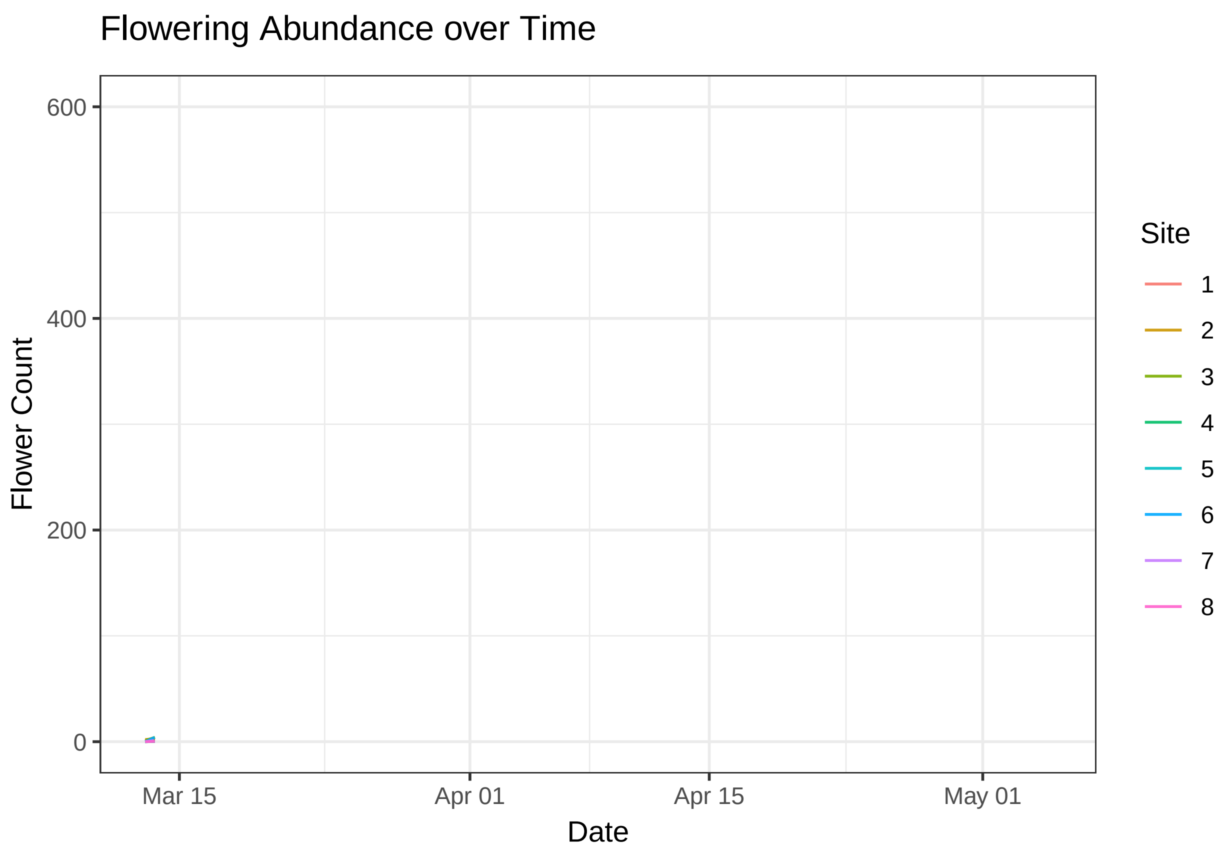
```
  transition_reveal(date)
)
```

```
## `geom_line()`: Each group consists of only one observation.
## i Do you need to adjust the group aesthetic?

## Warning in formals(fun): argument is not a function

## `geom_line()`: Each group consists of only one observation.
## i Do you need to adjust the group aesthetic?
```




Flowering Abundance over Time
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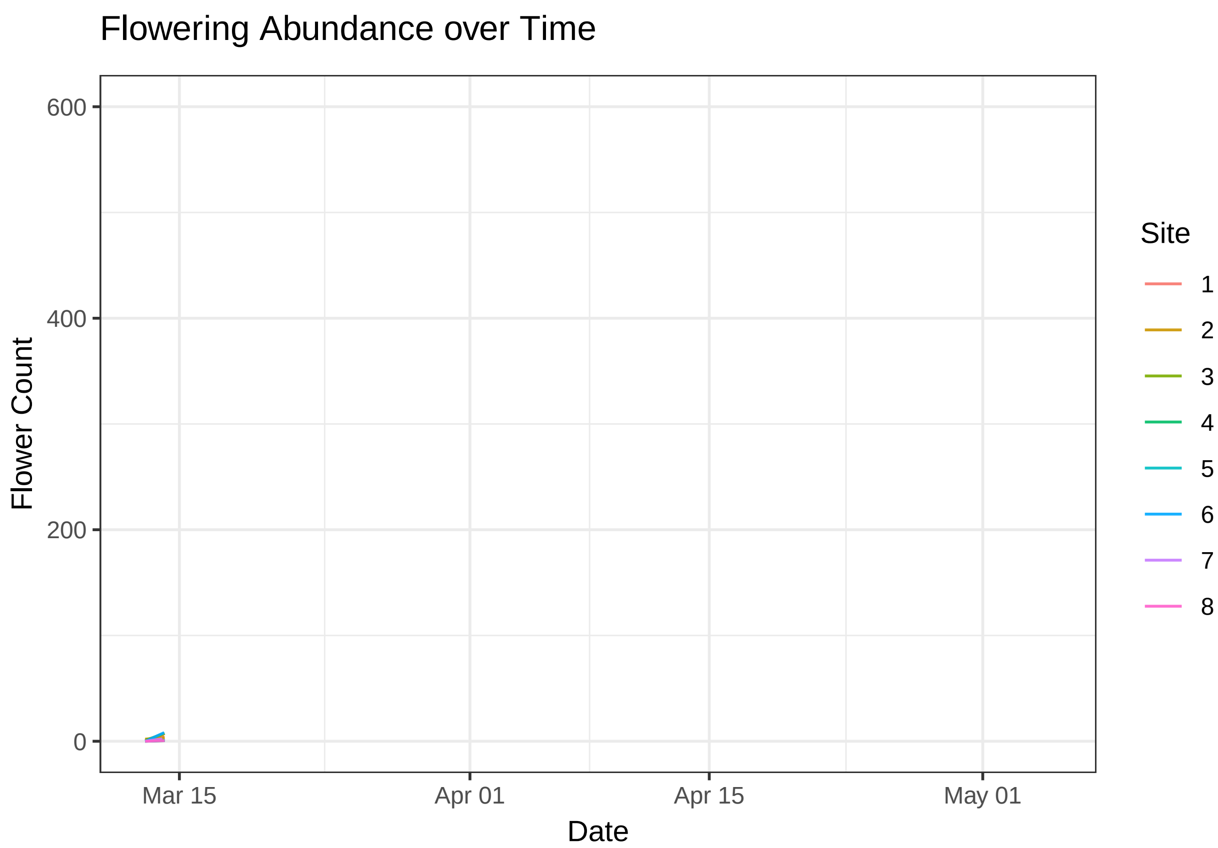
Flowering Abundance over Time
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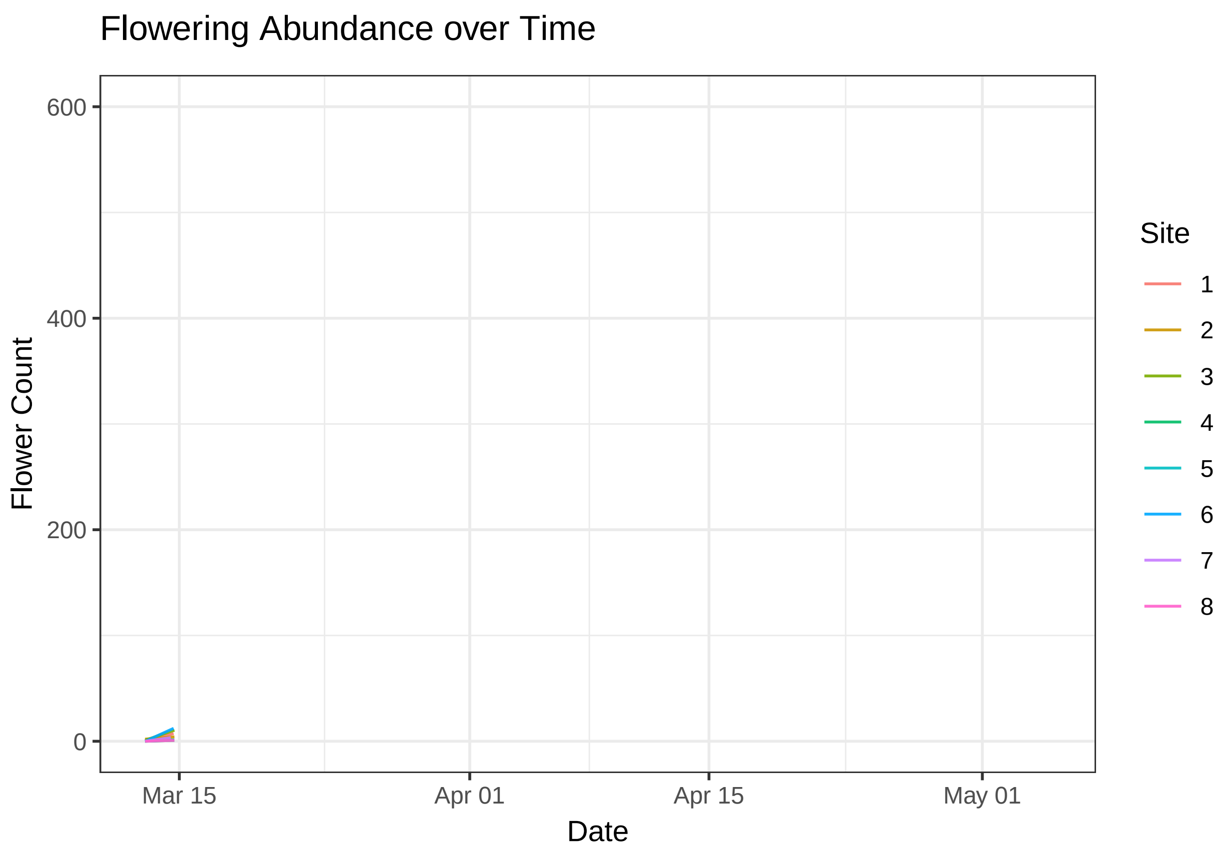
Flowering Abundance over Time
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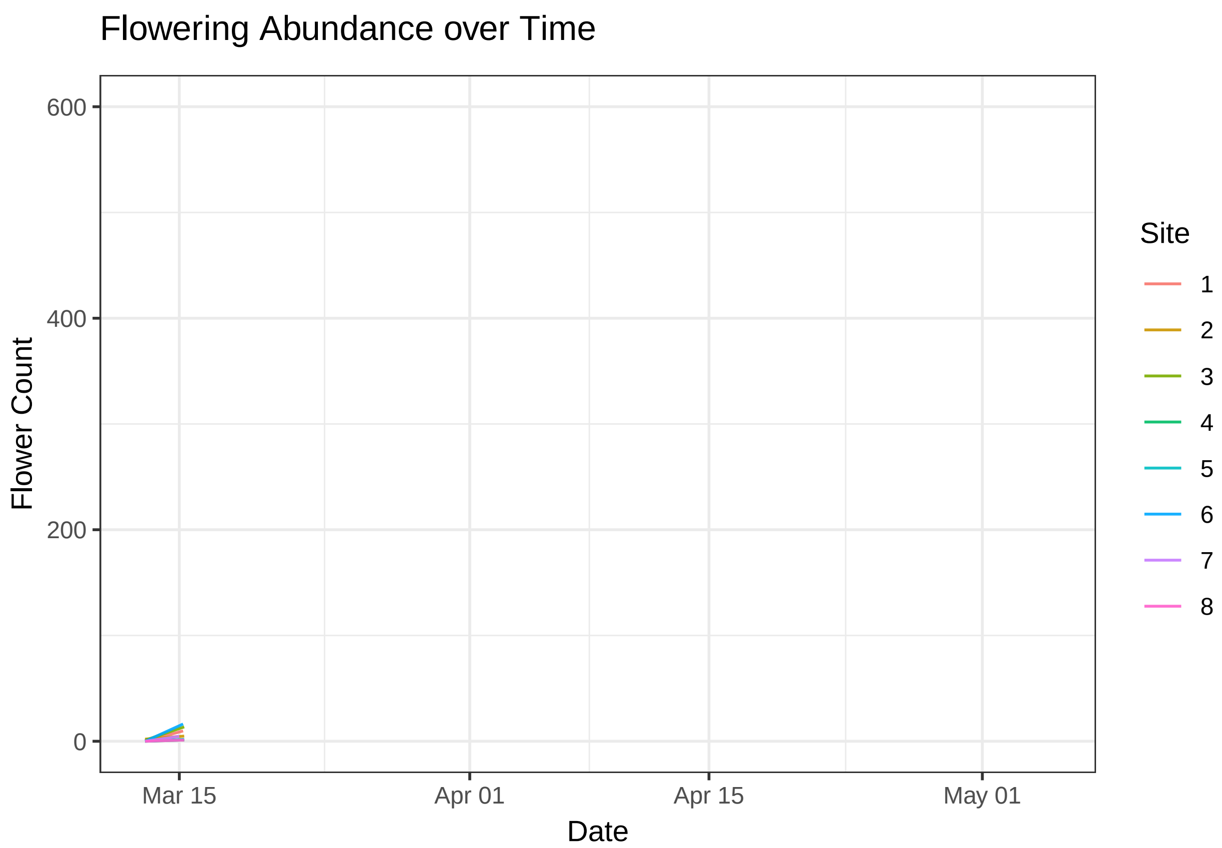

Flowering Abundance over Time
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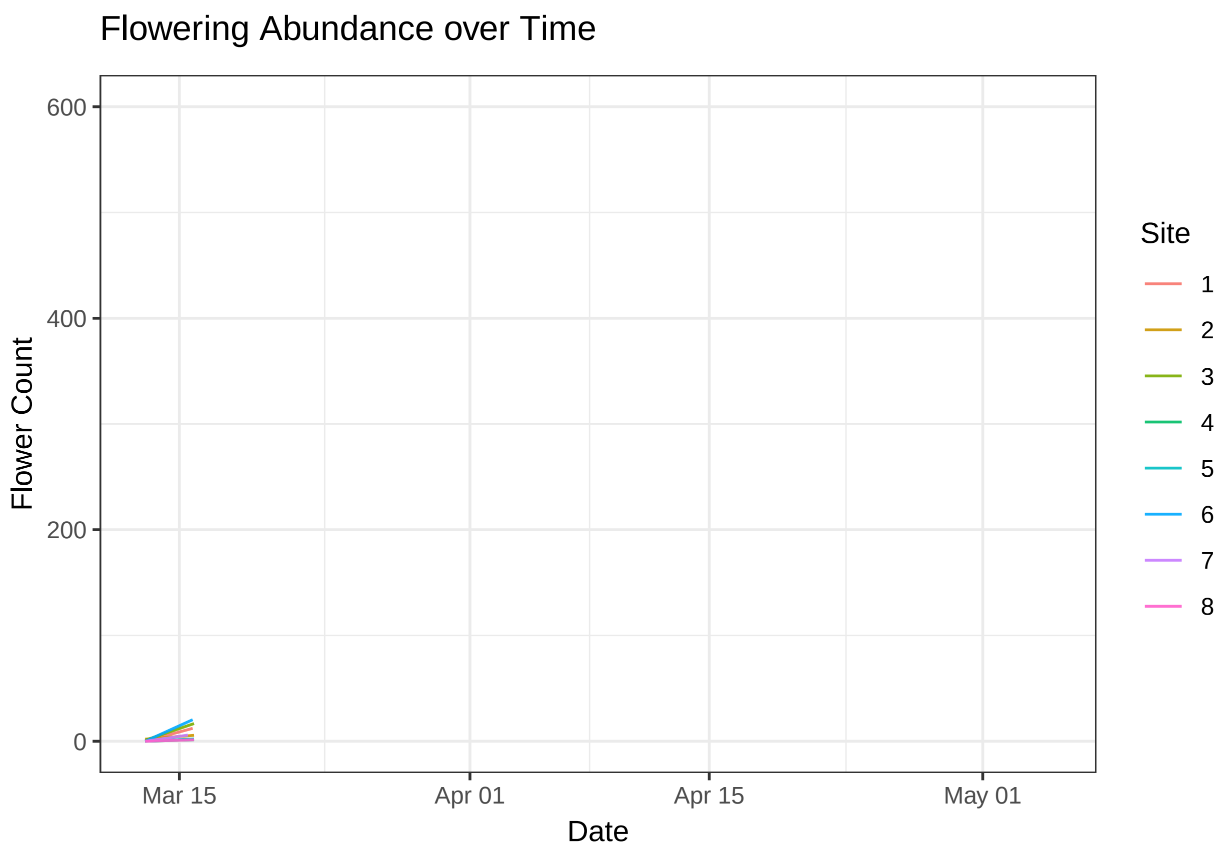

Flowering Abundance over Time
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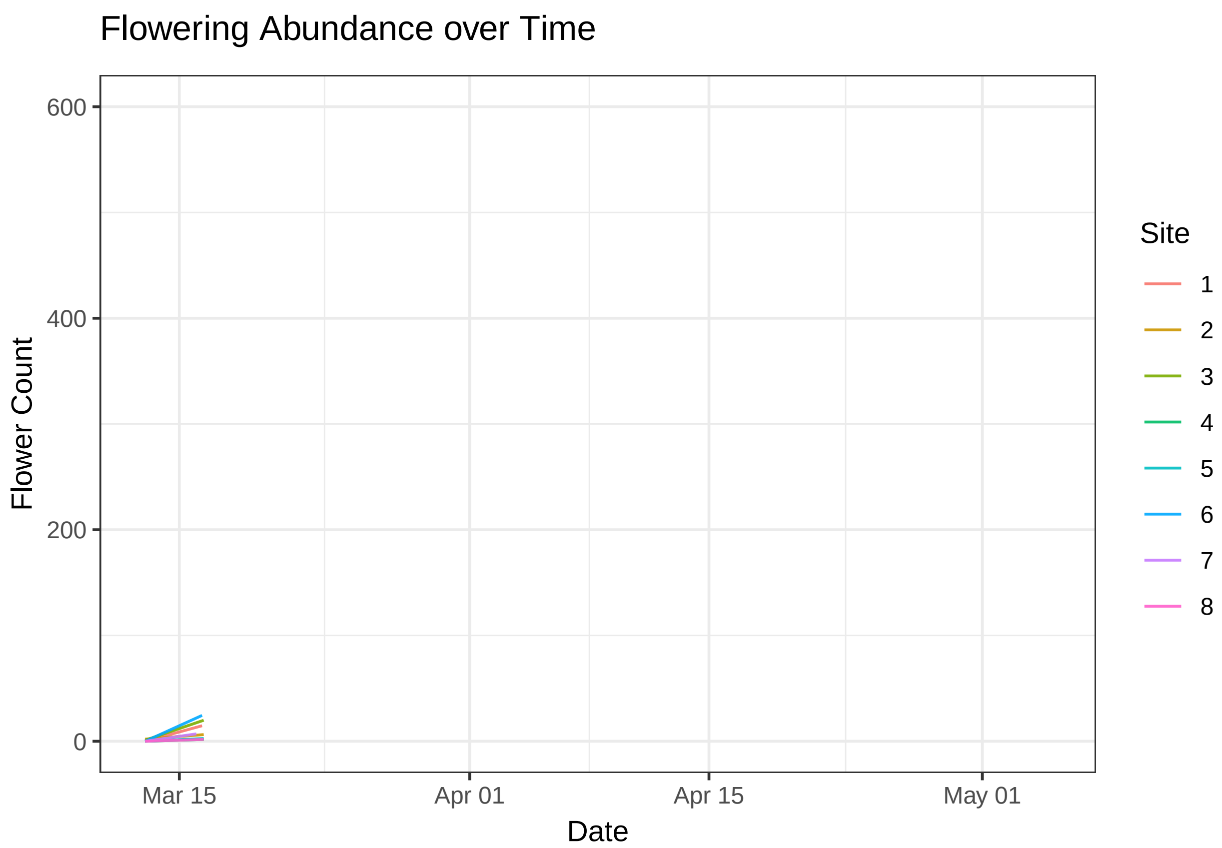
Flowering Abundance over Time
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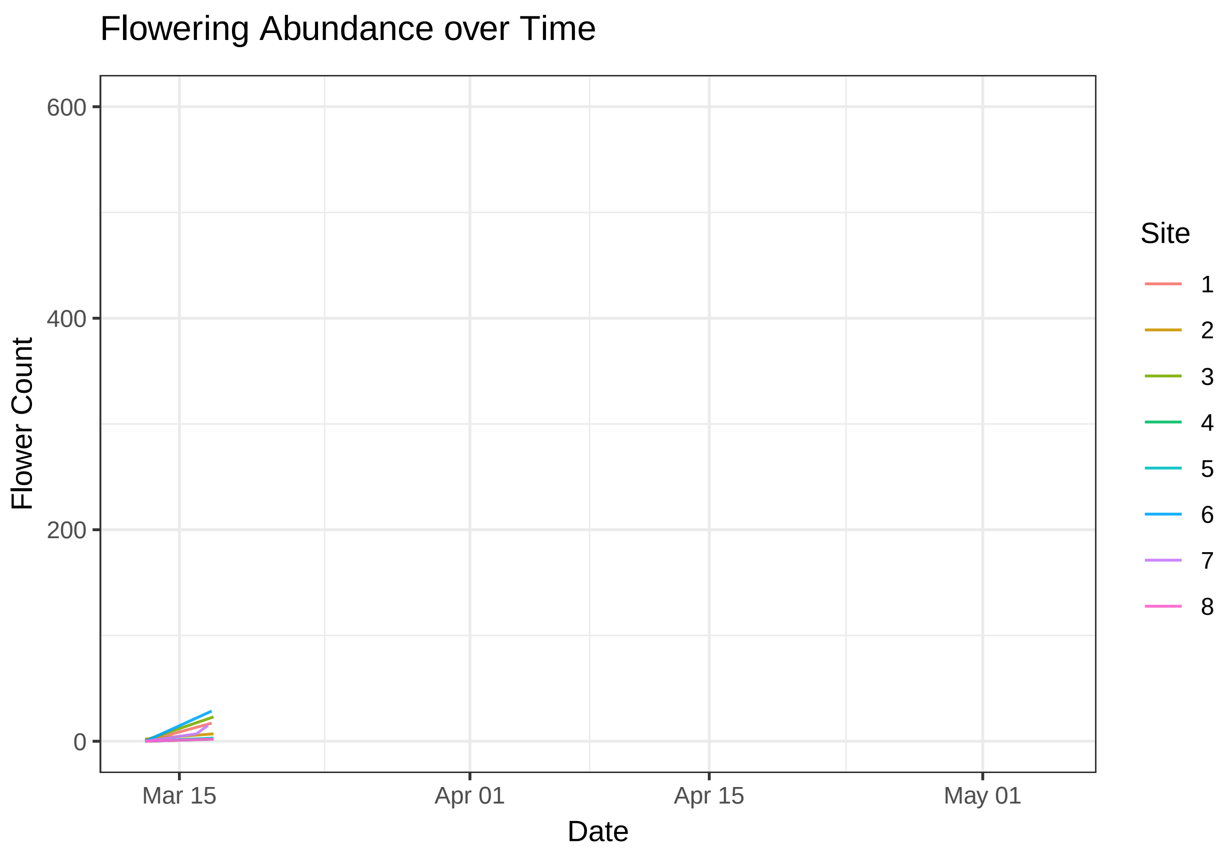
Flowering Abundance over Time
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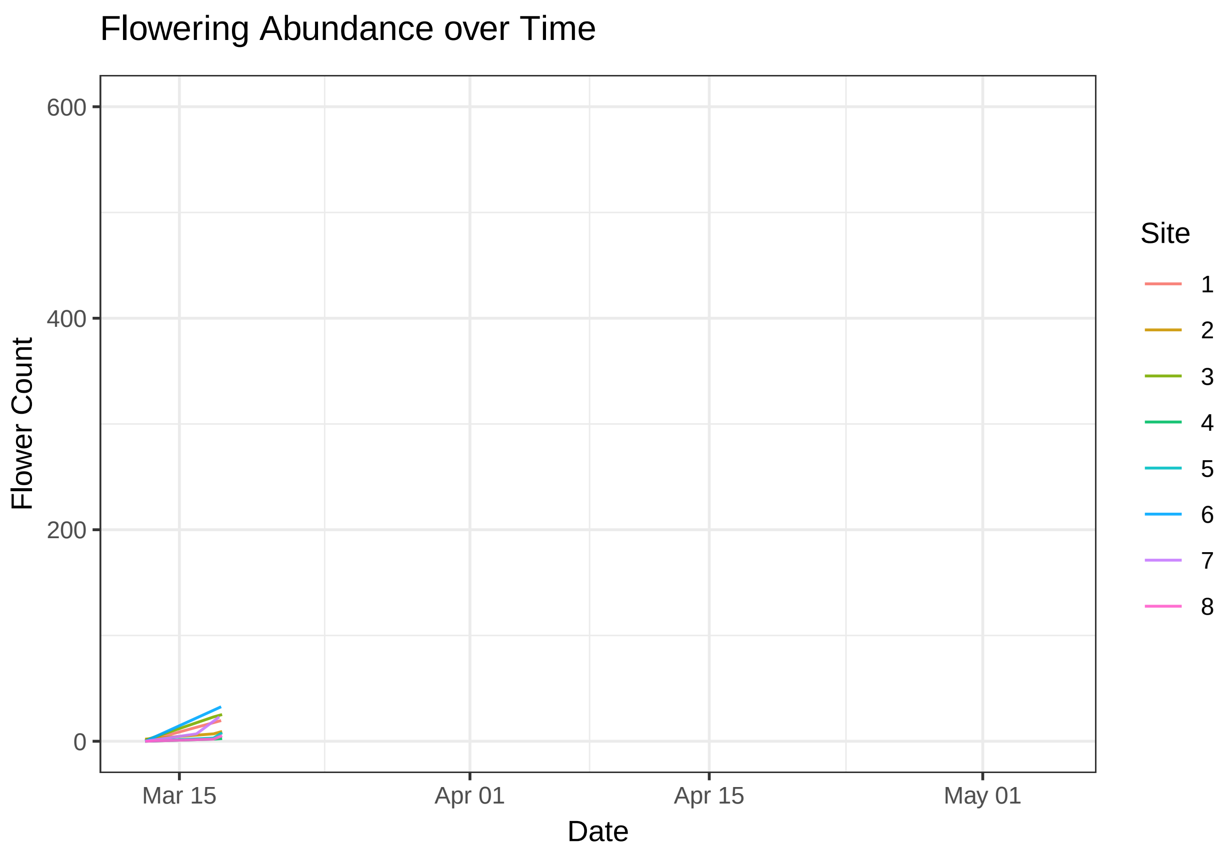
Flowering Abundance over Time
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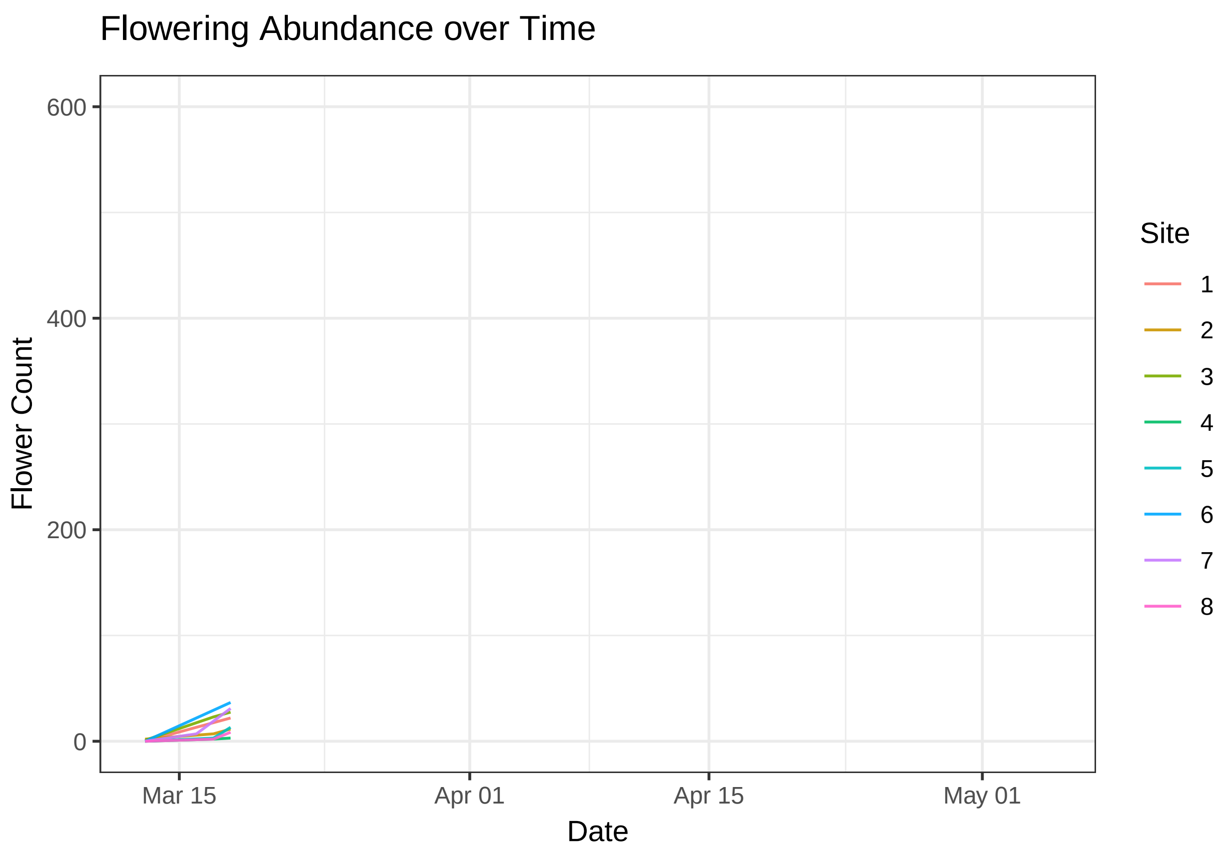
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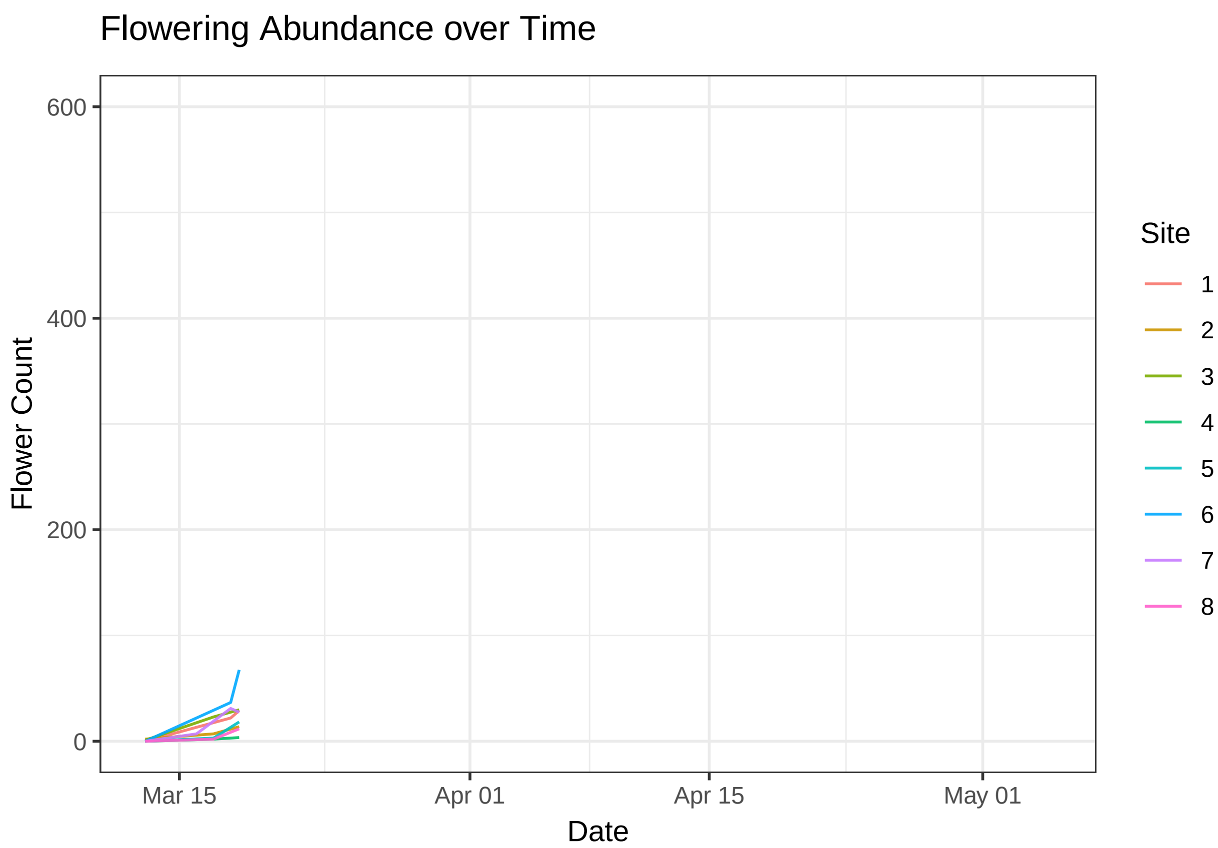
Flowering Abundance over Time
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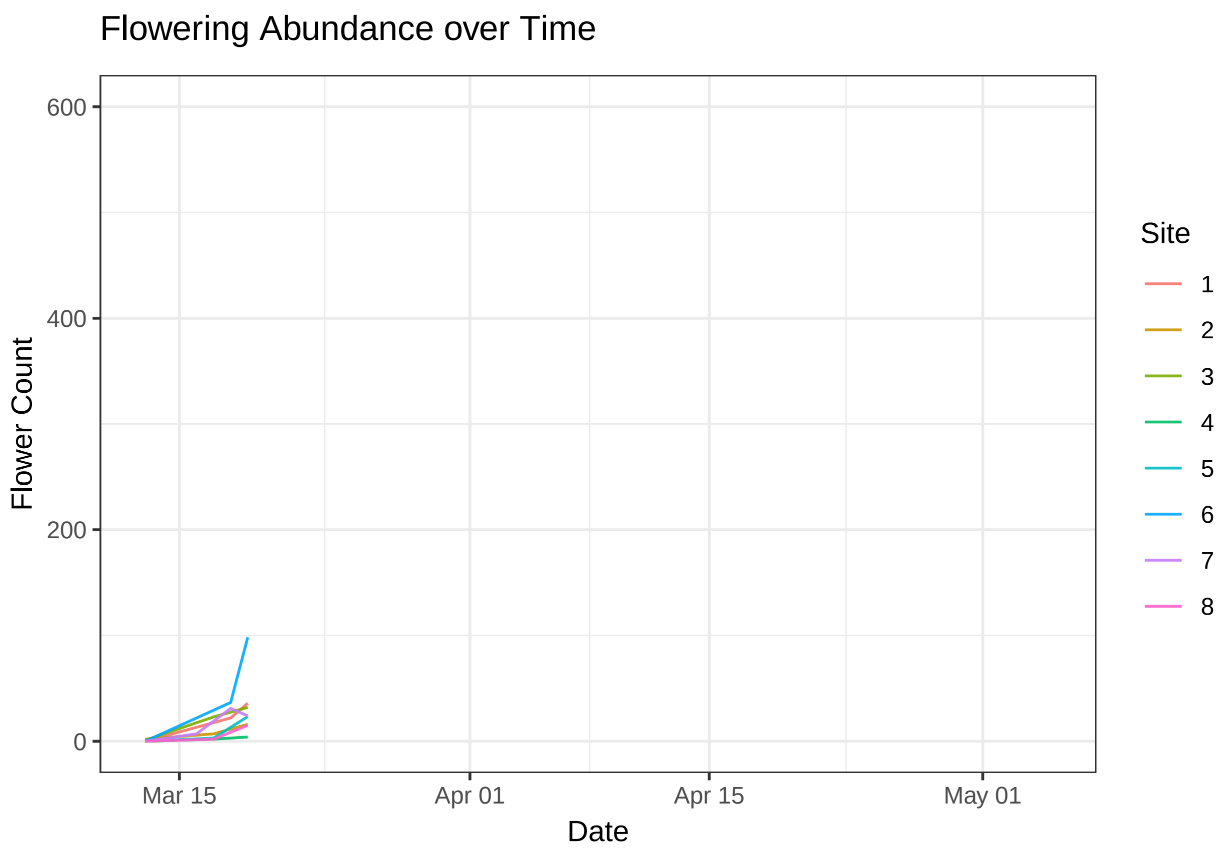
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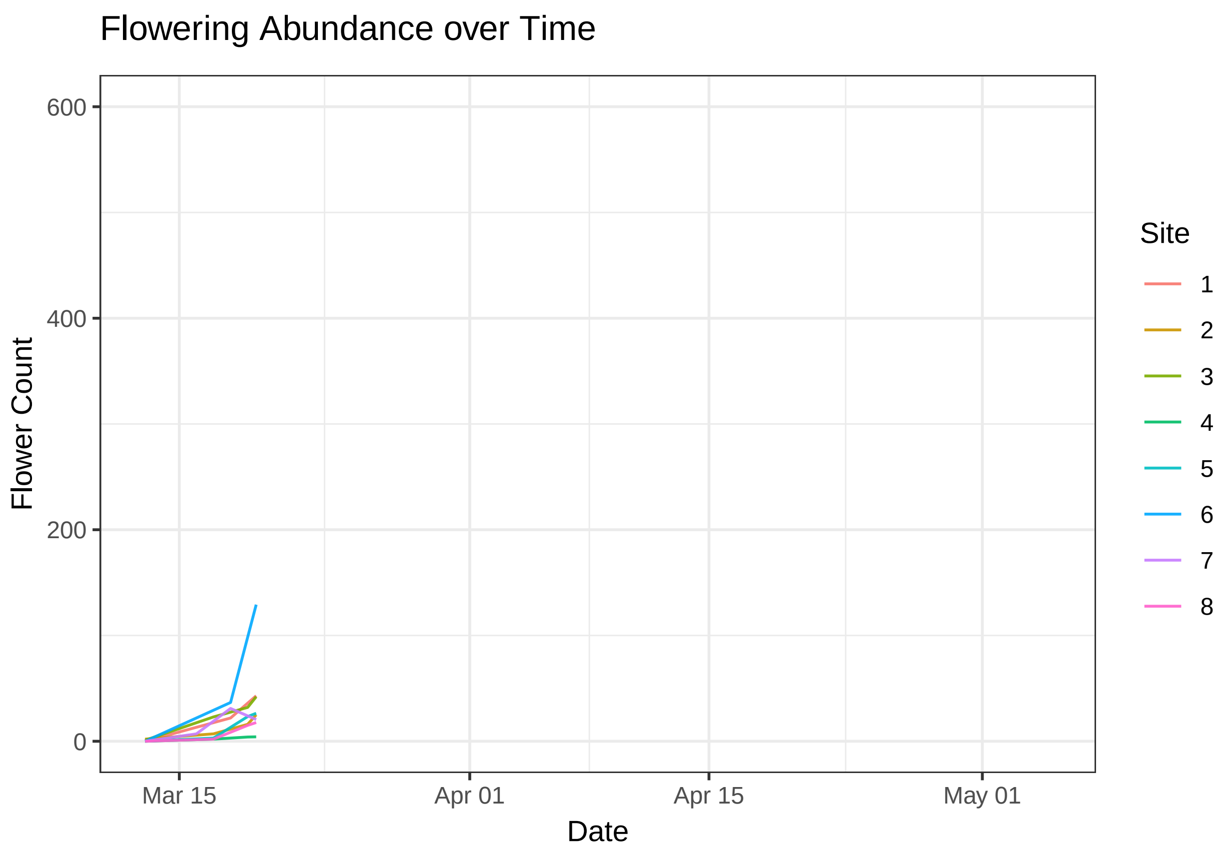

Flowering Abundance over Time
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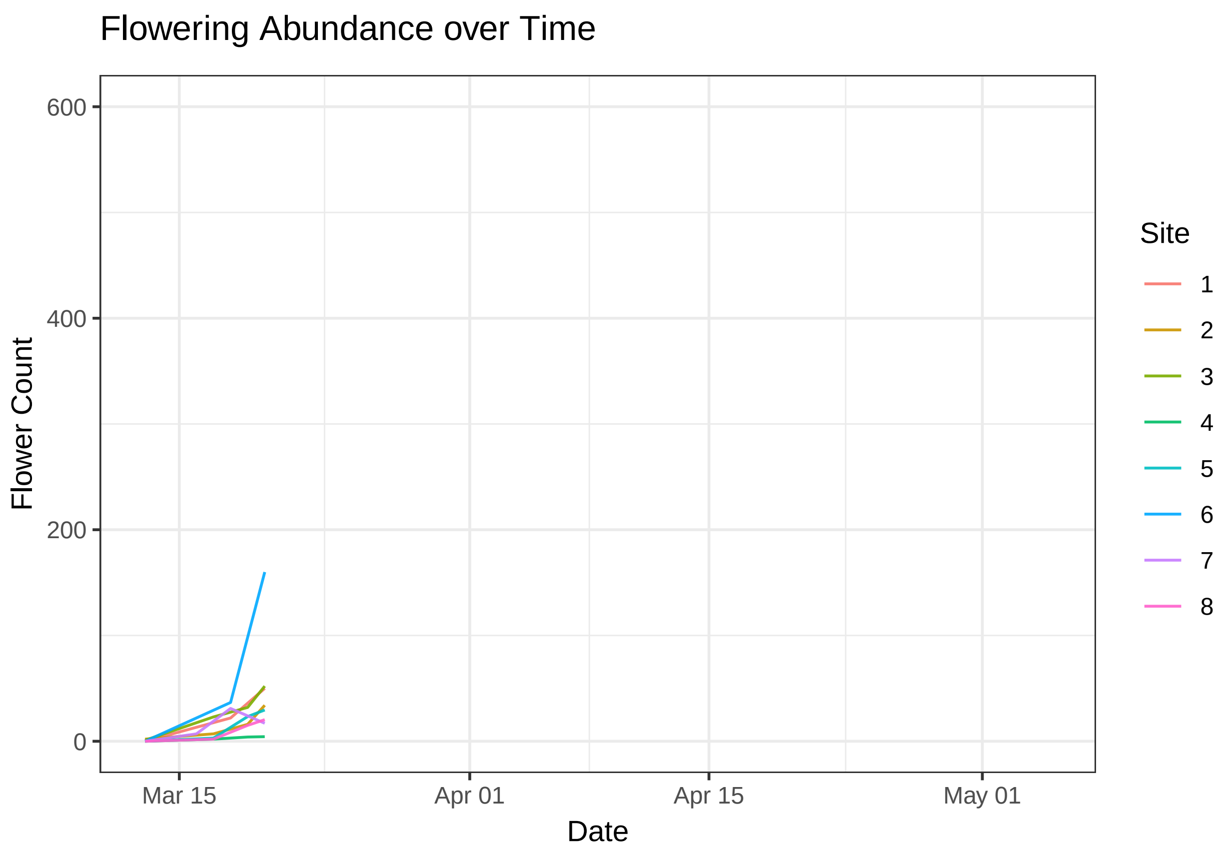
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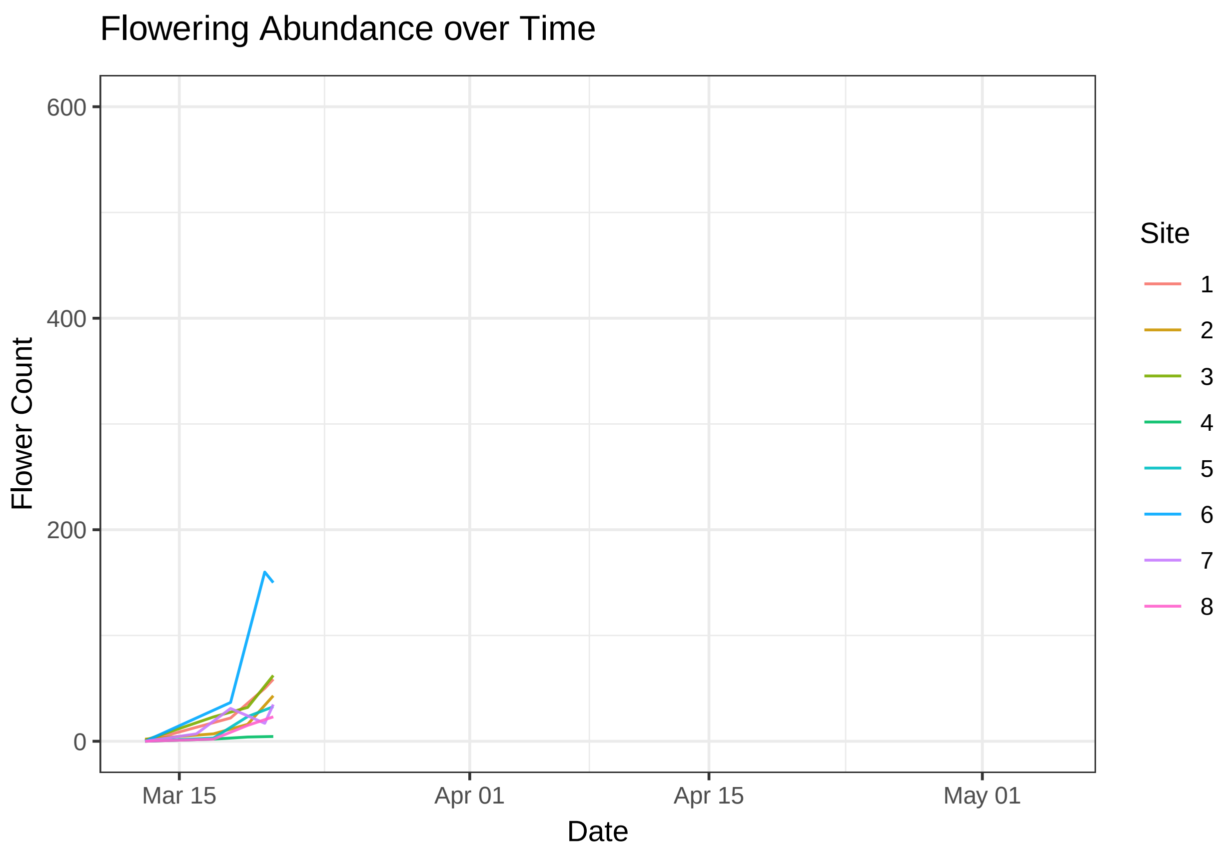
Flowering Abundance over Time
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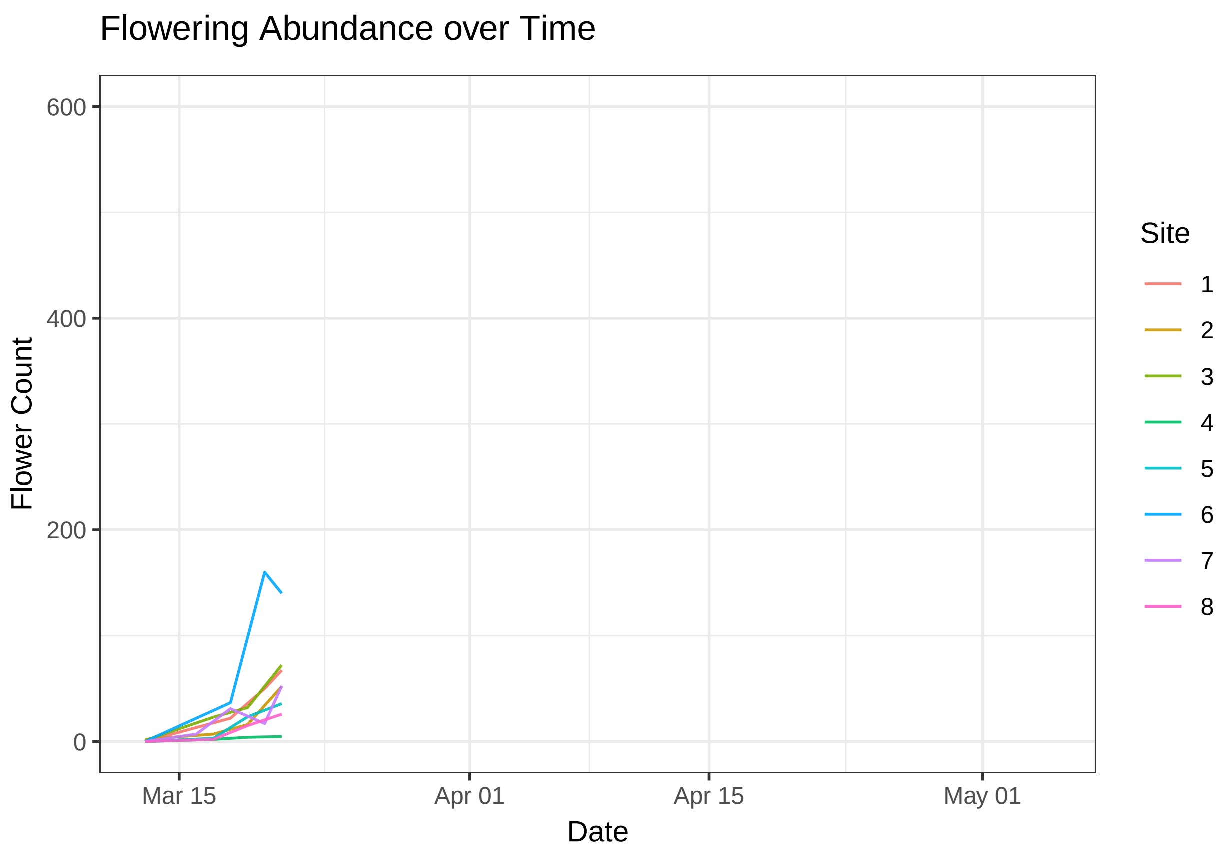

Flowering Abundance over Time
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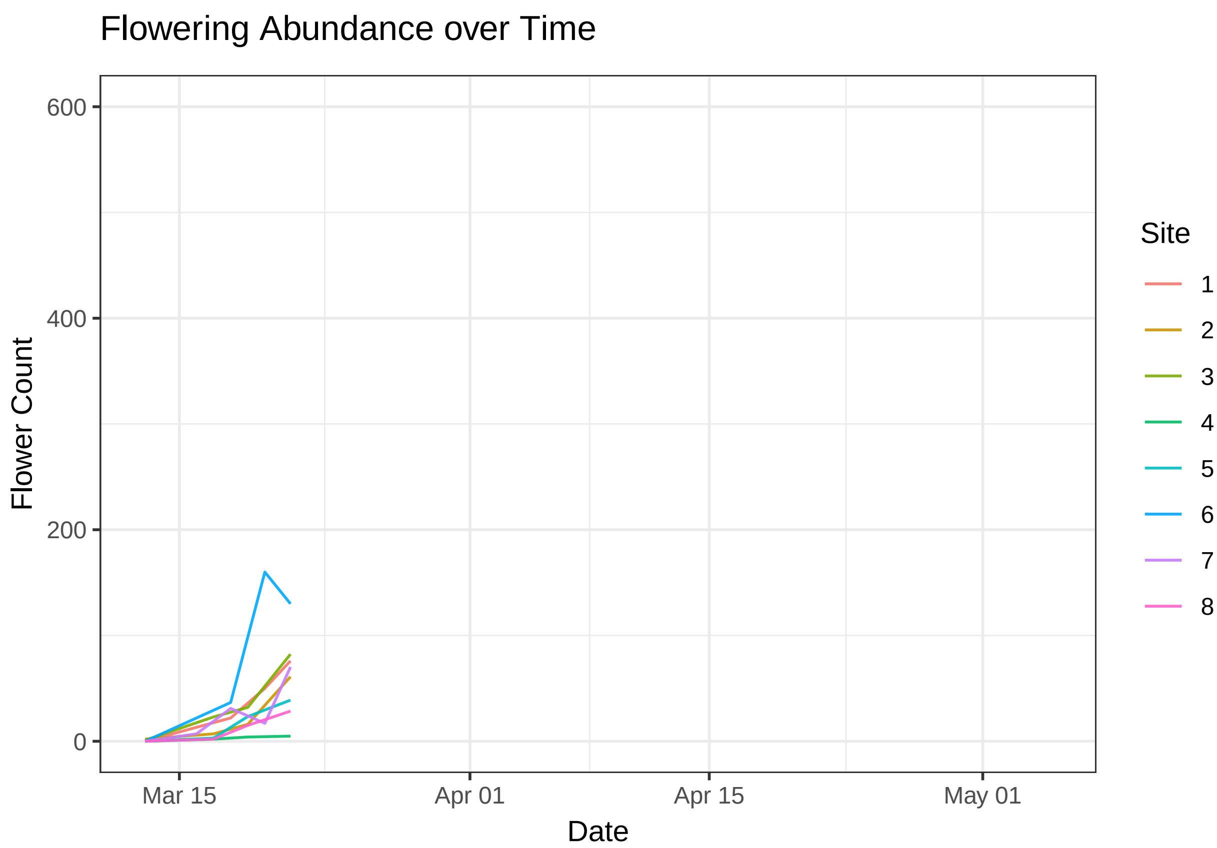
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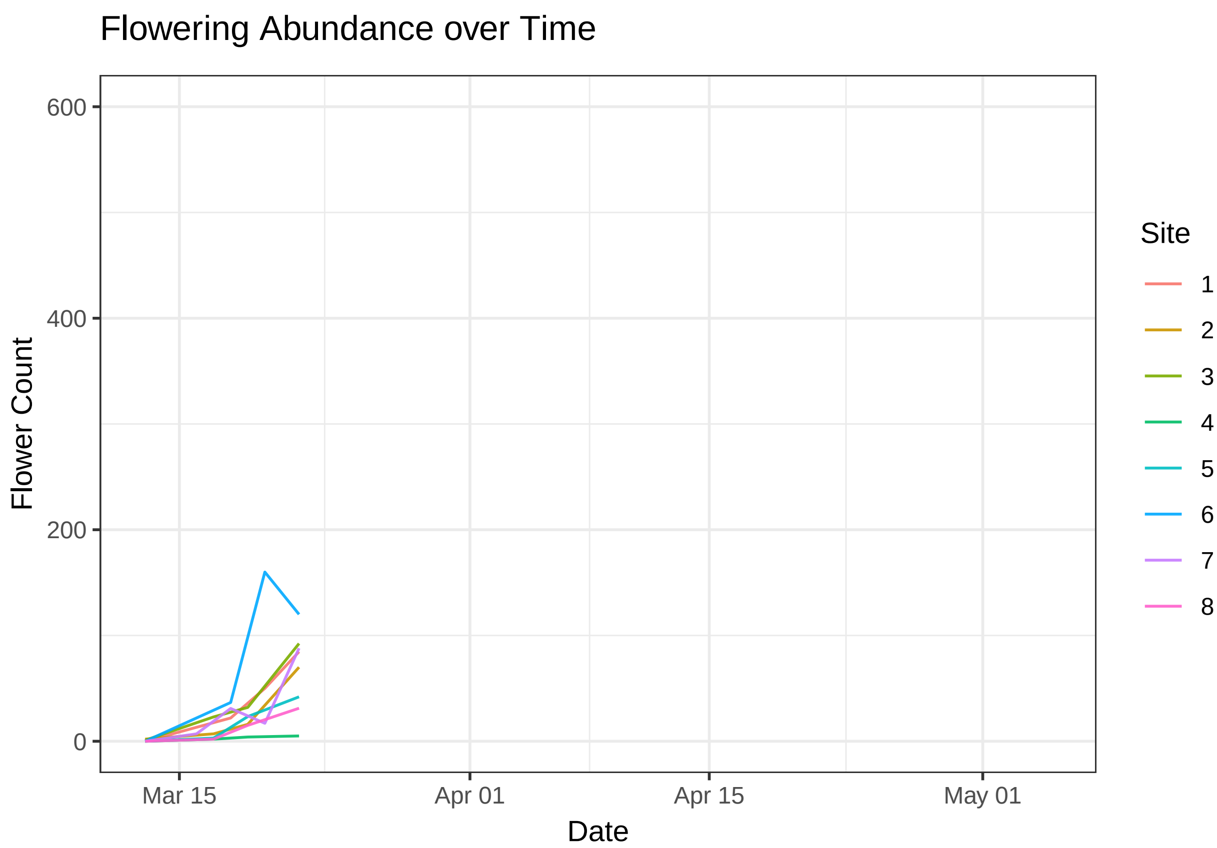
Flowering Abundance over Time
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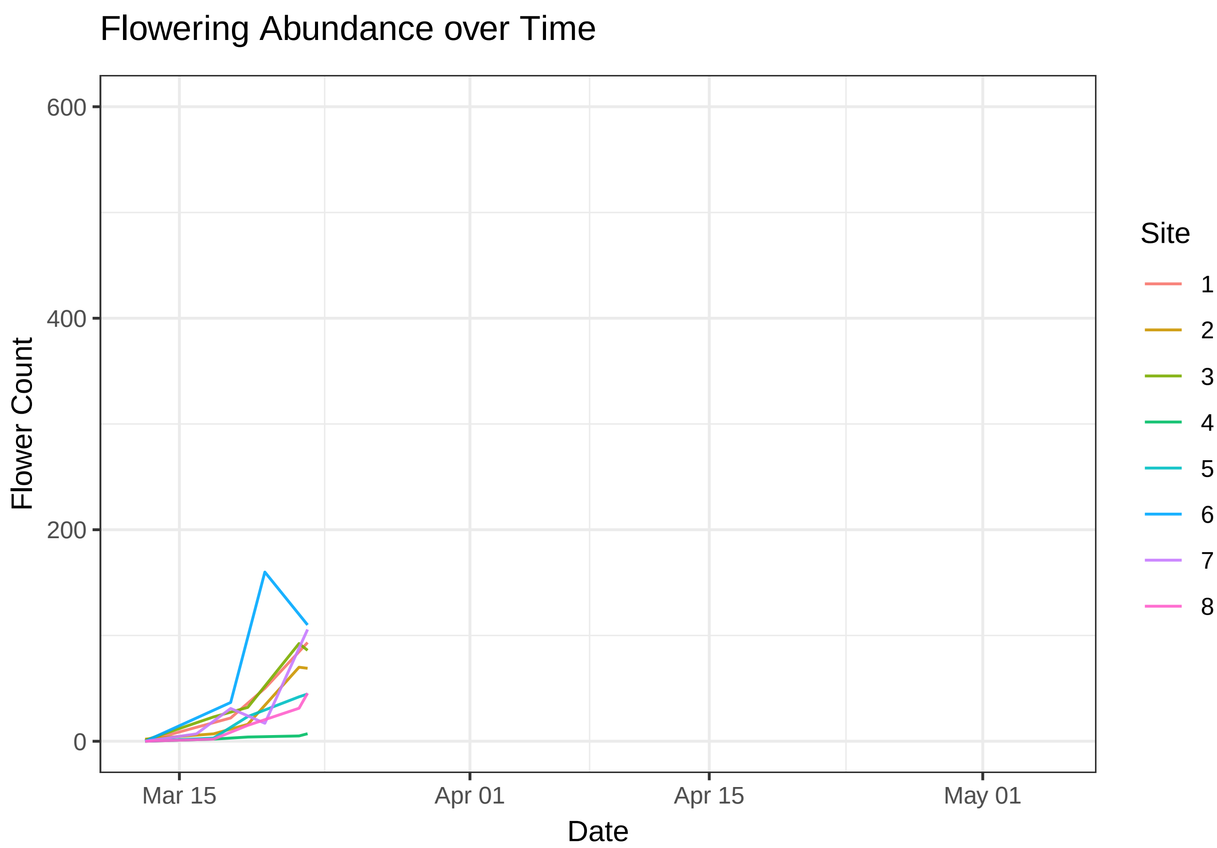
Flowering Abundance over Time
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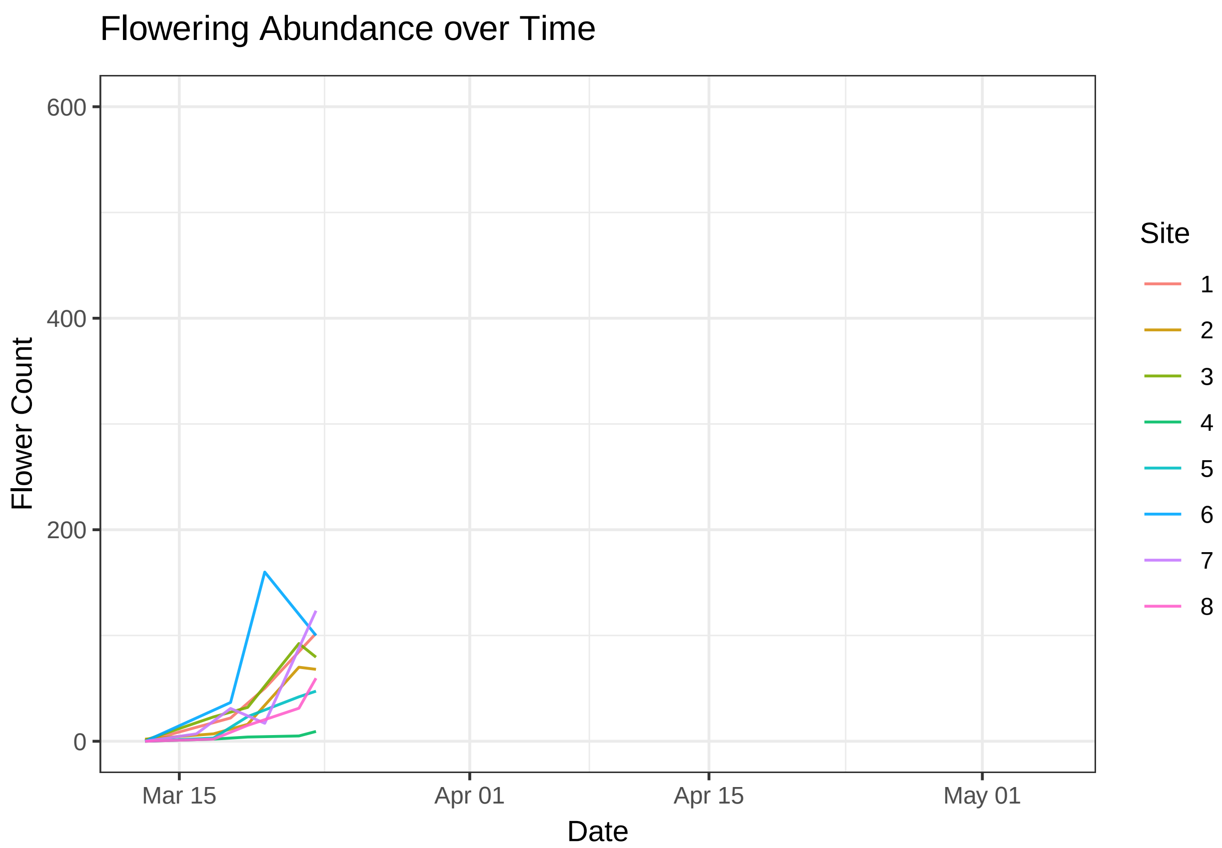
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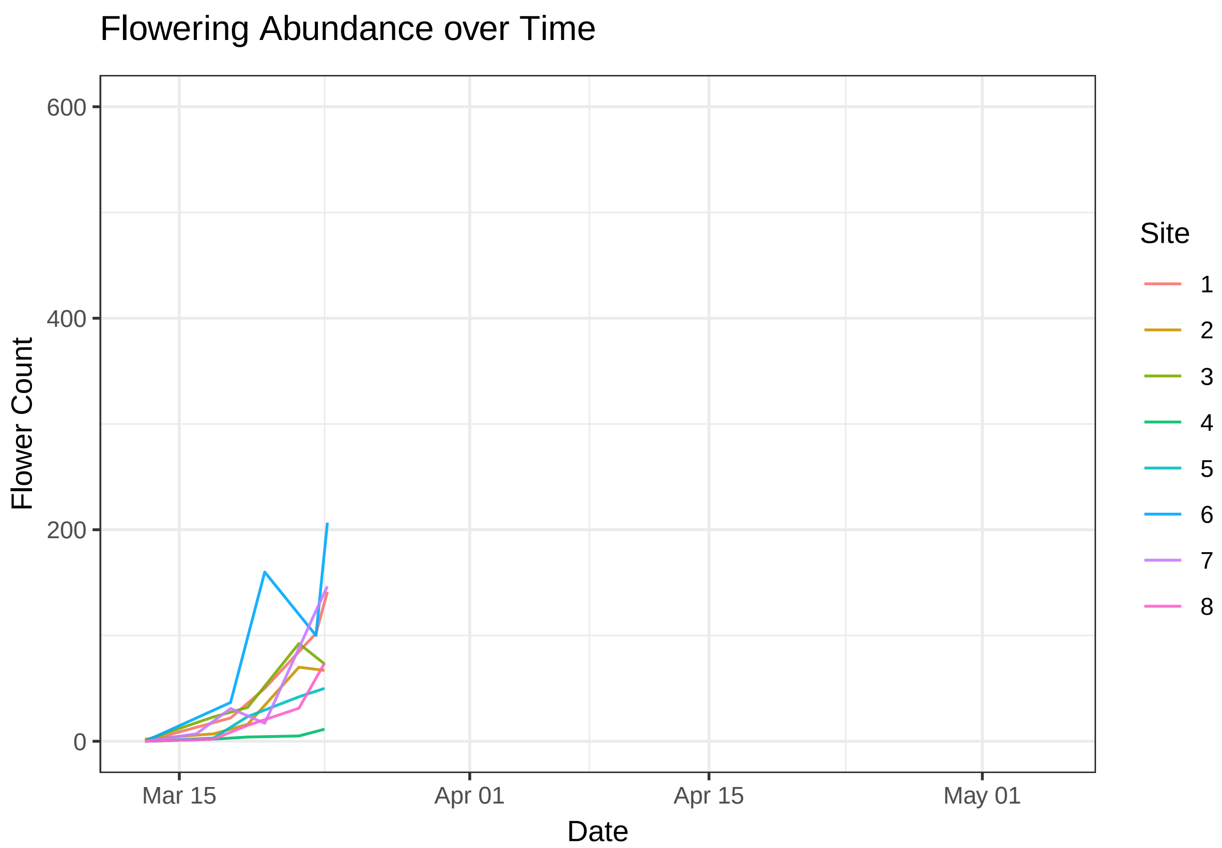
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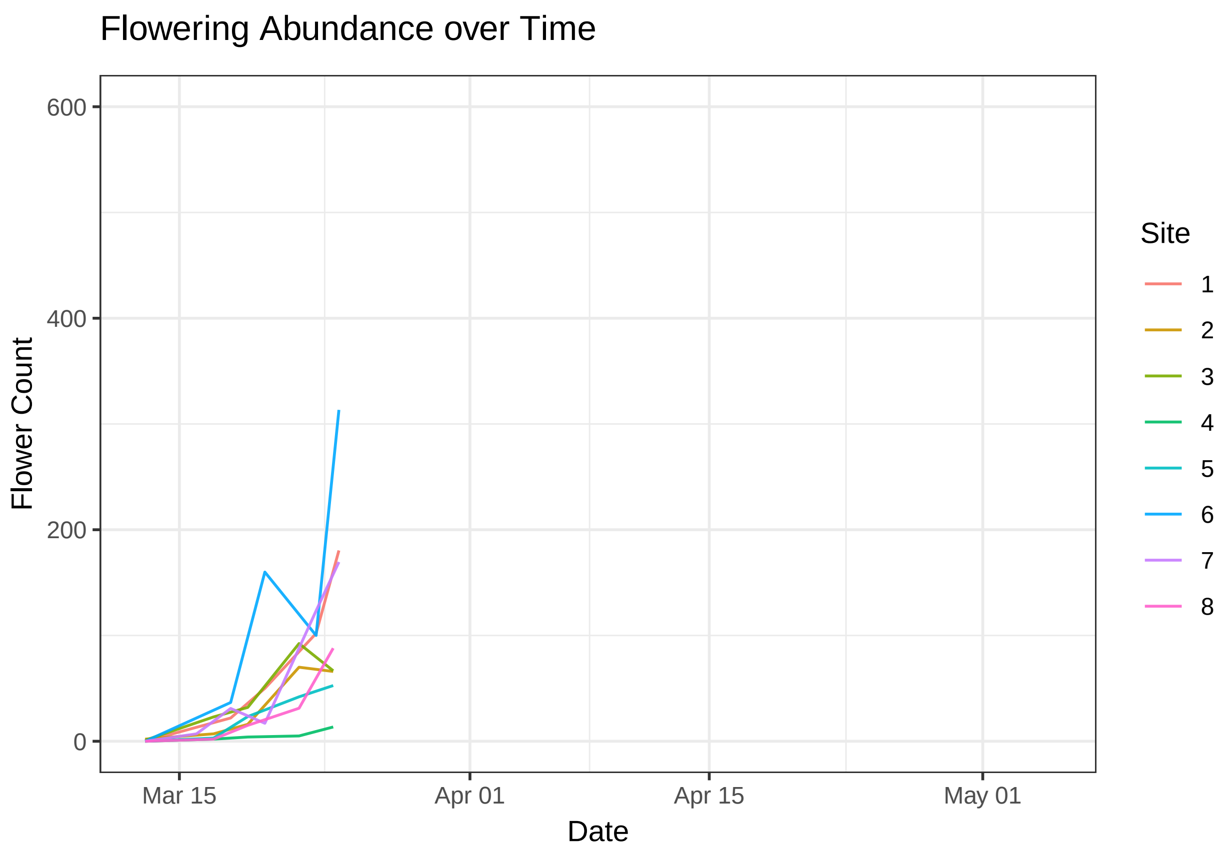
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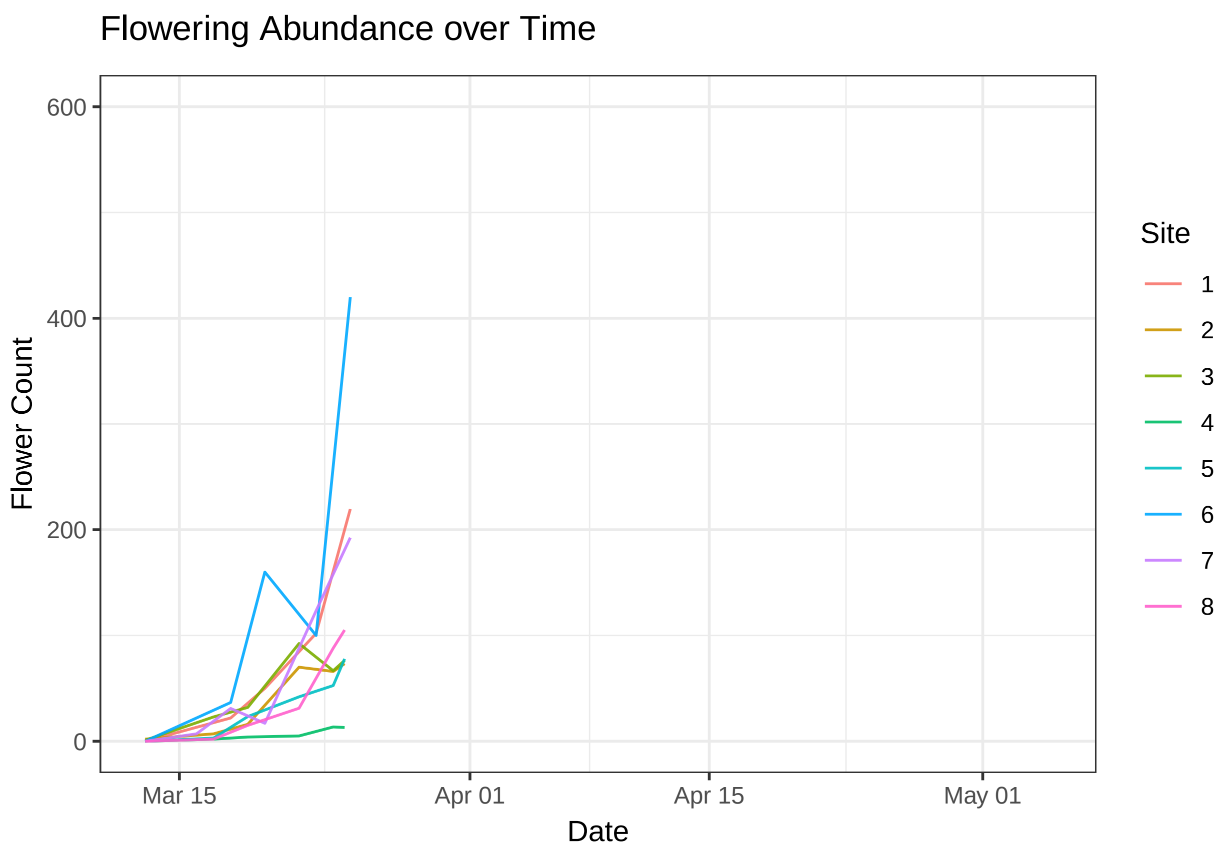

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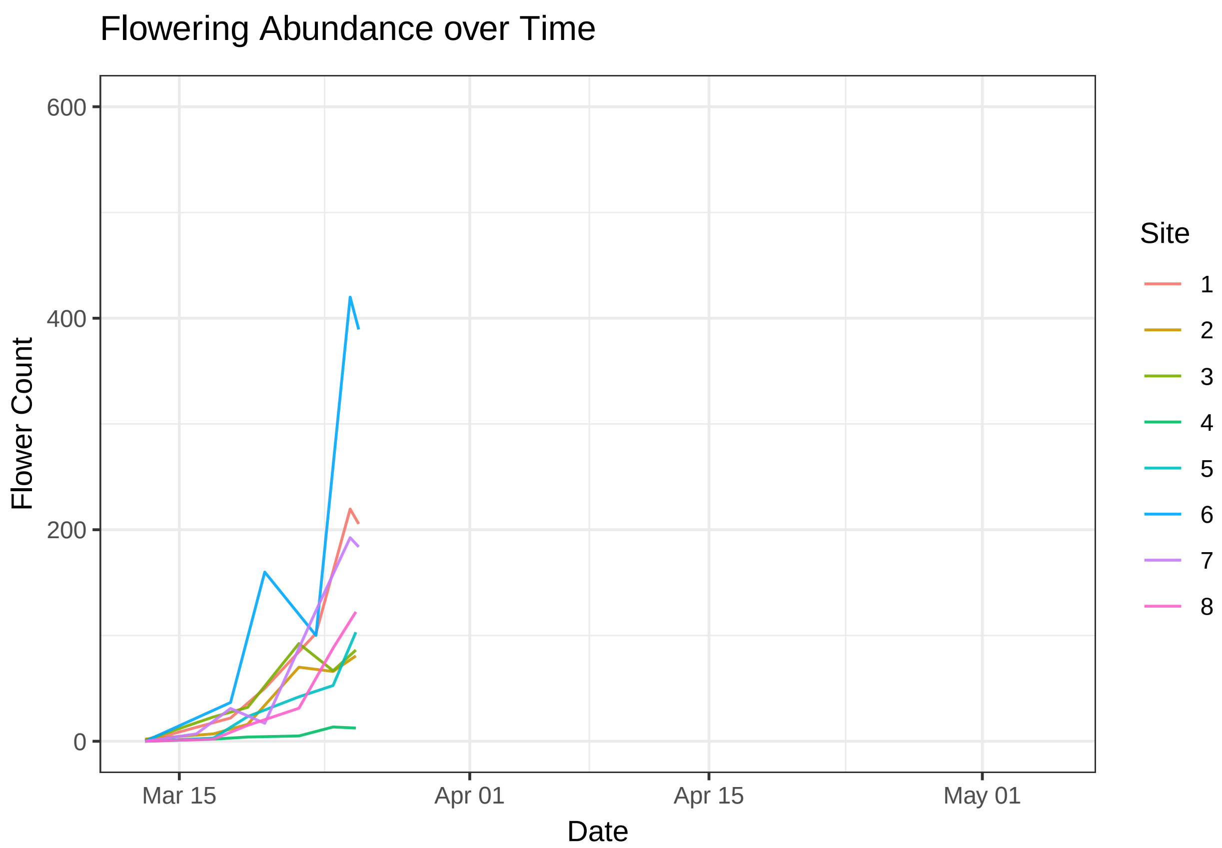

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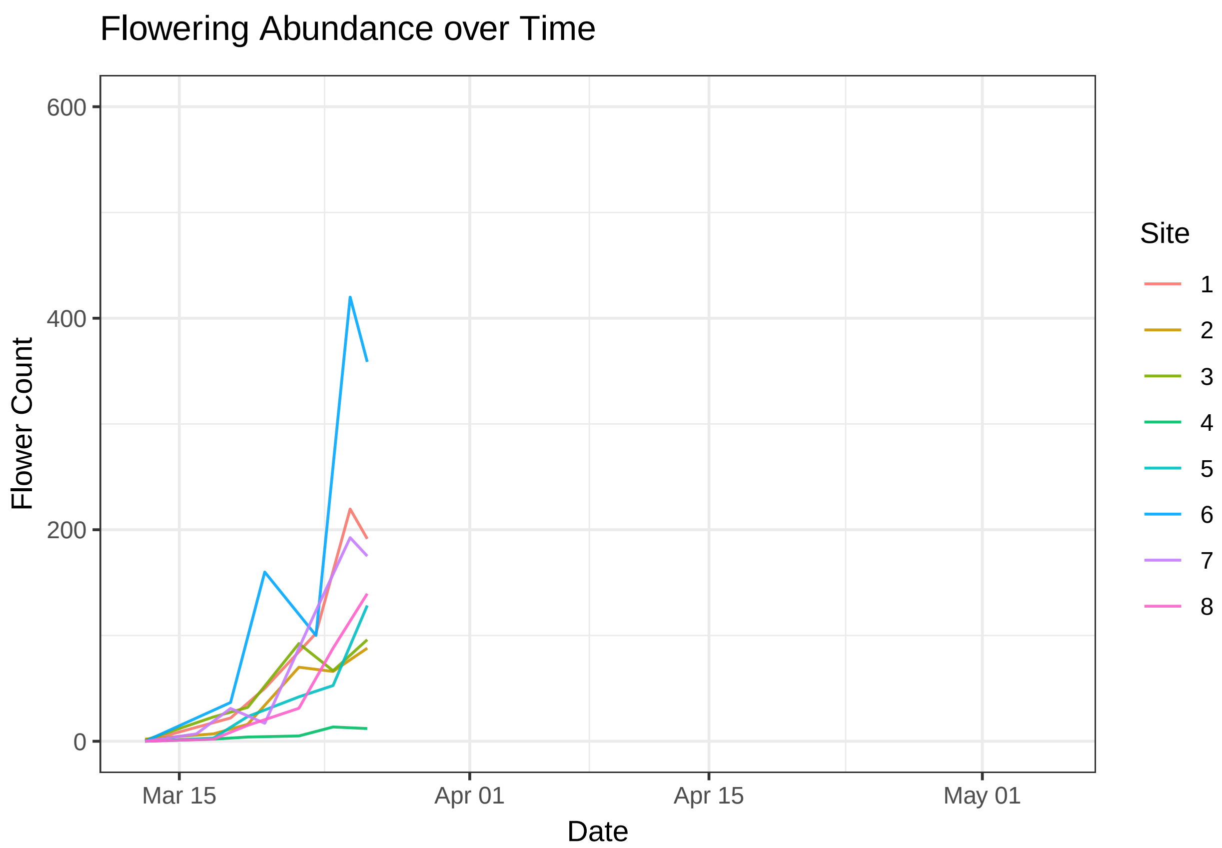

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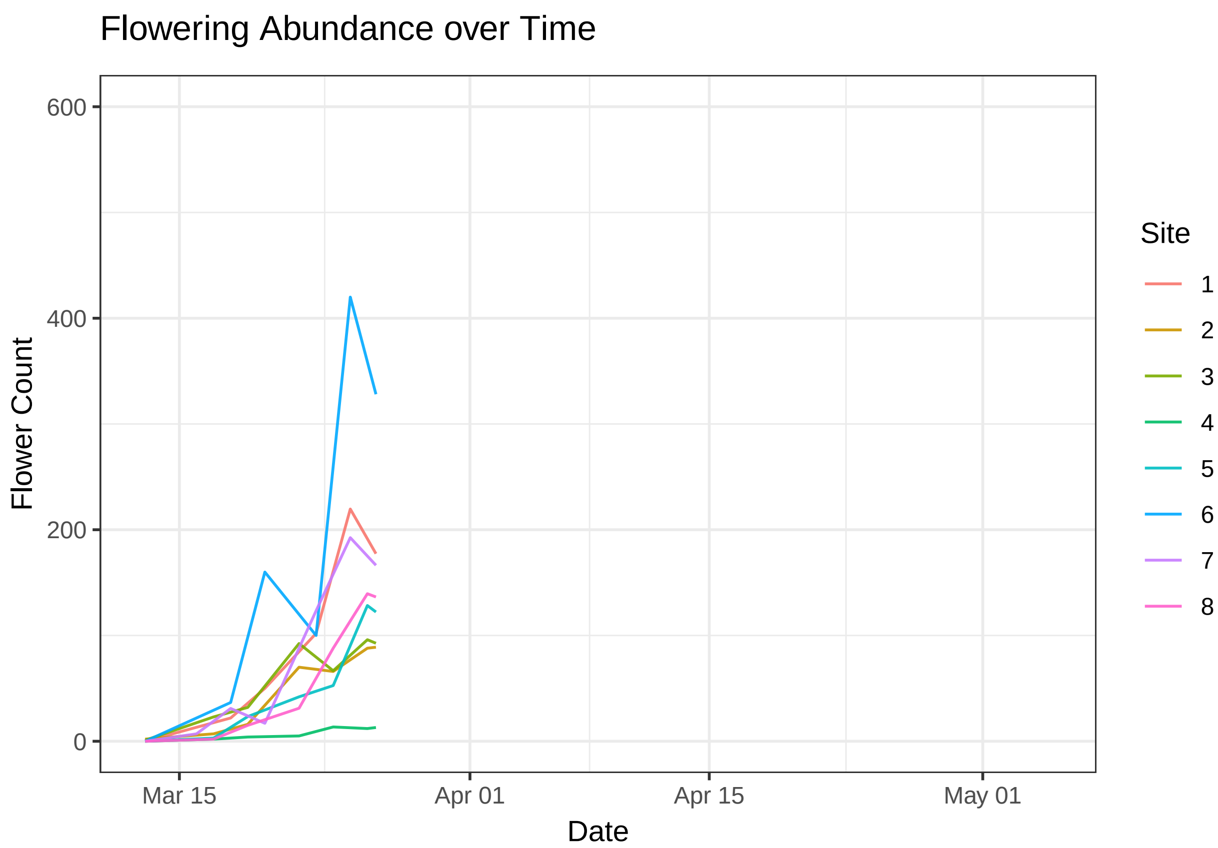
Flowering Abundance over Time
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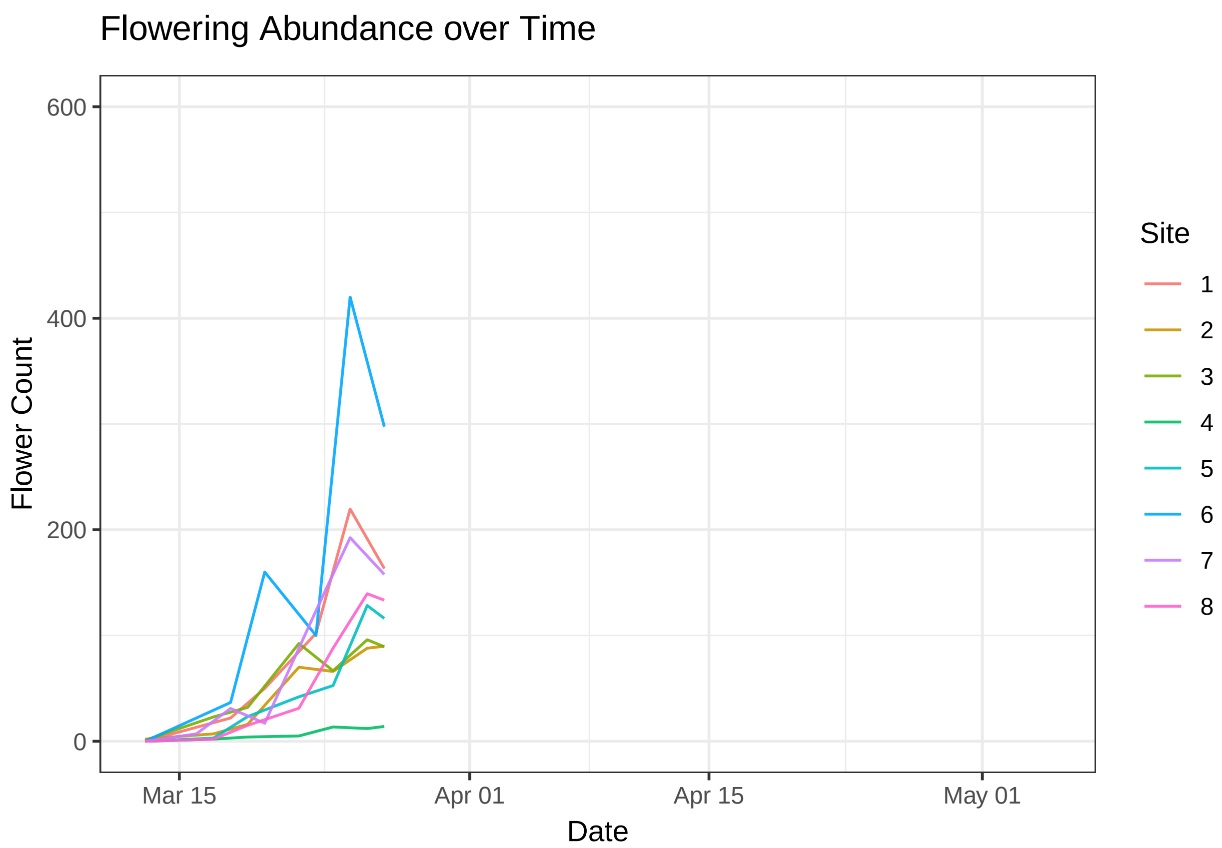
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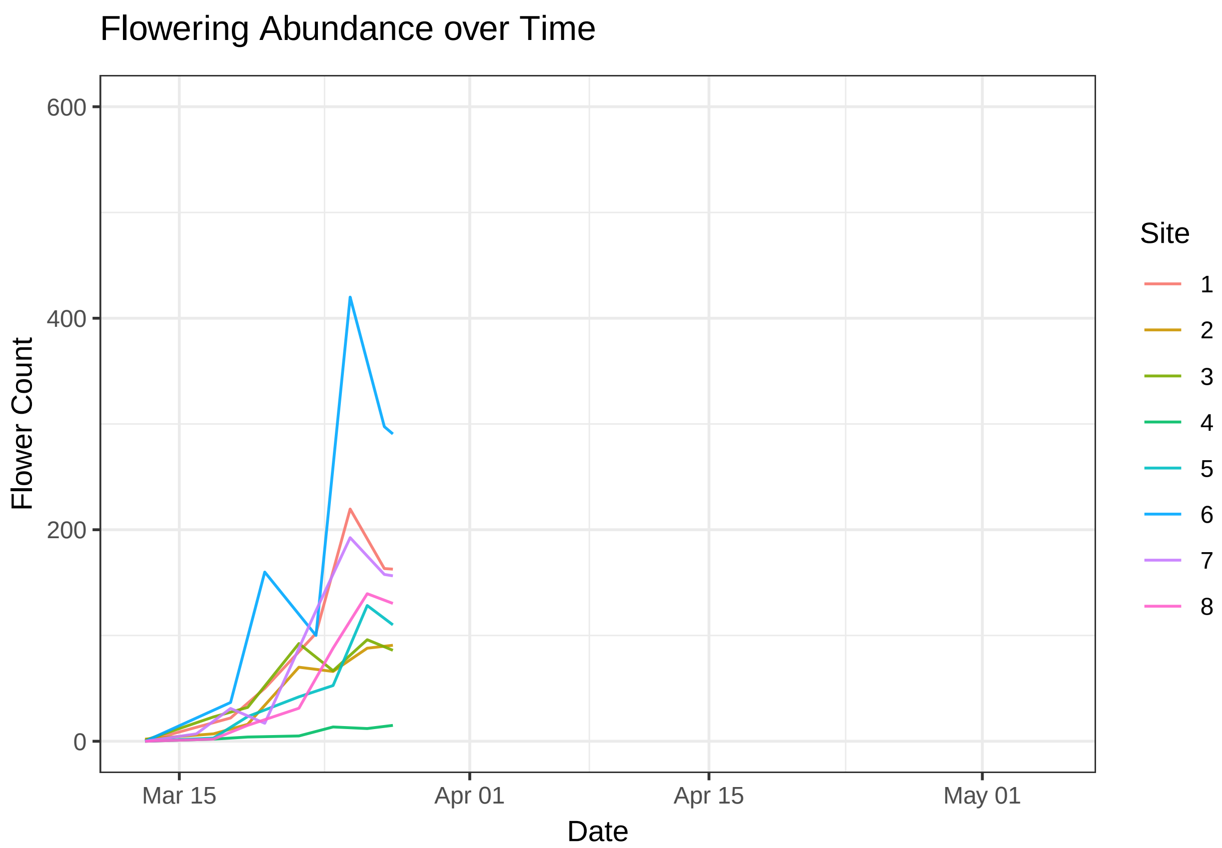
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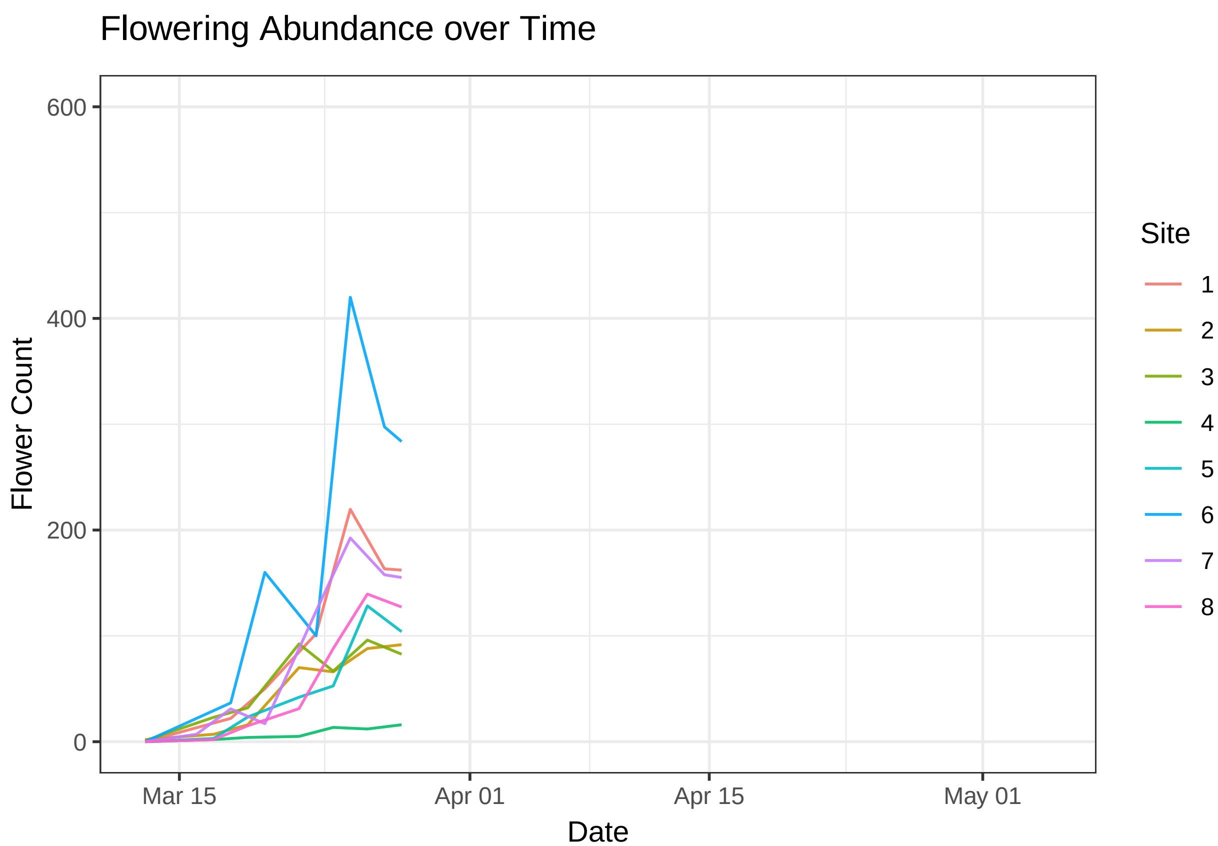

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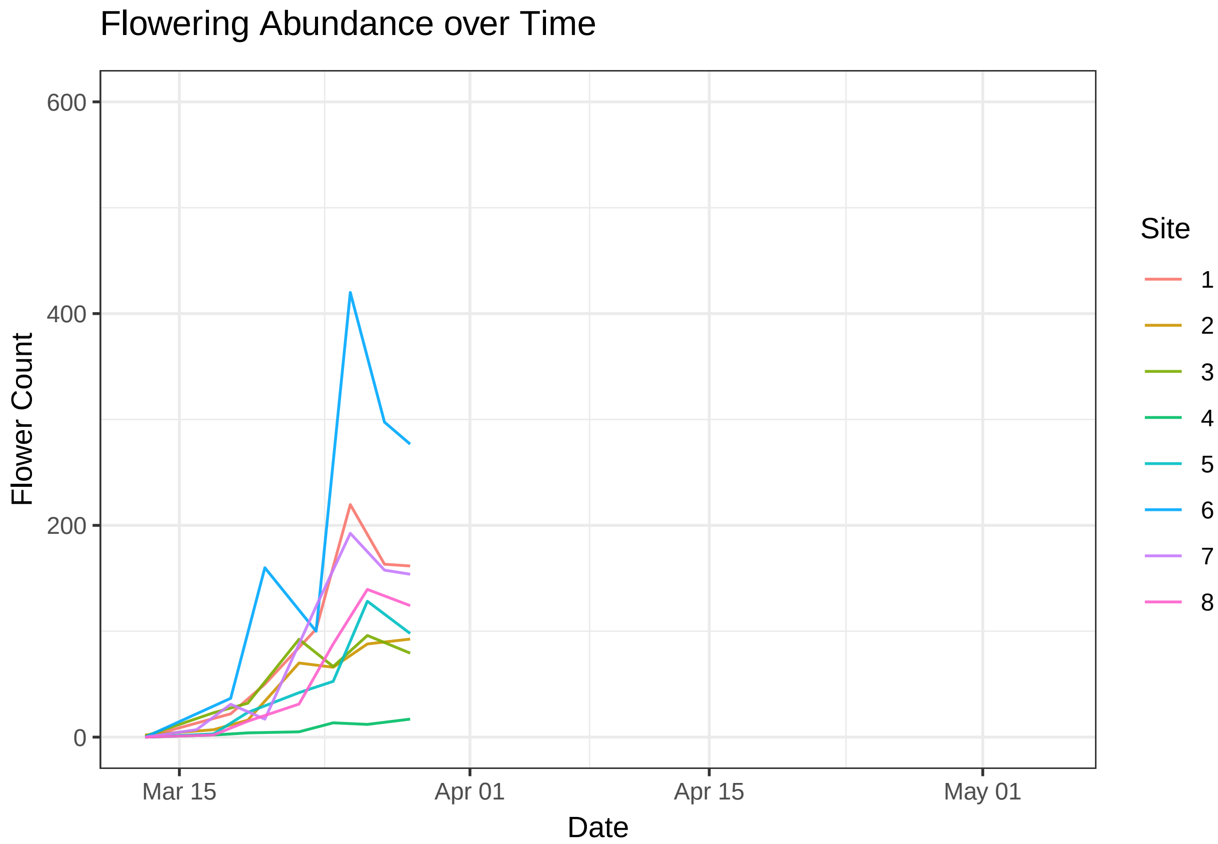
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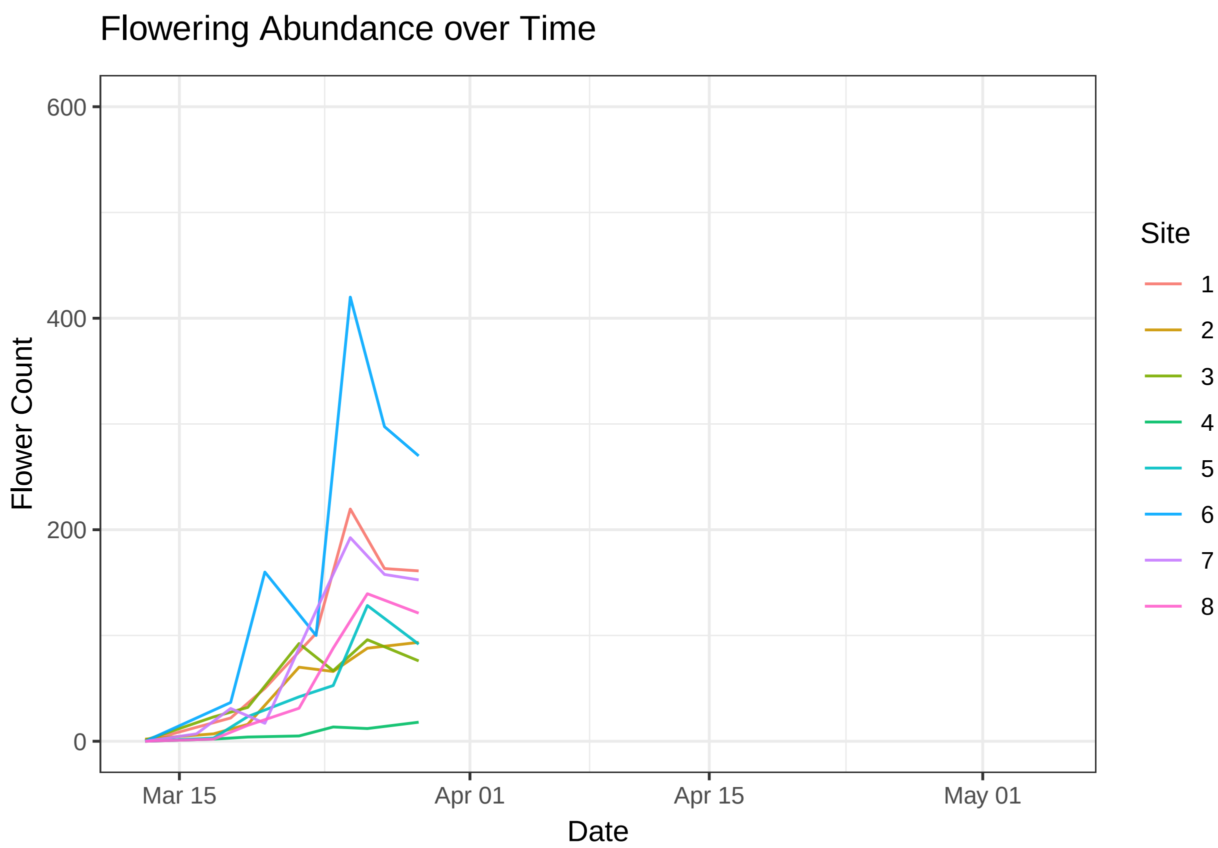
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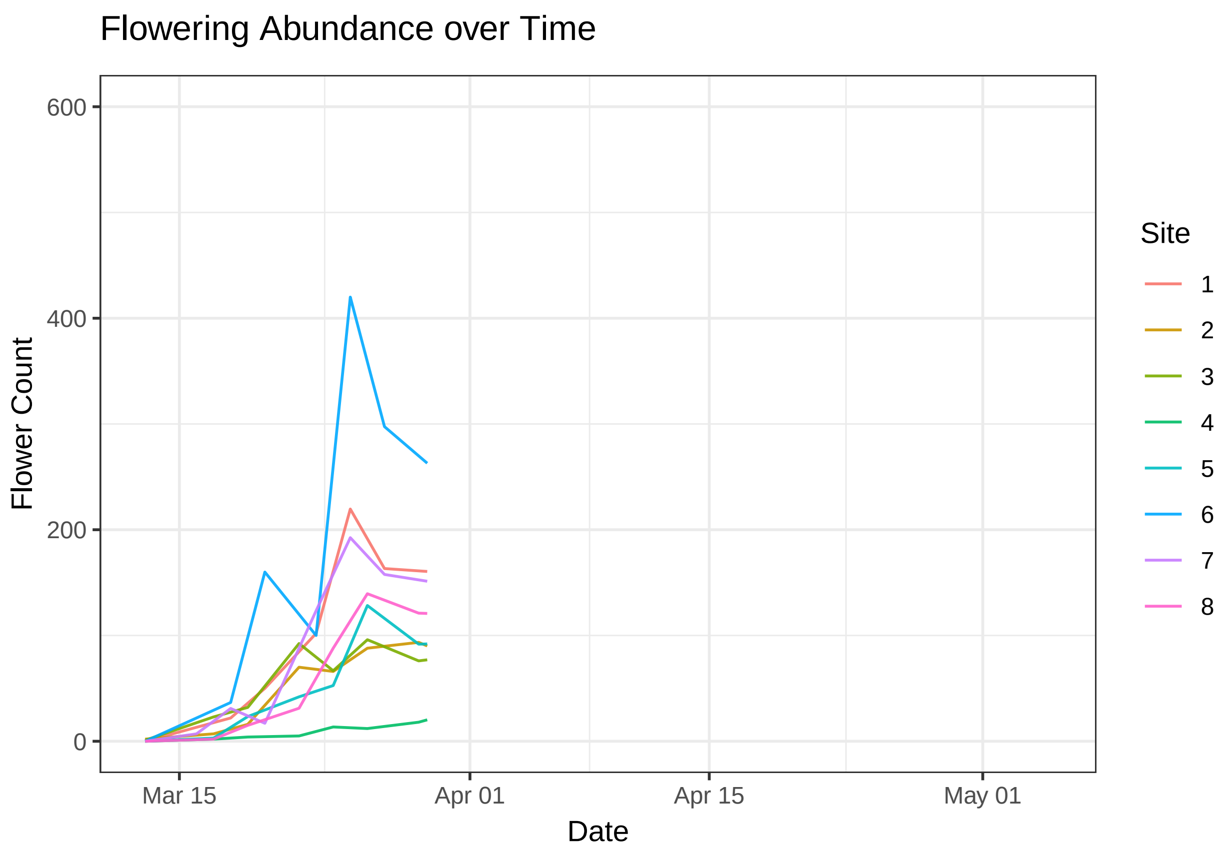
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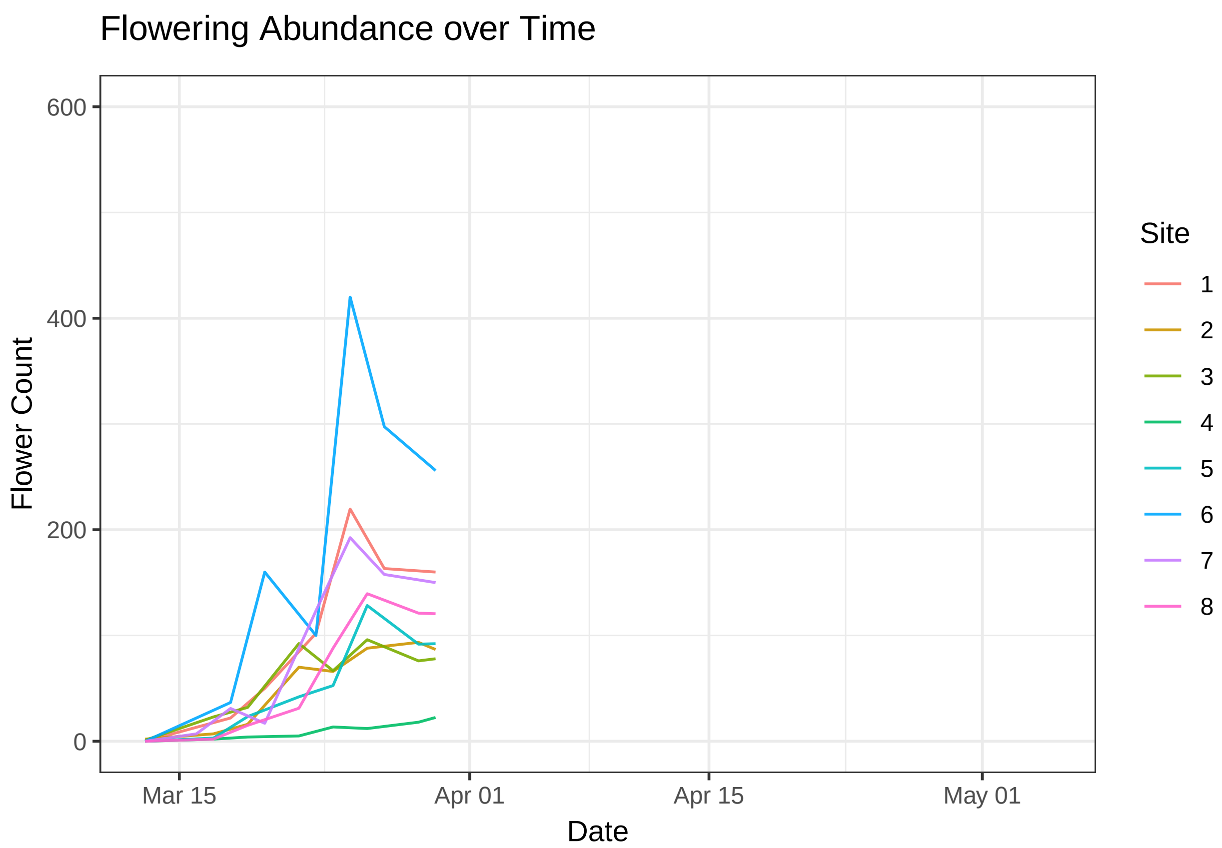
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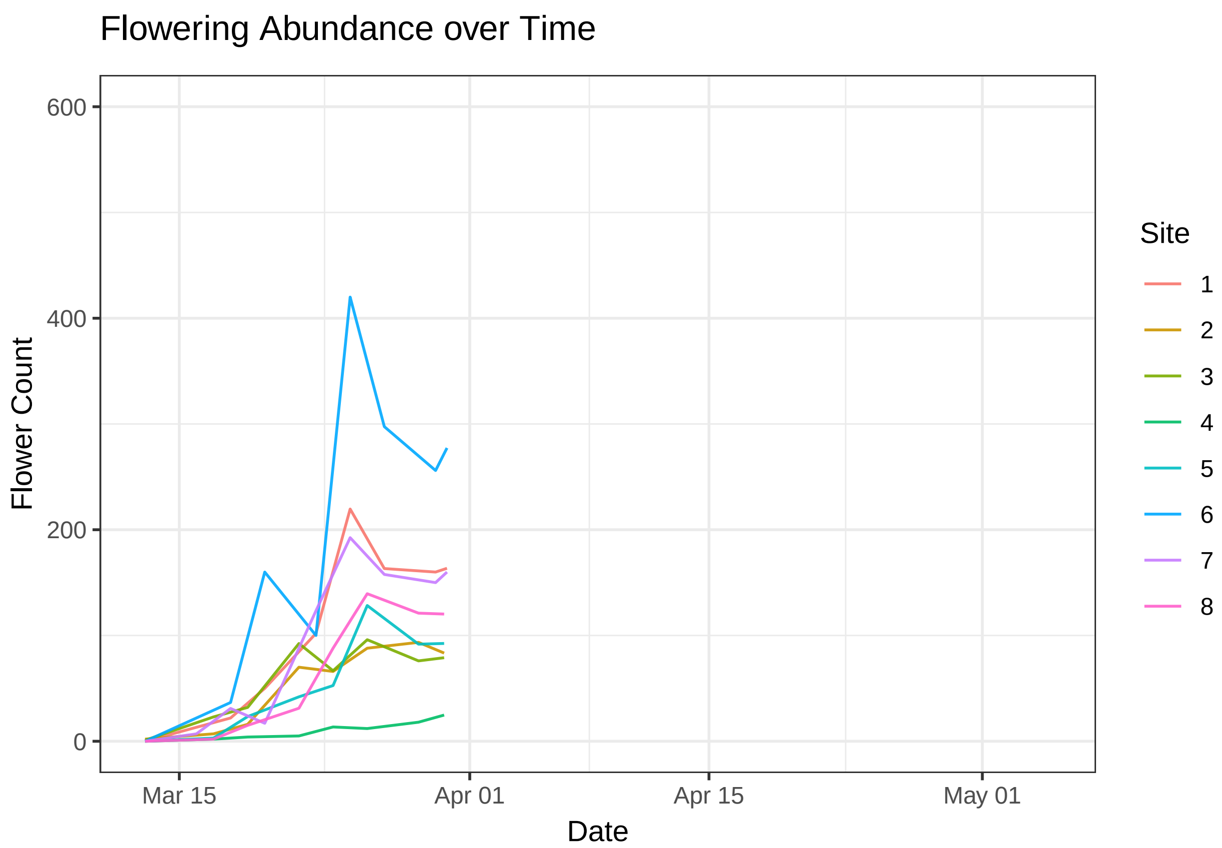

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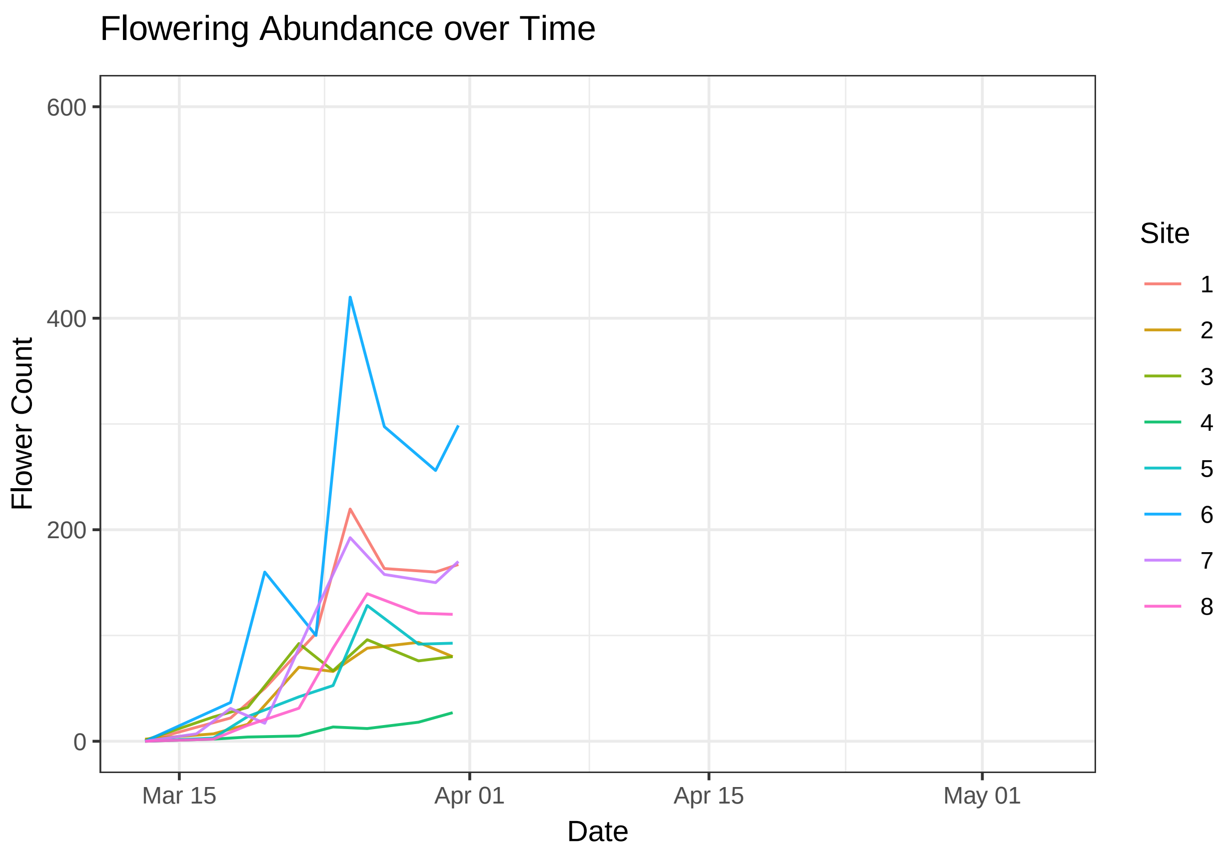
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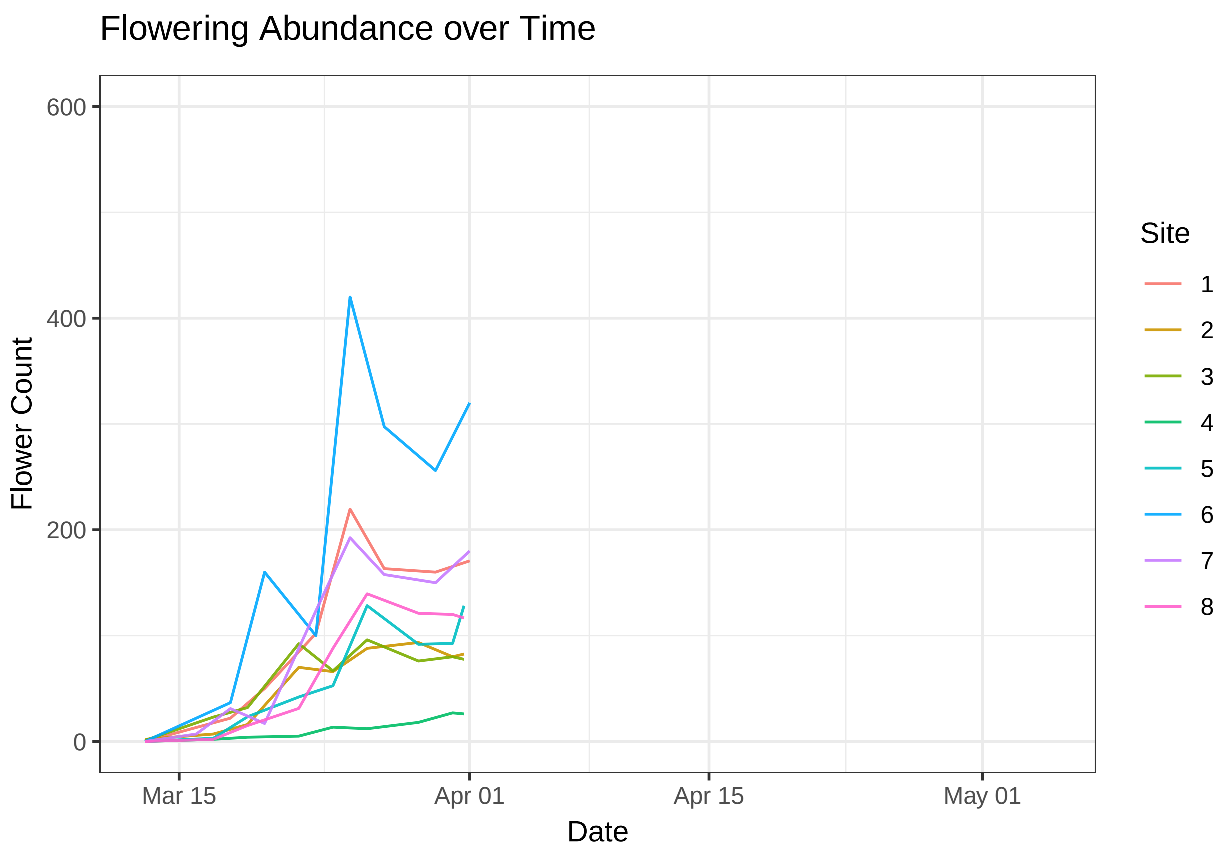

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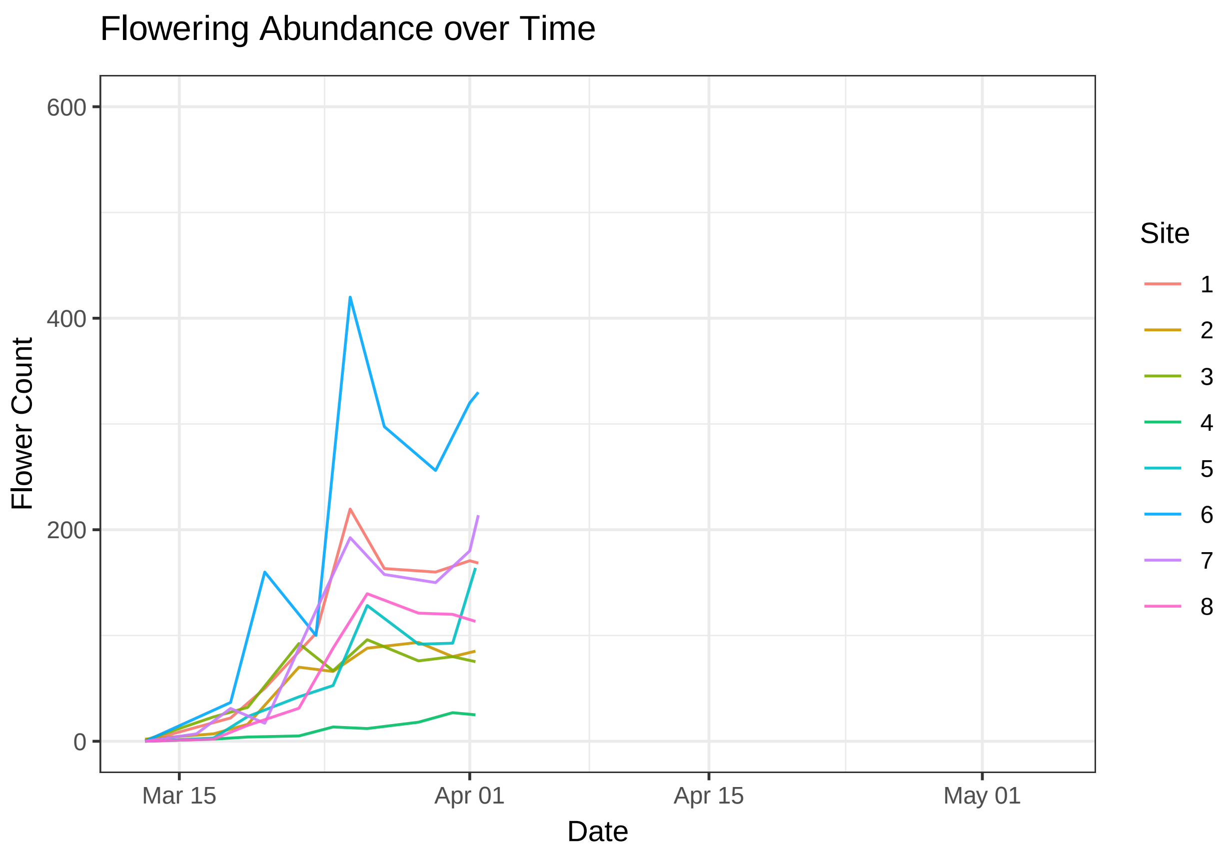
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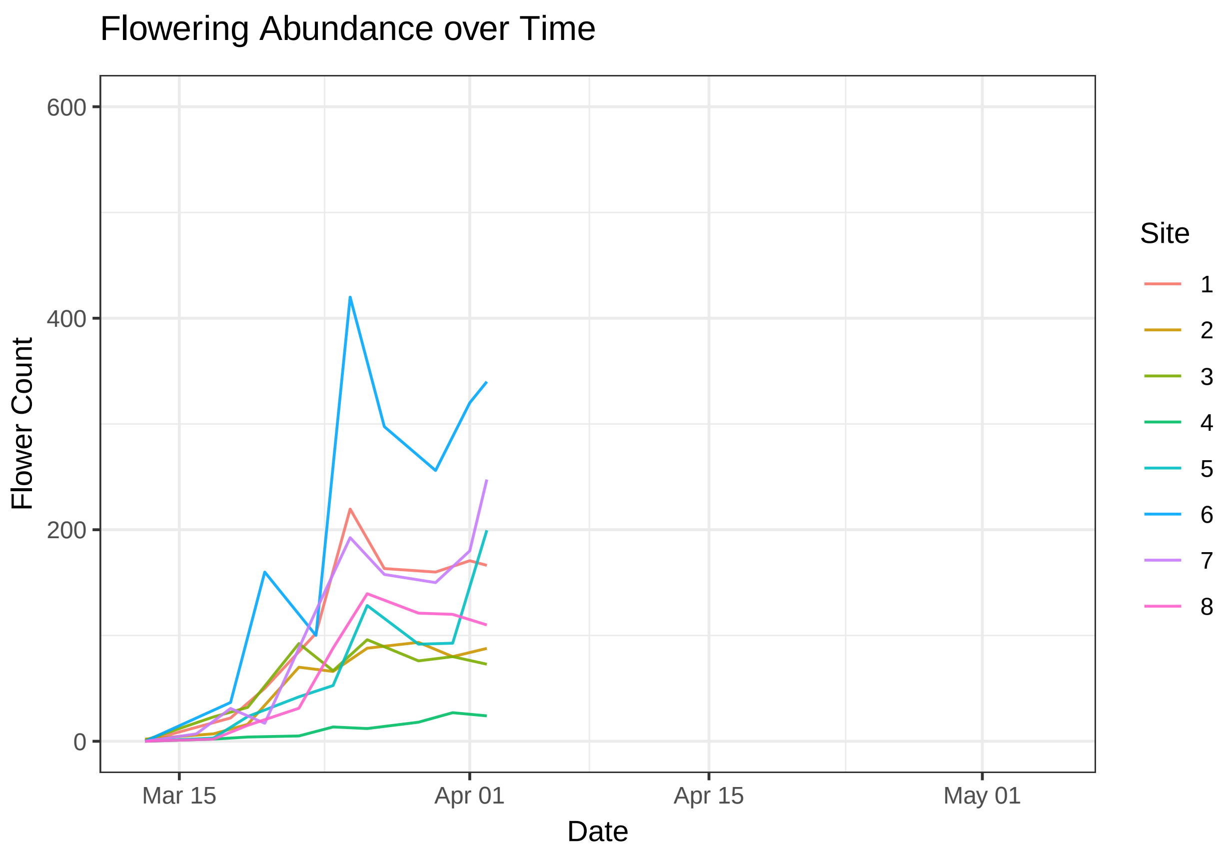
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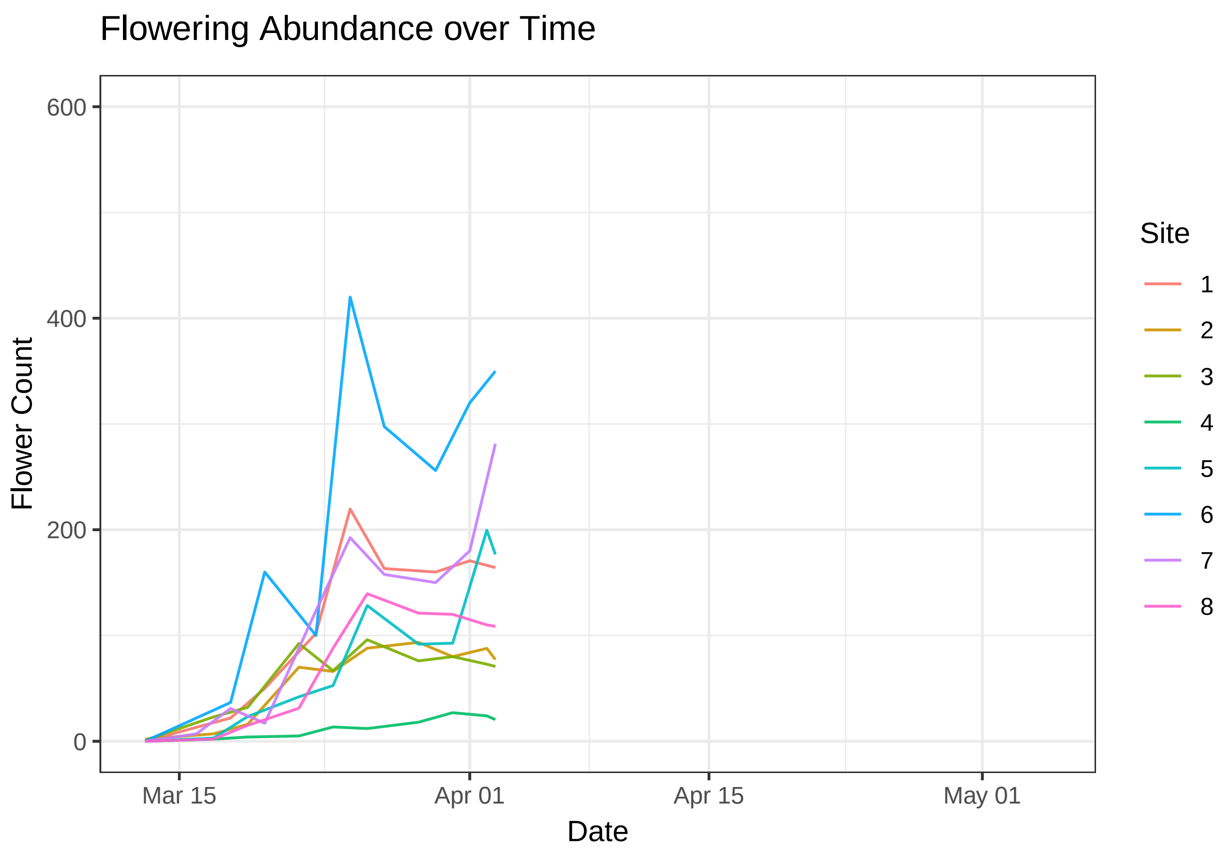
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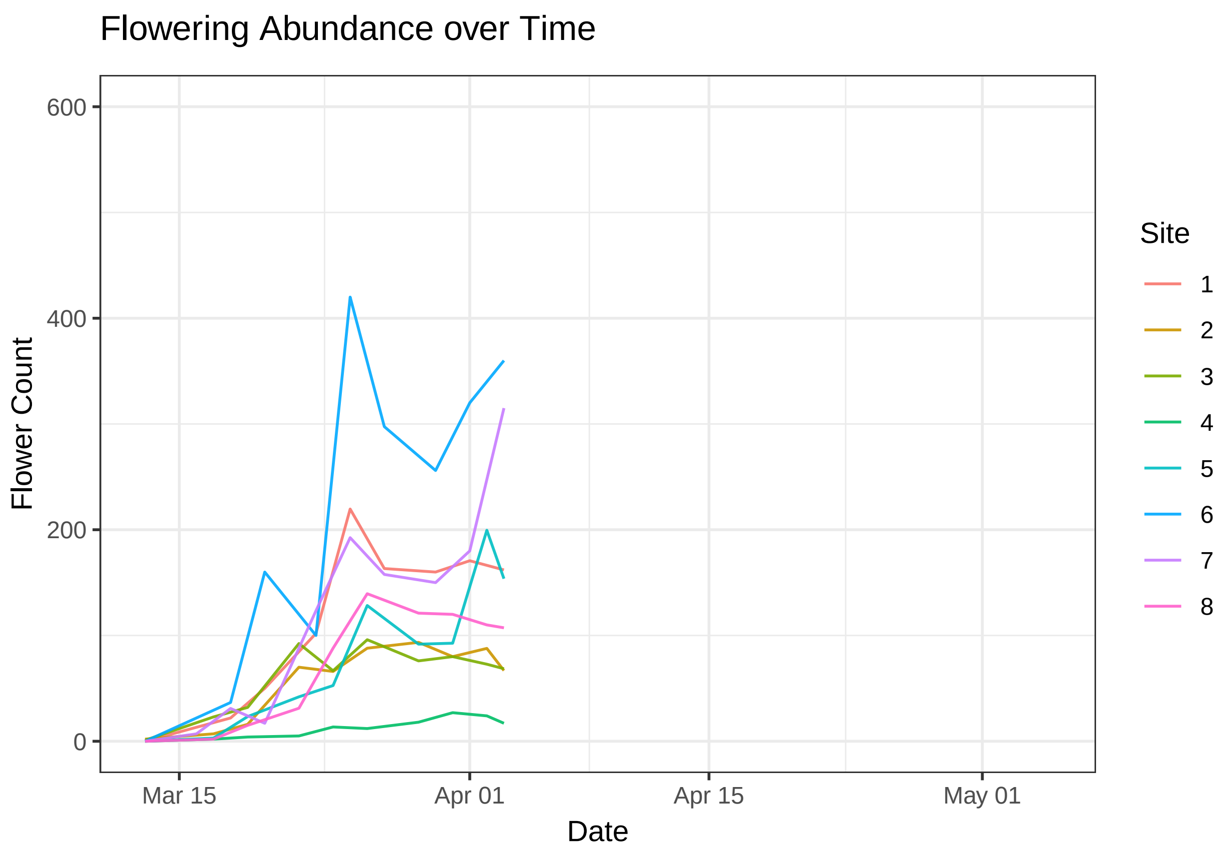

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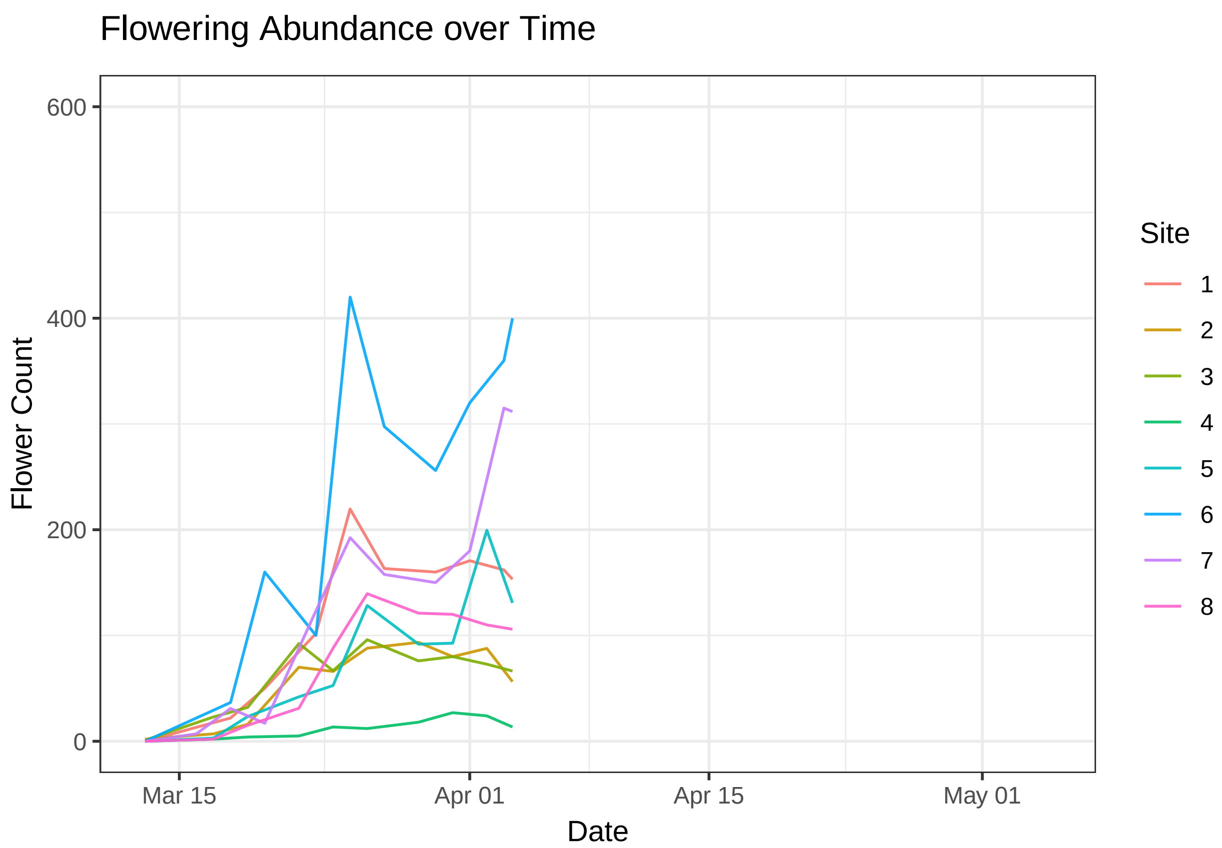
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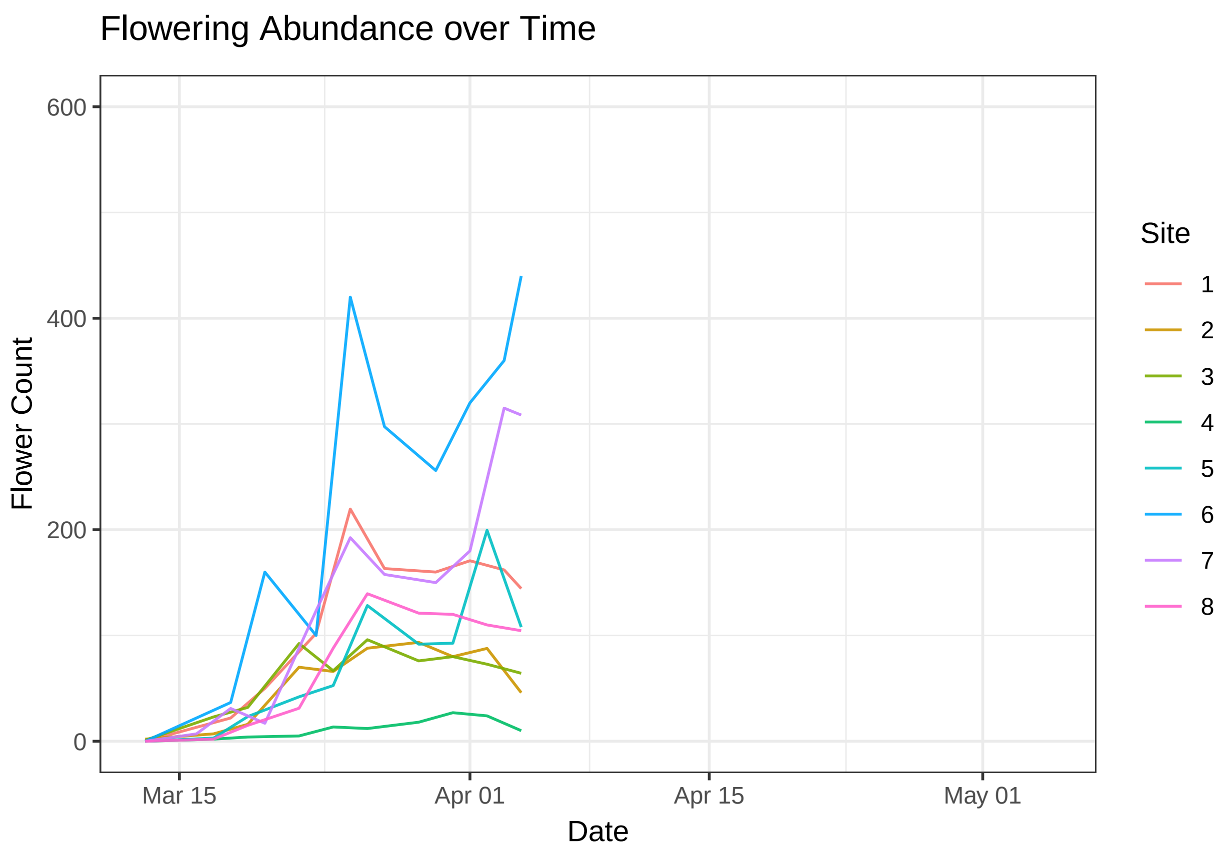
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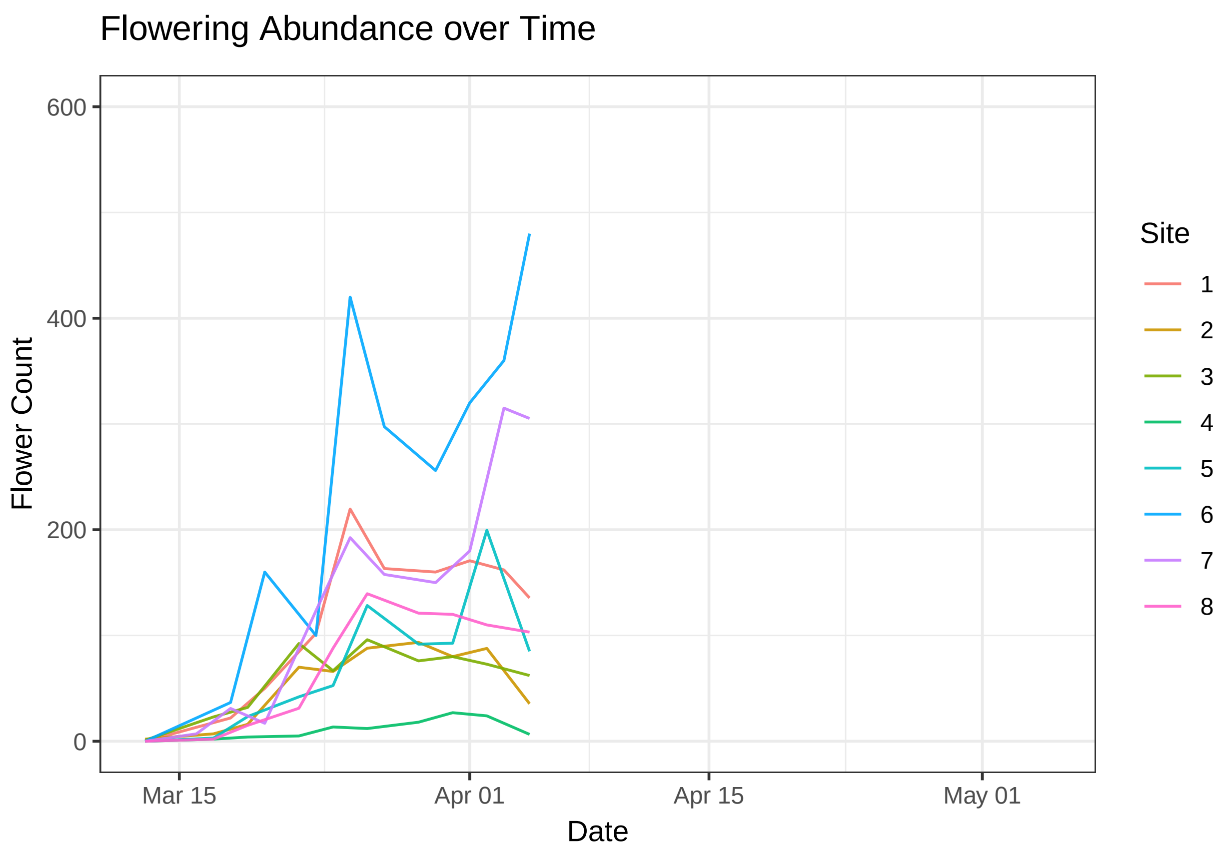

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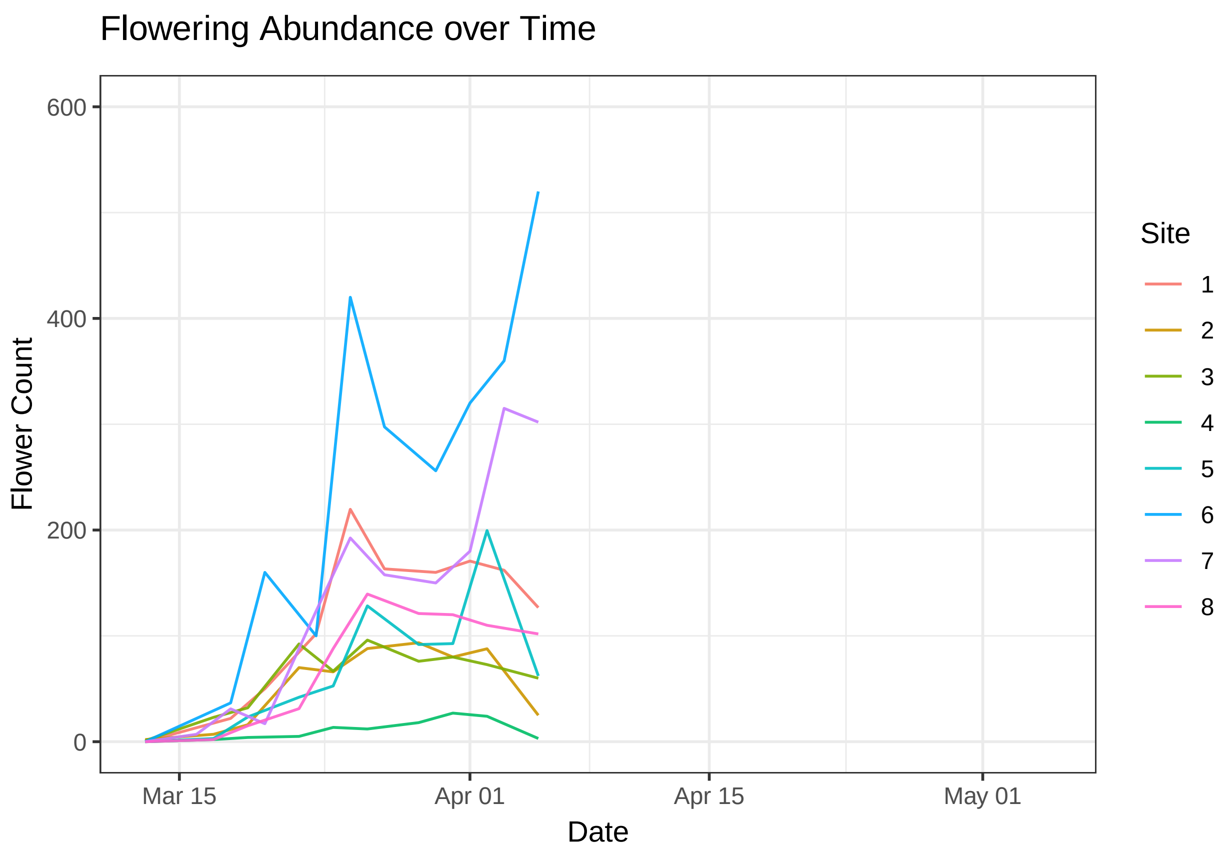

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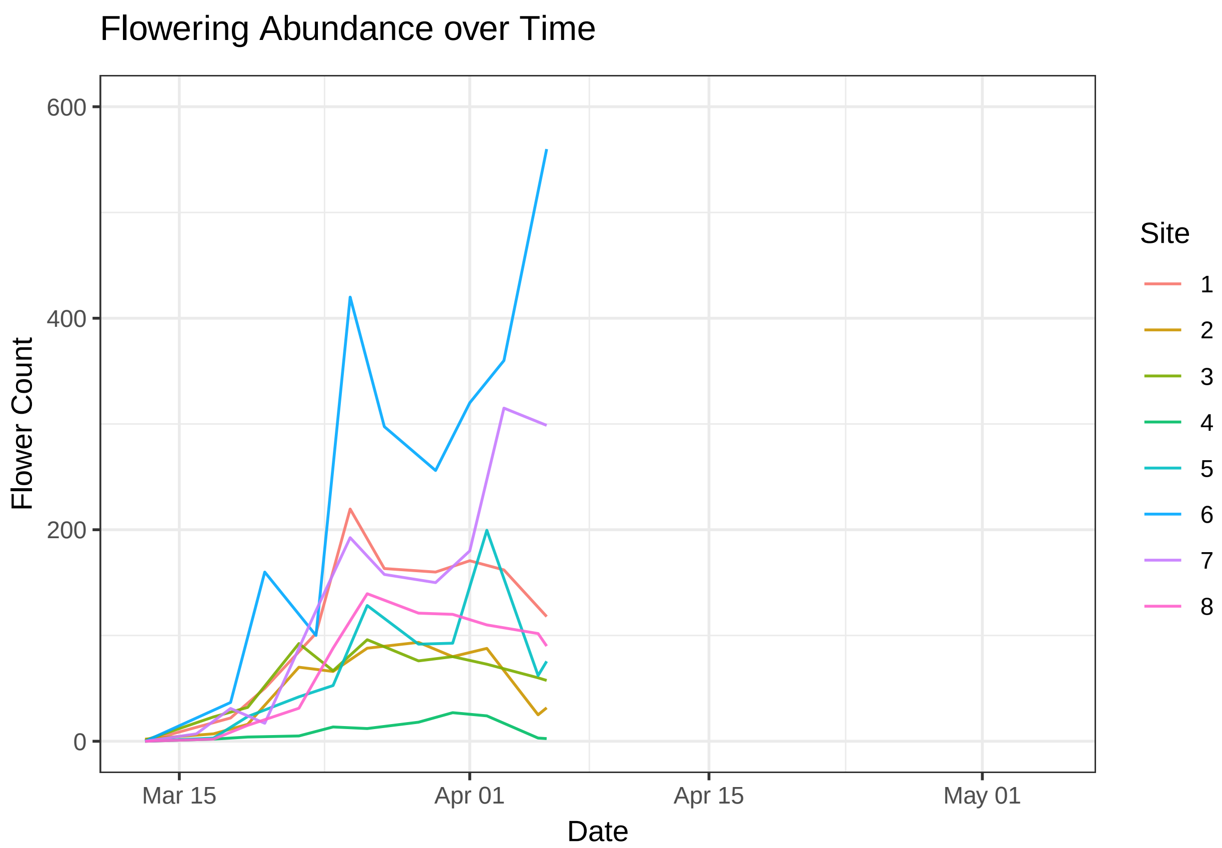
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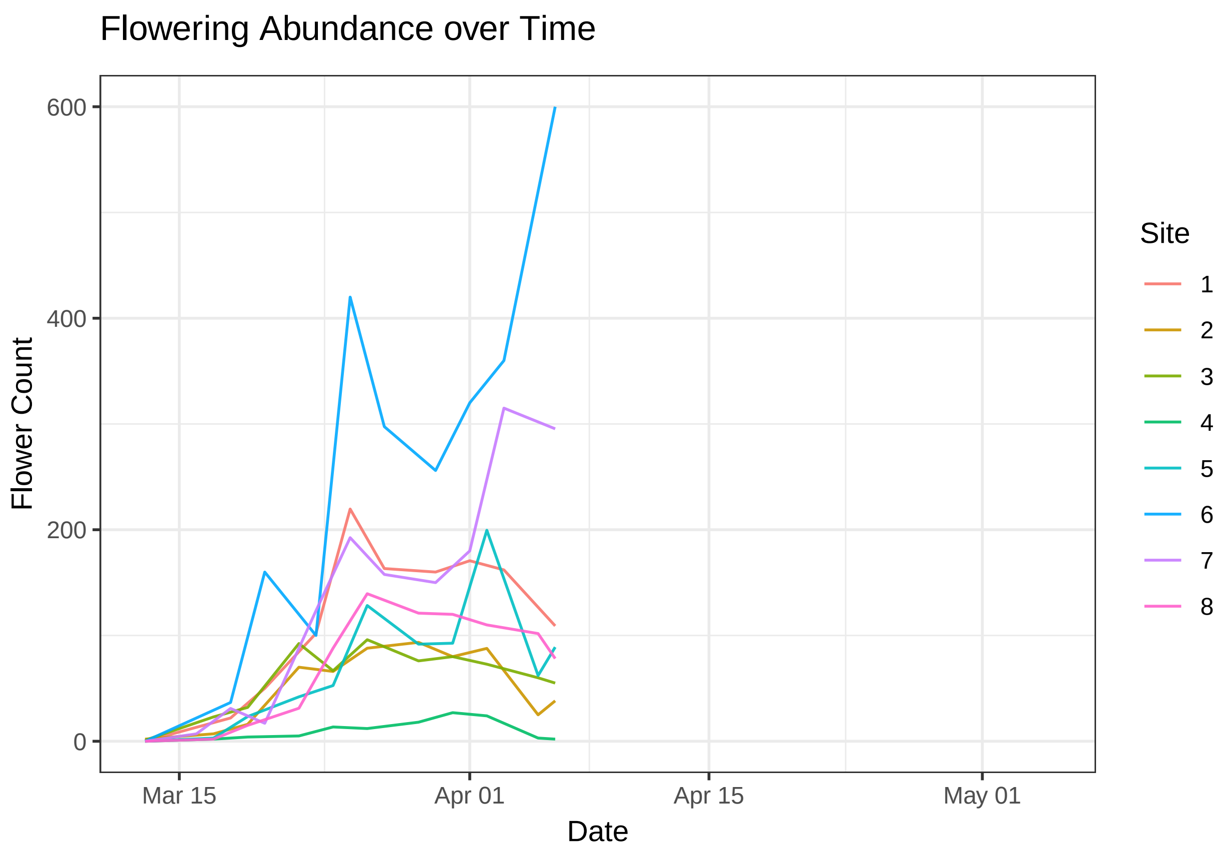
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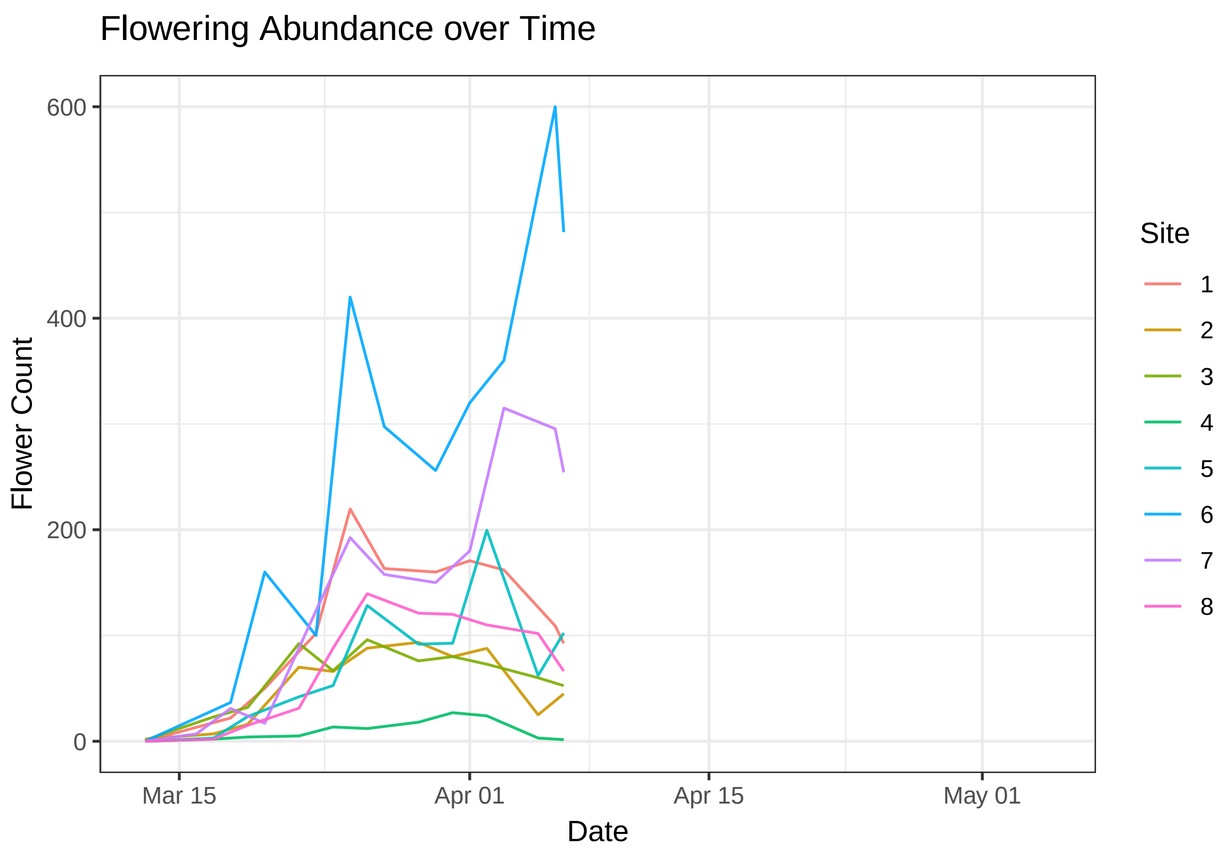
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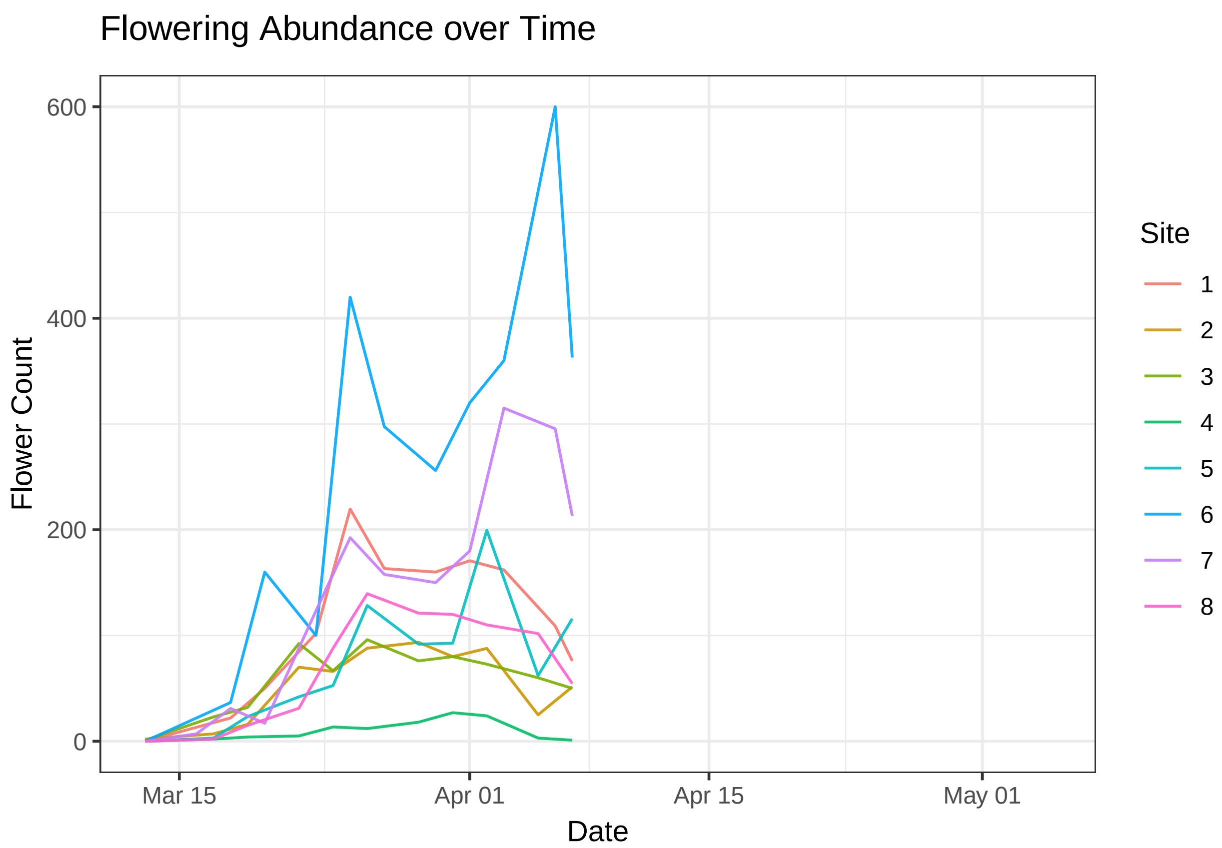
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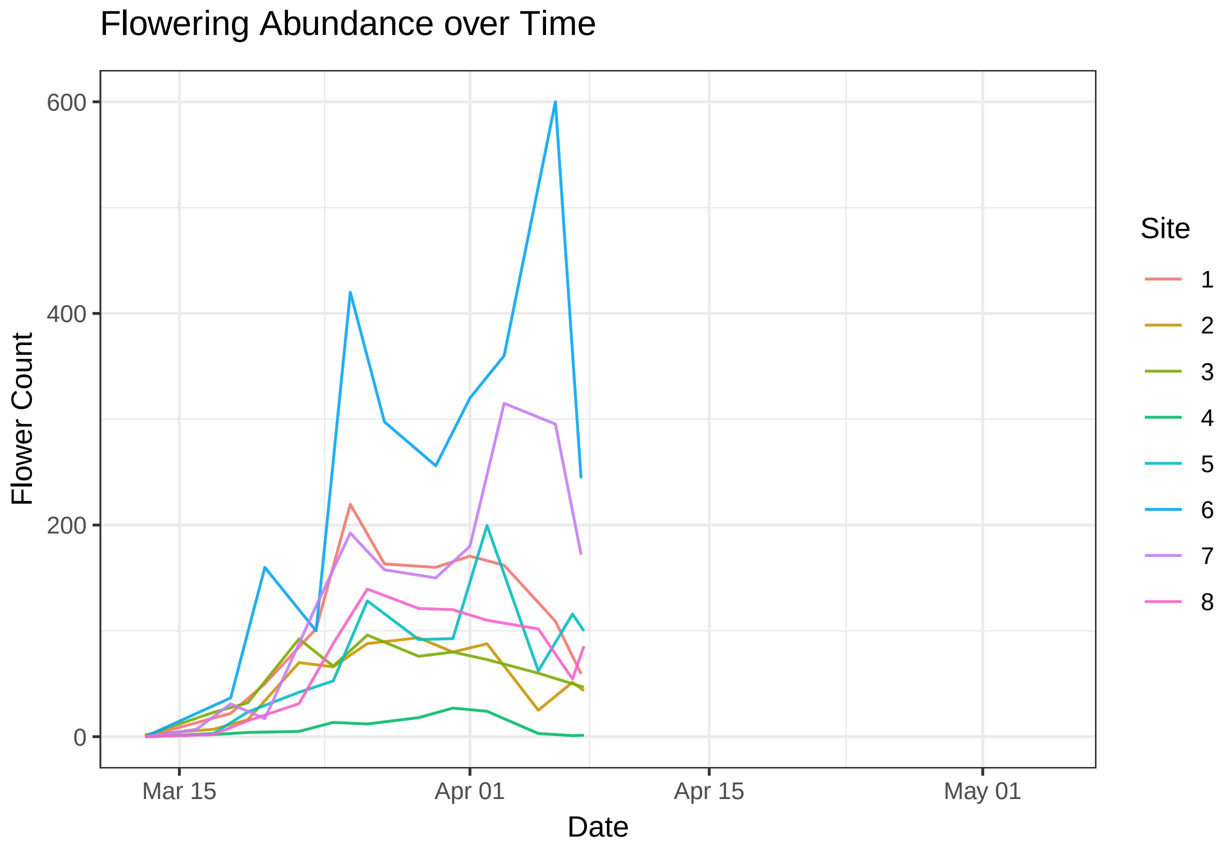
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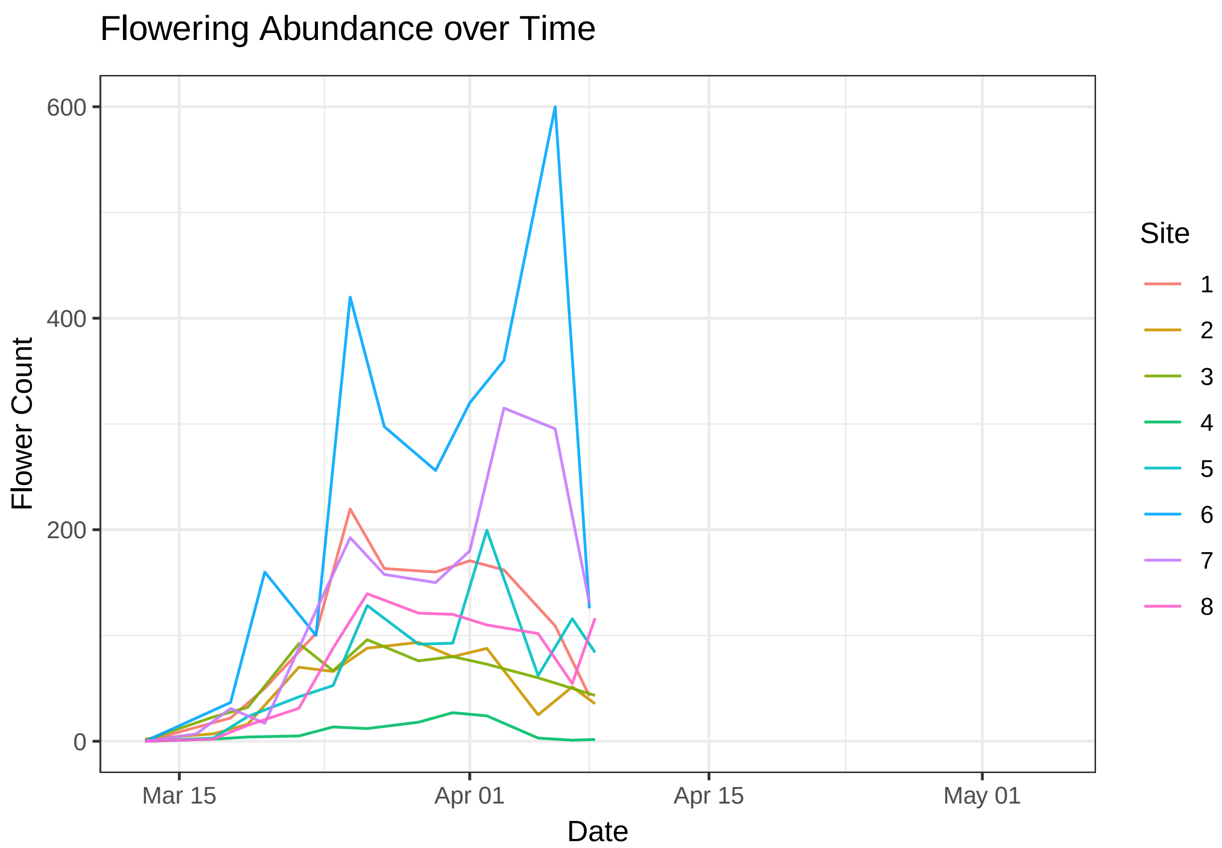
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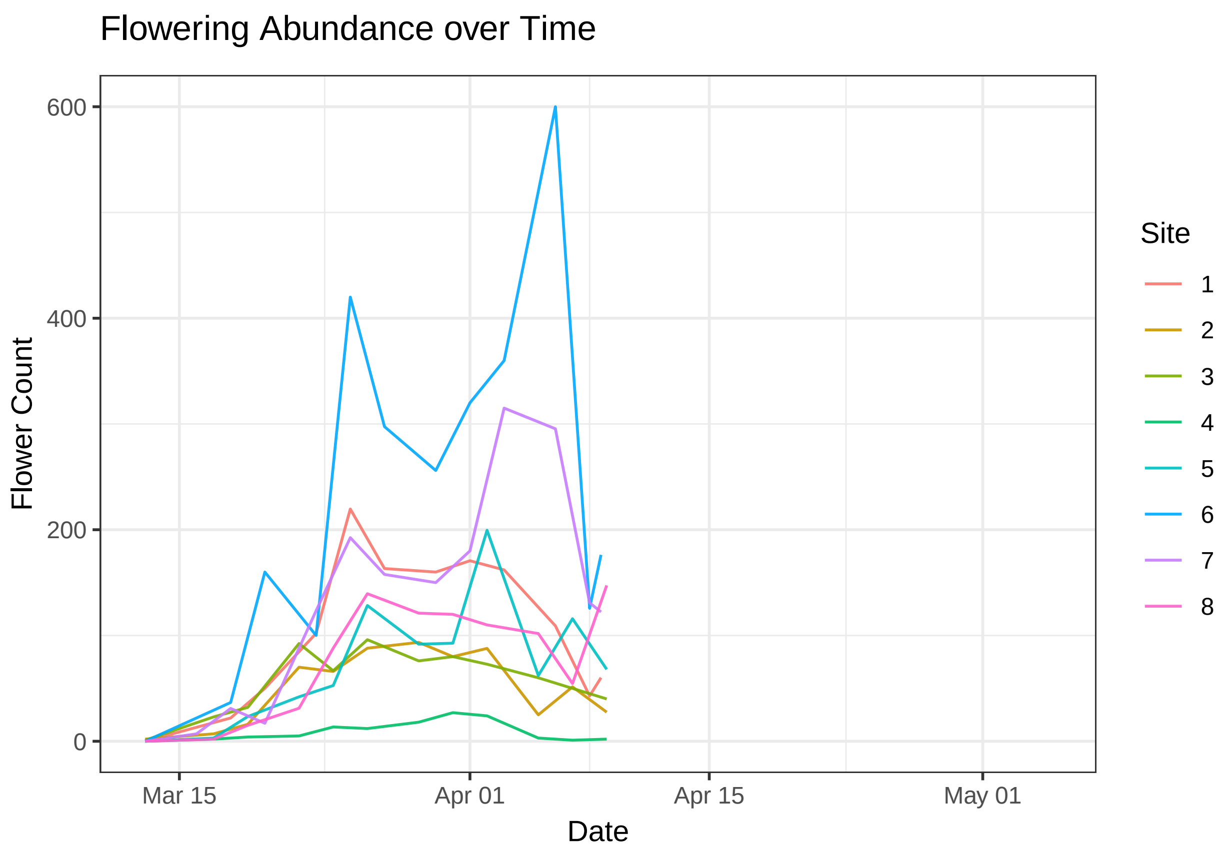

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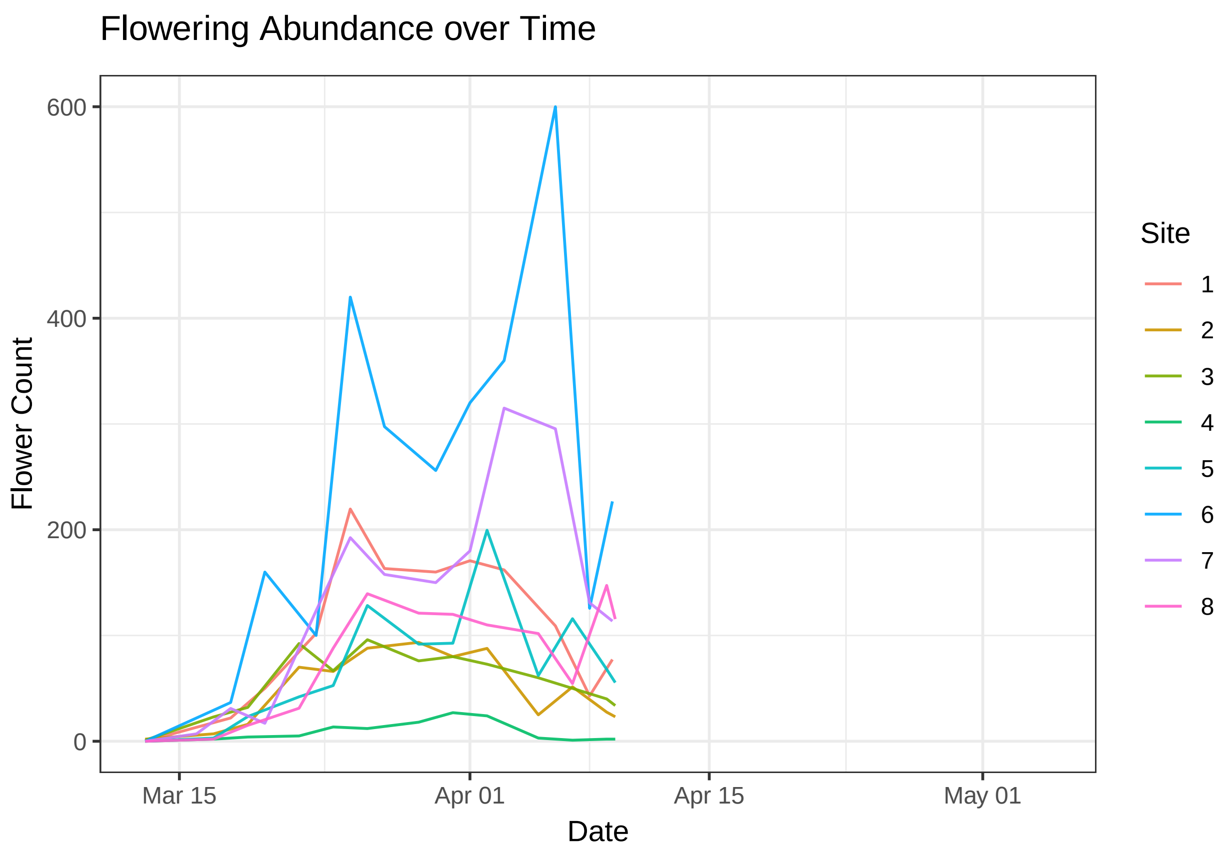
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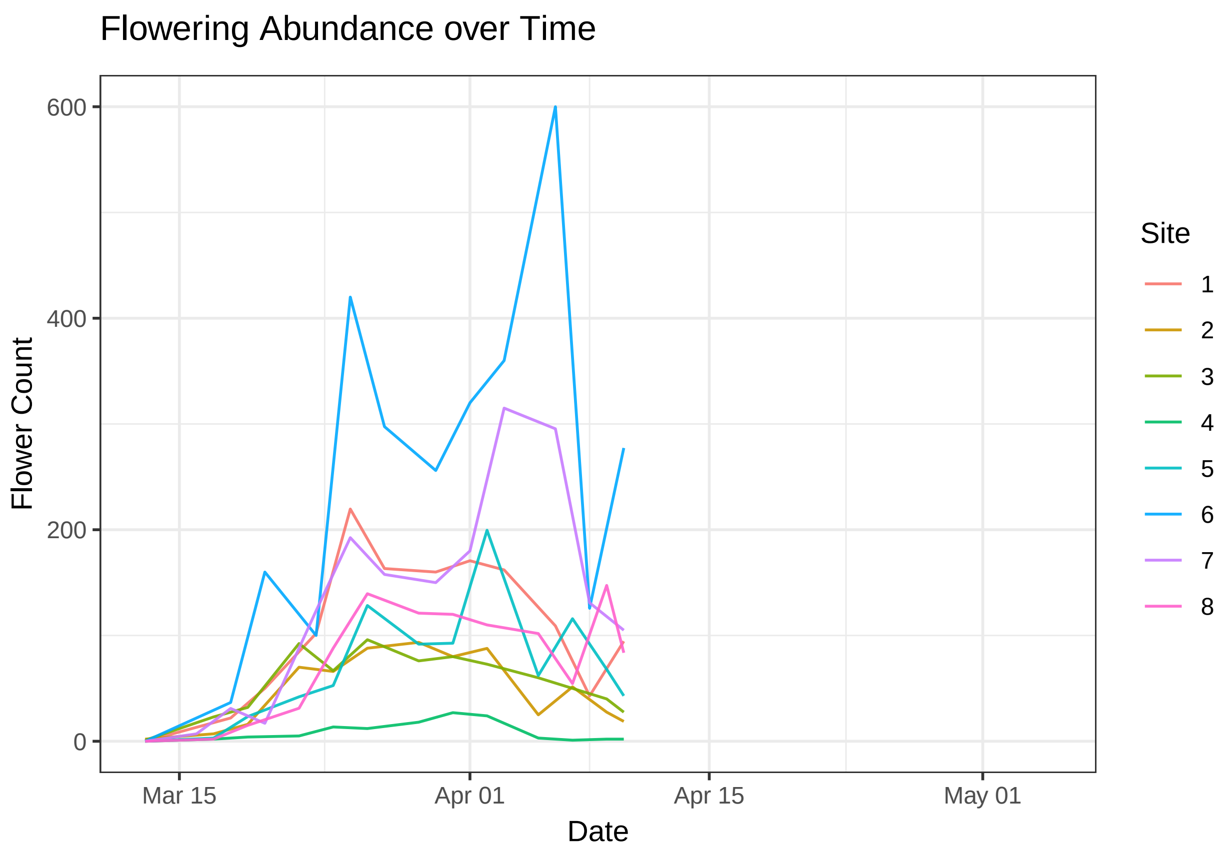
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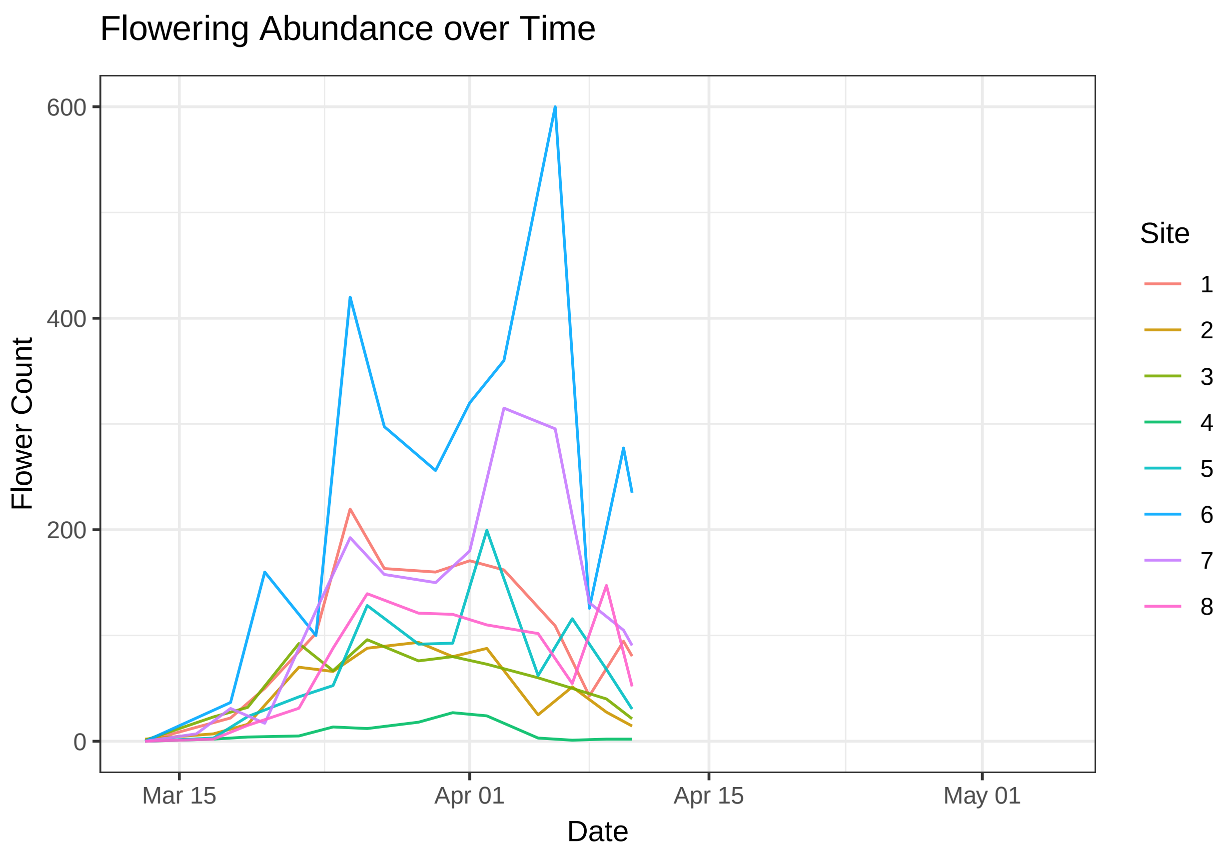
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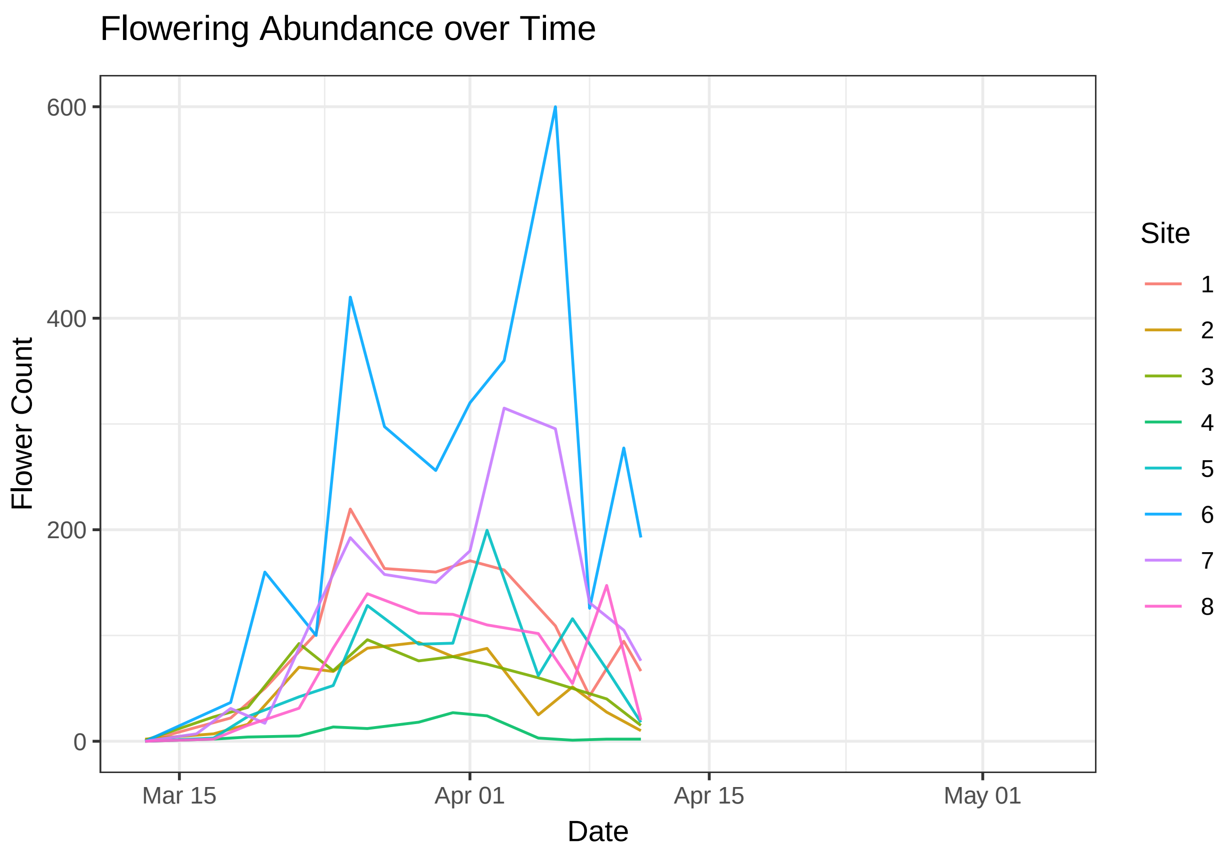
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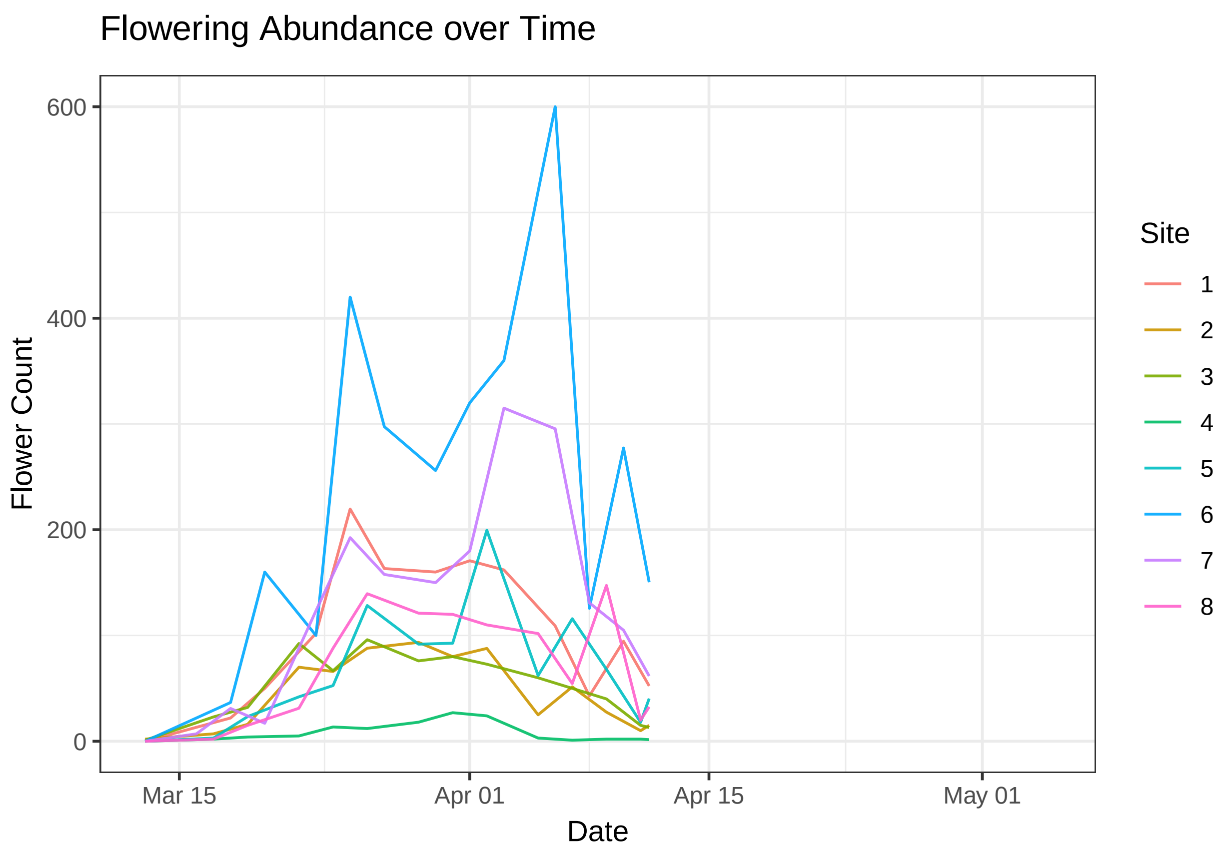
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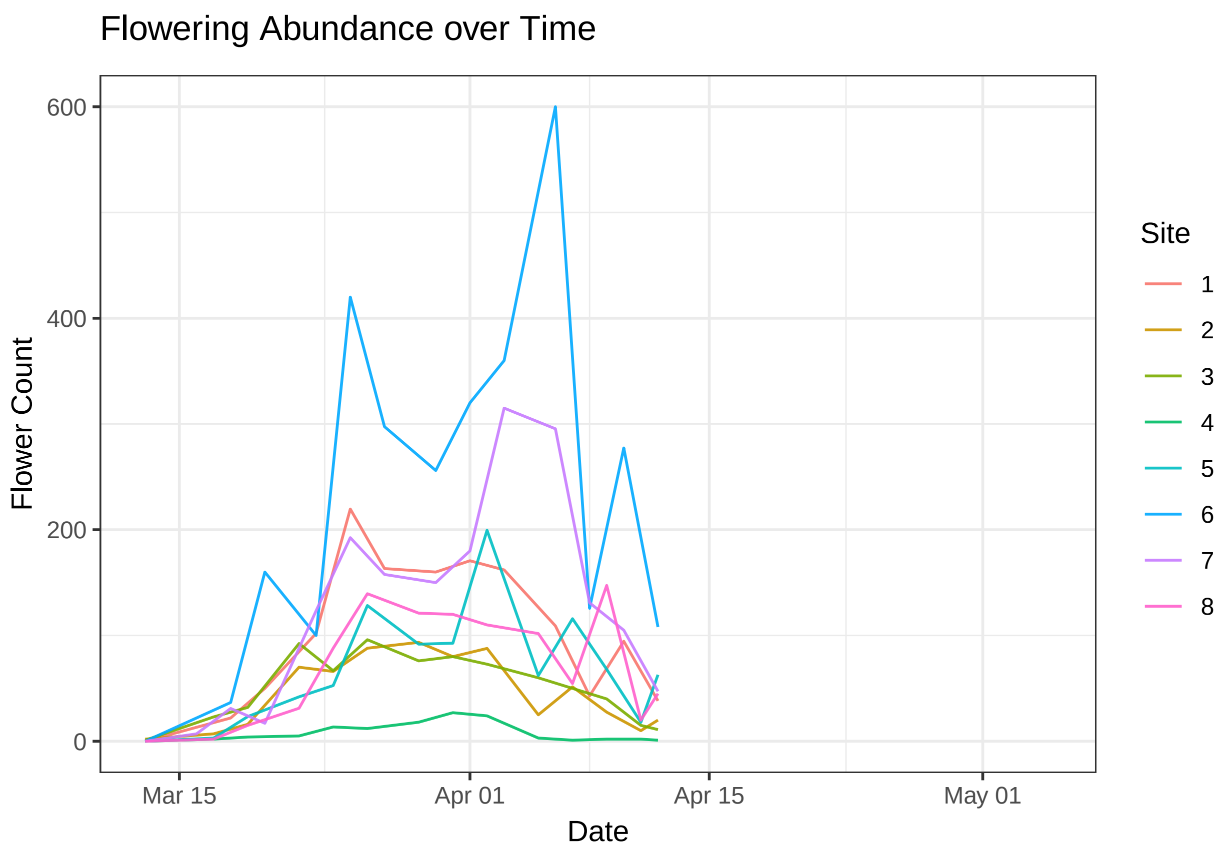
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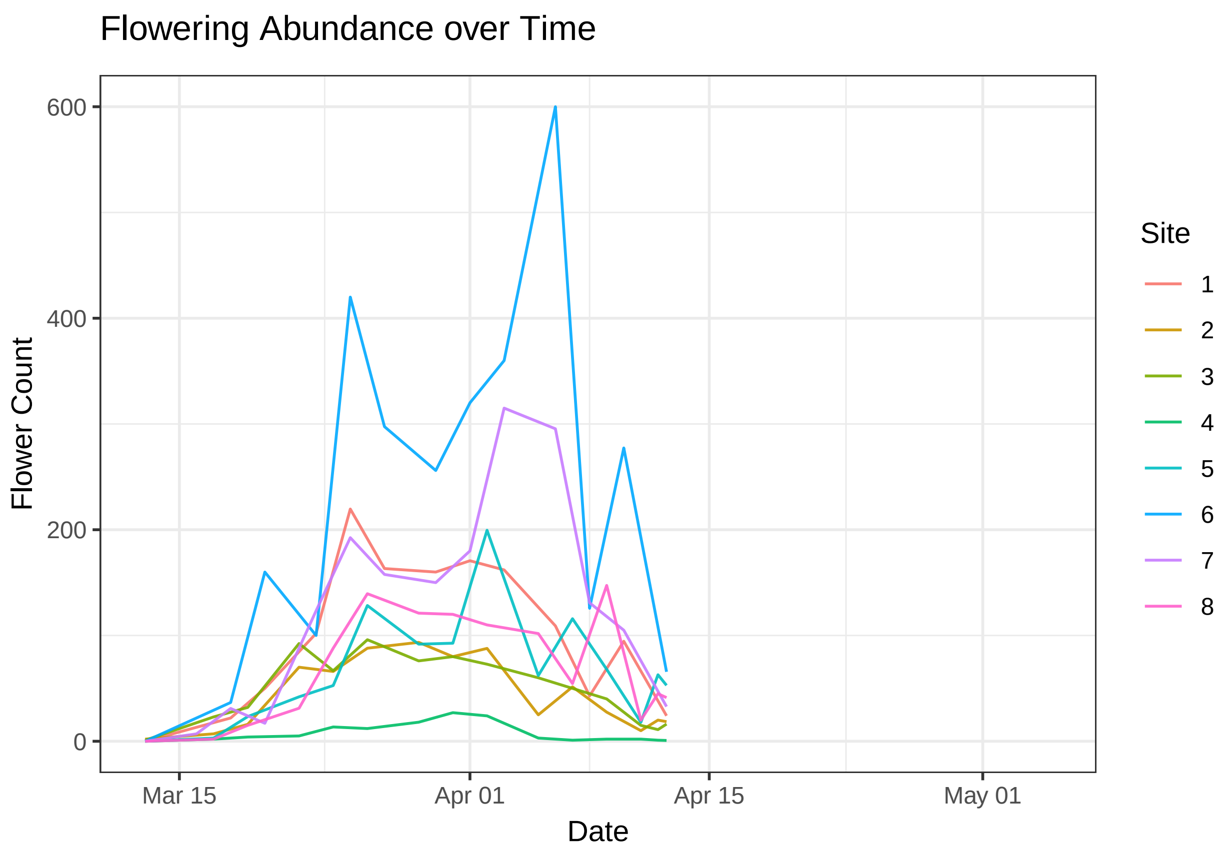
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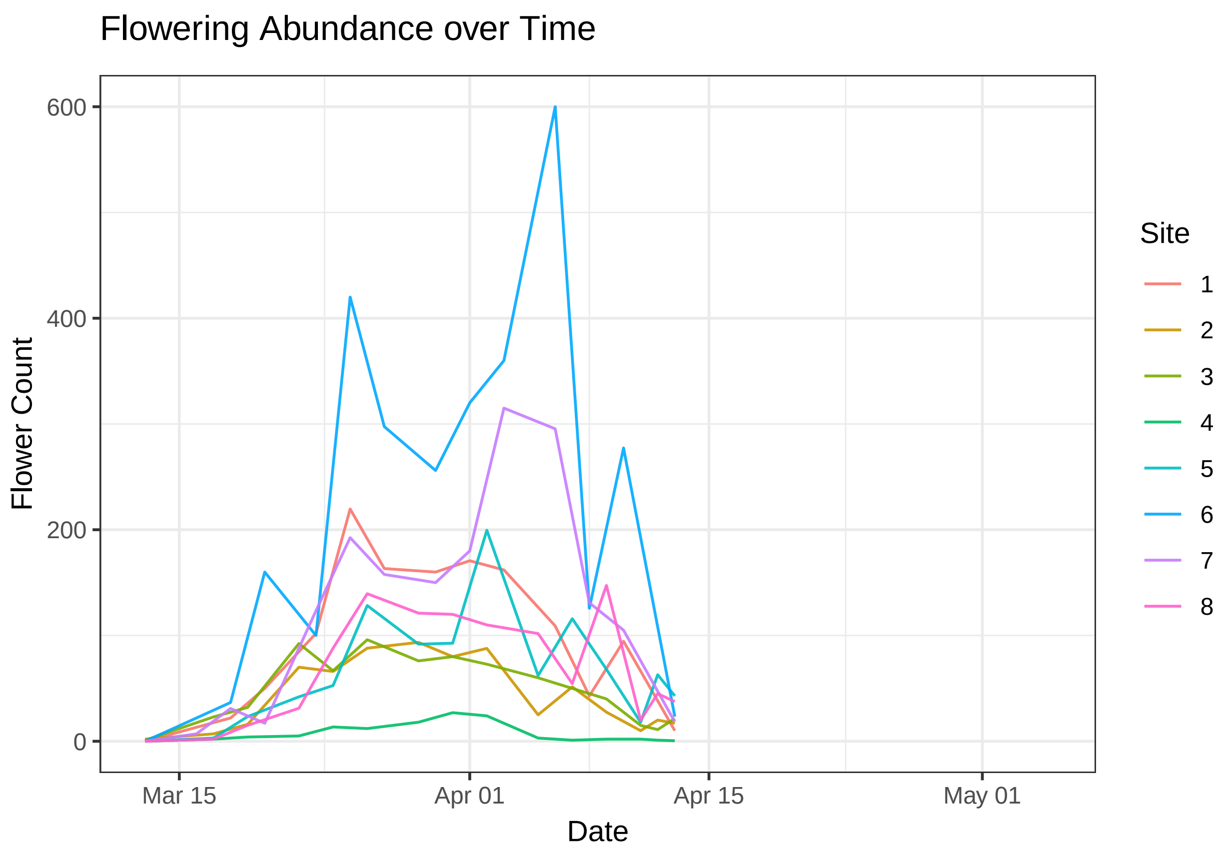

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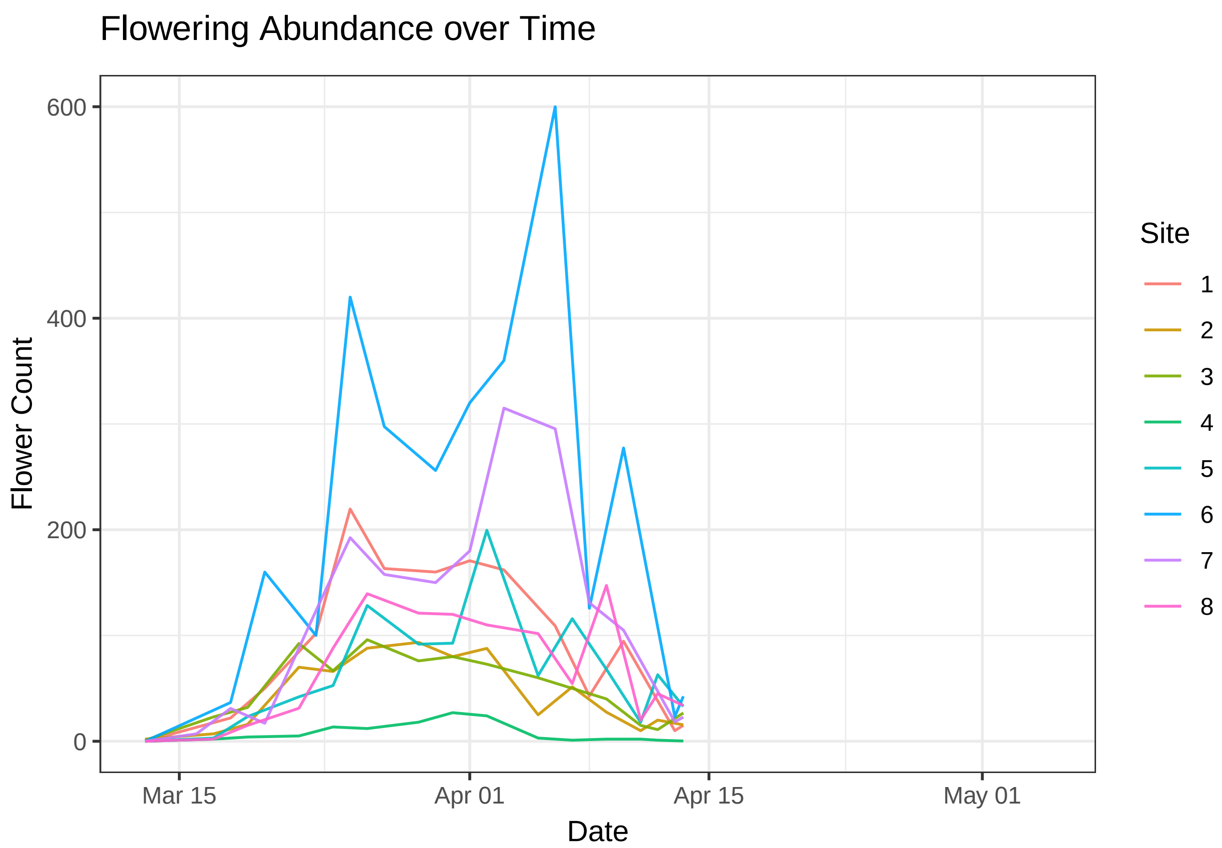

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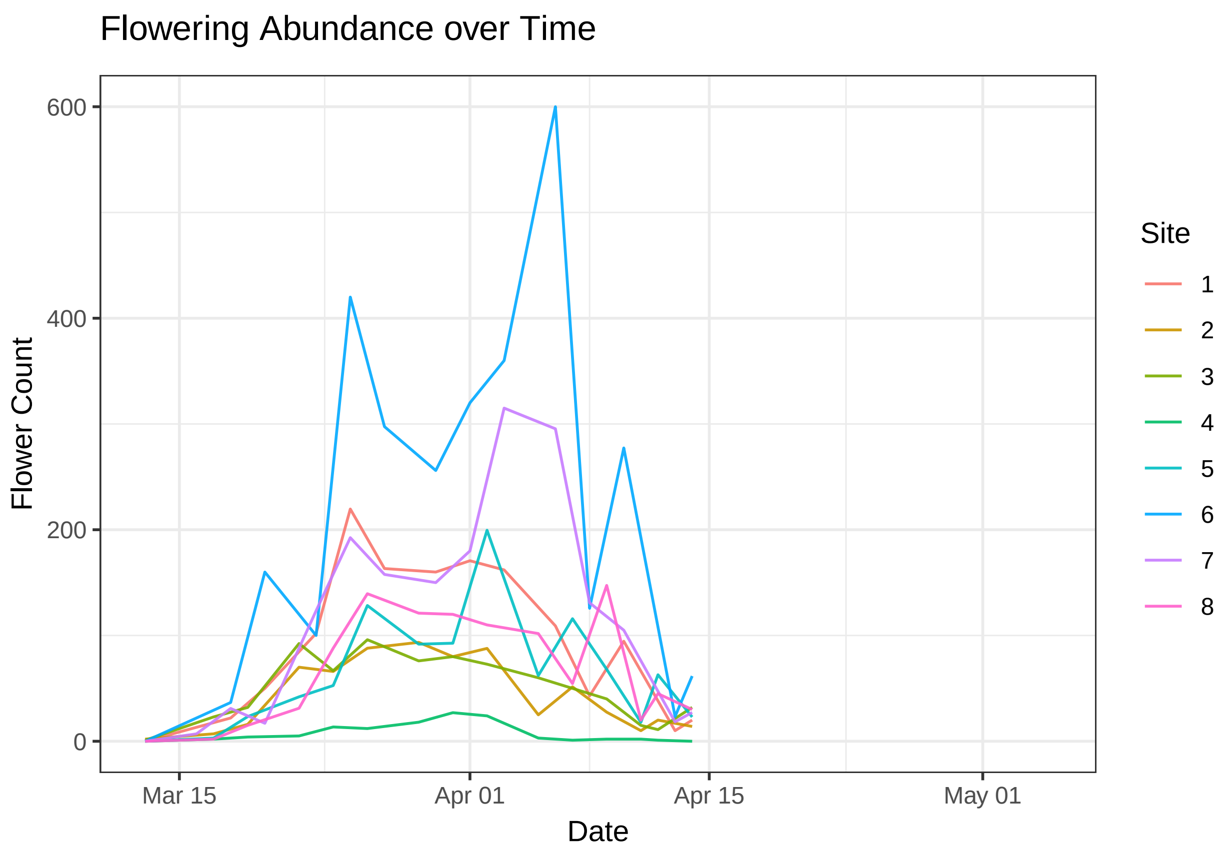

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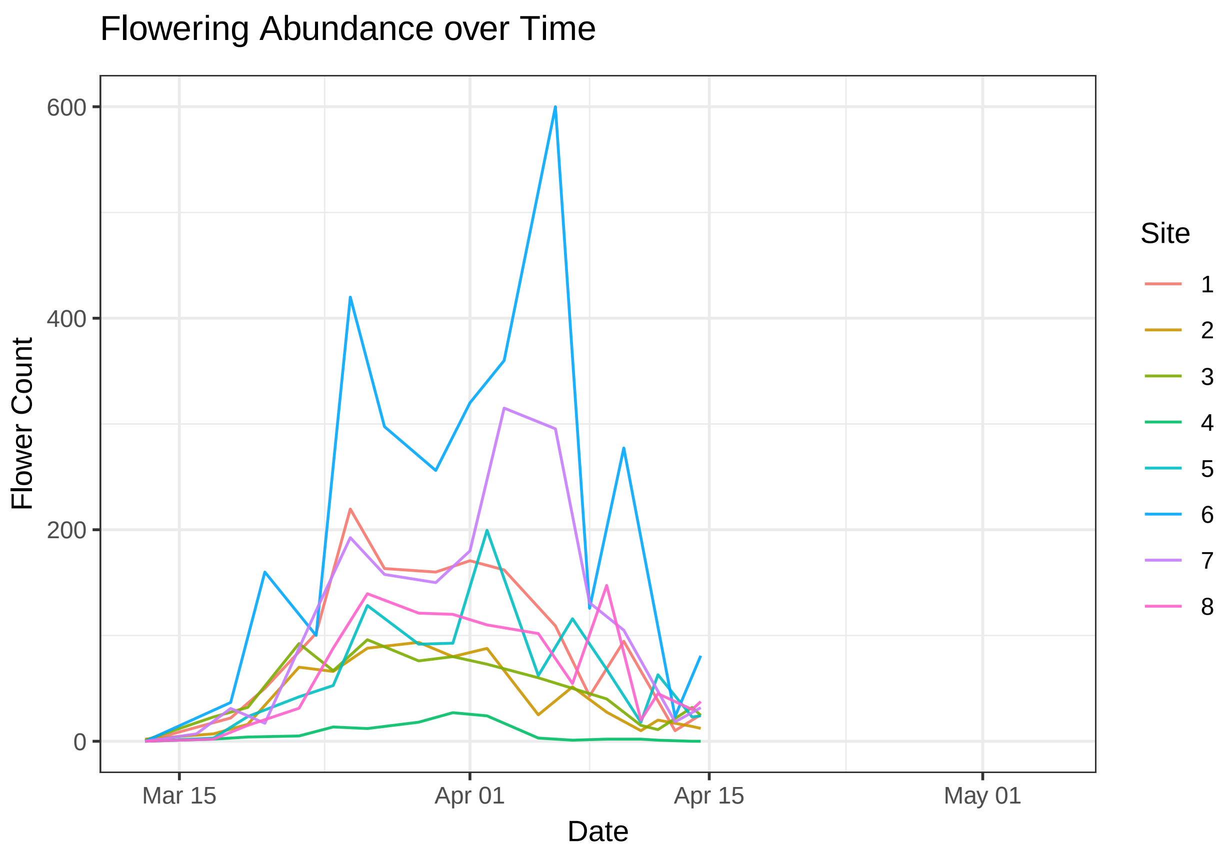
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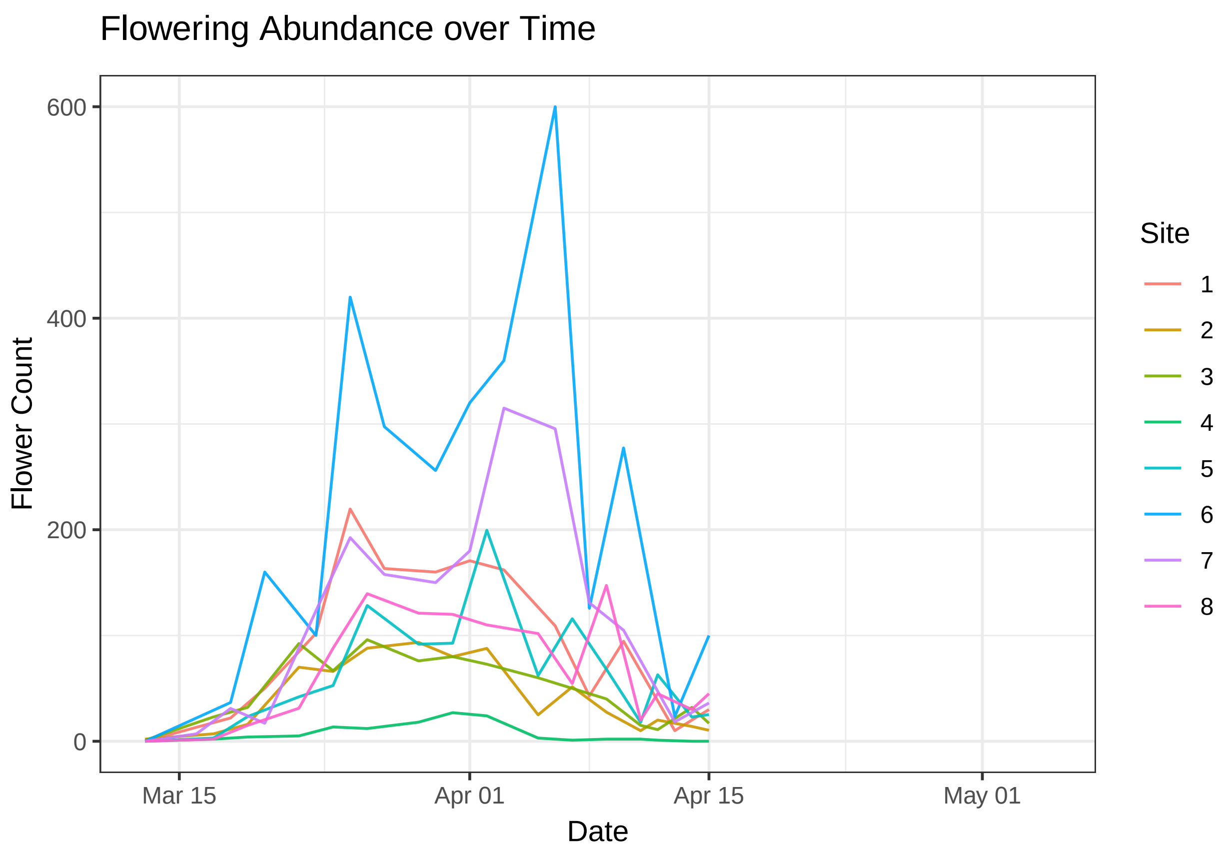

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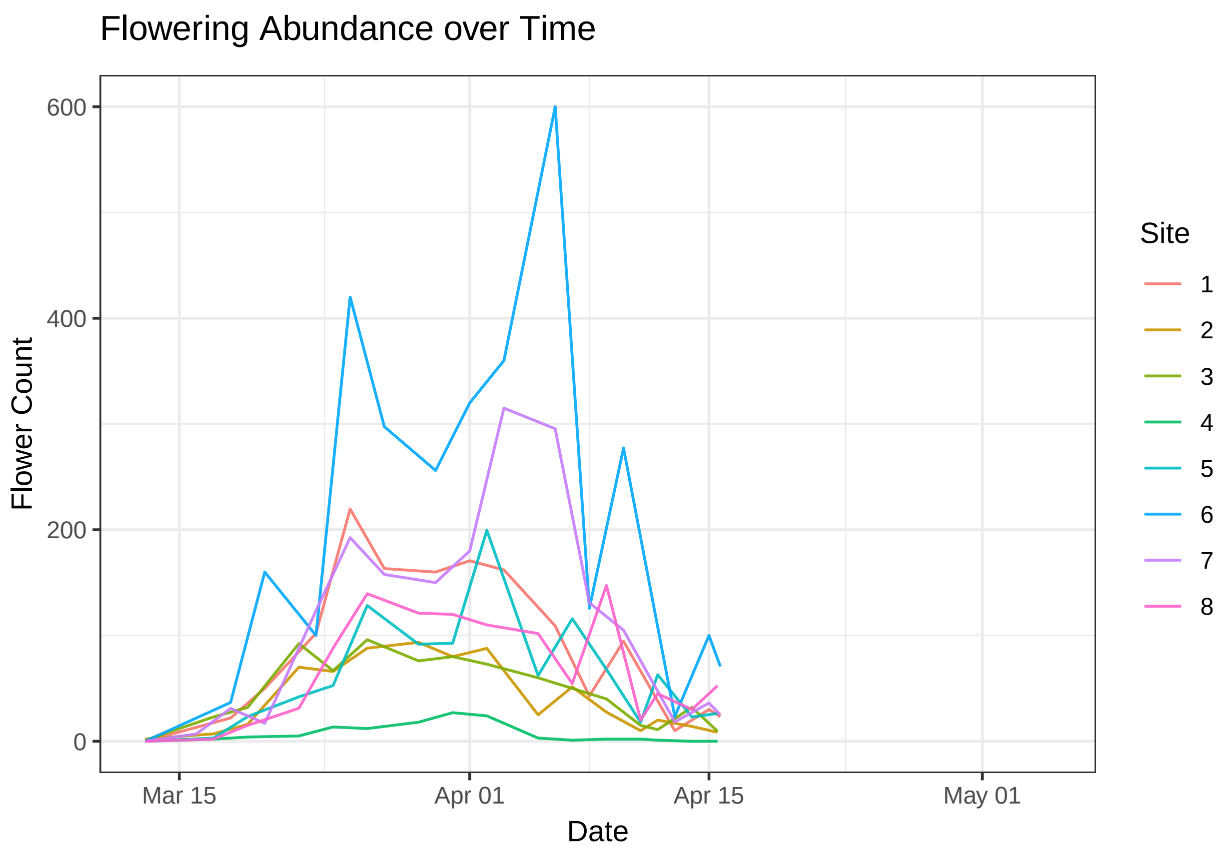
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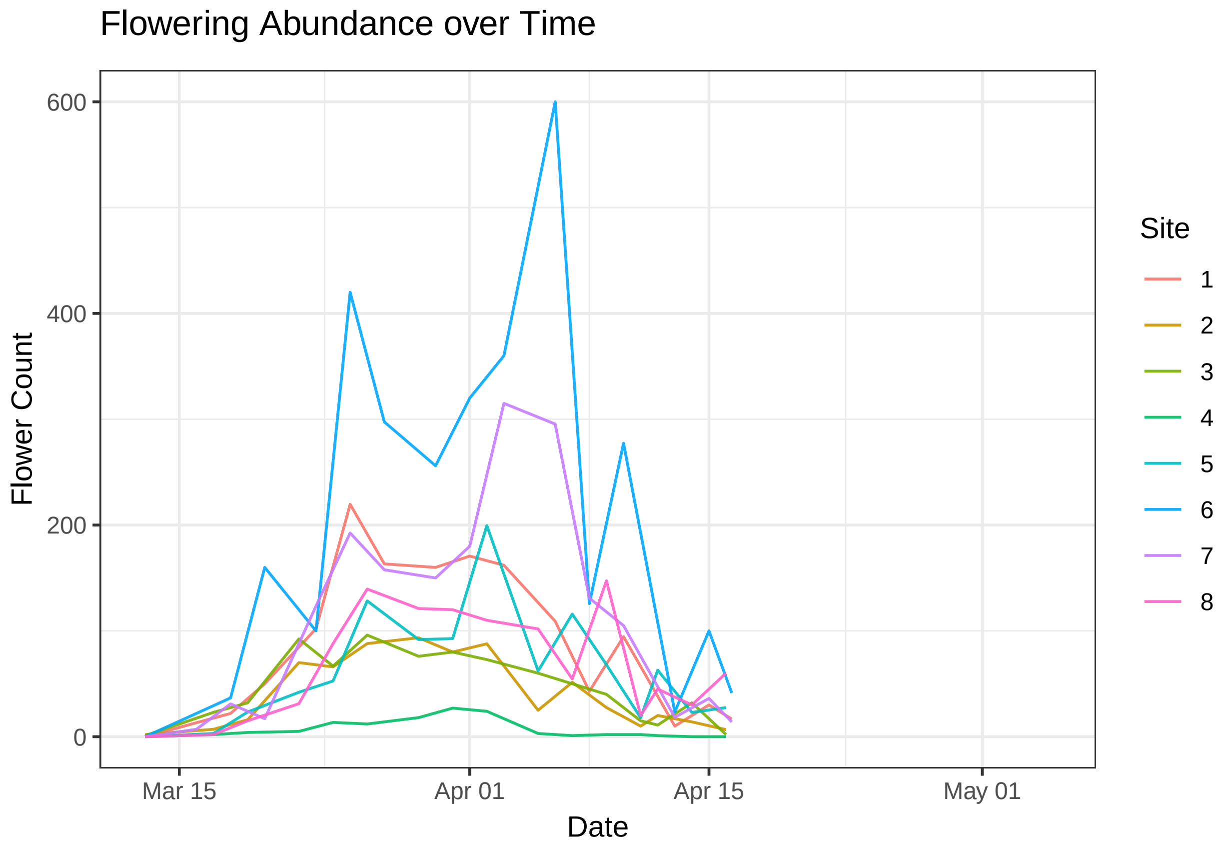
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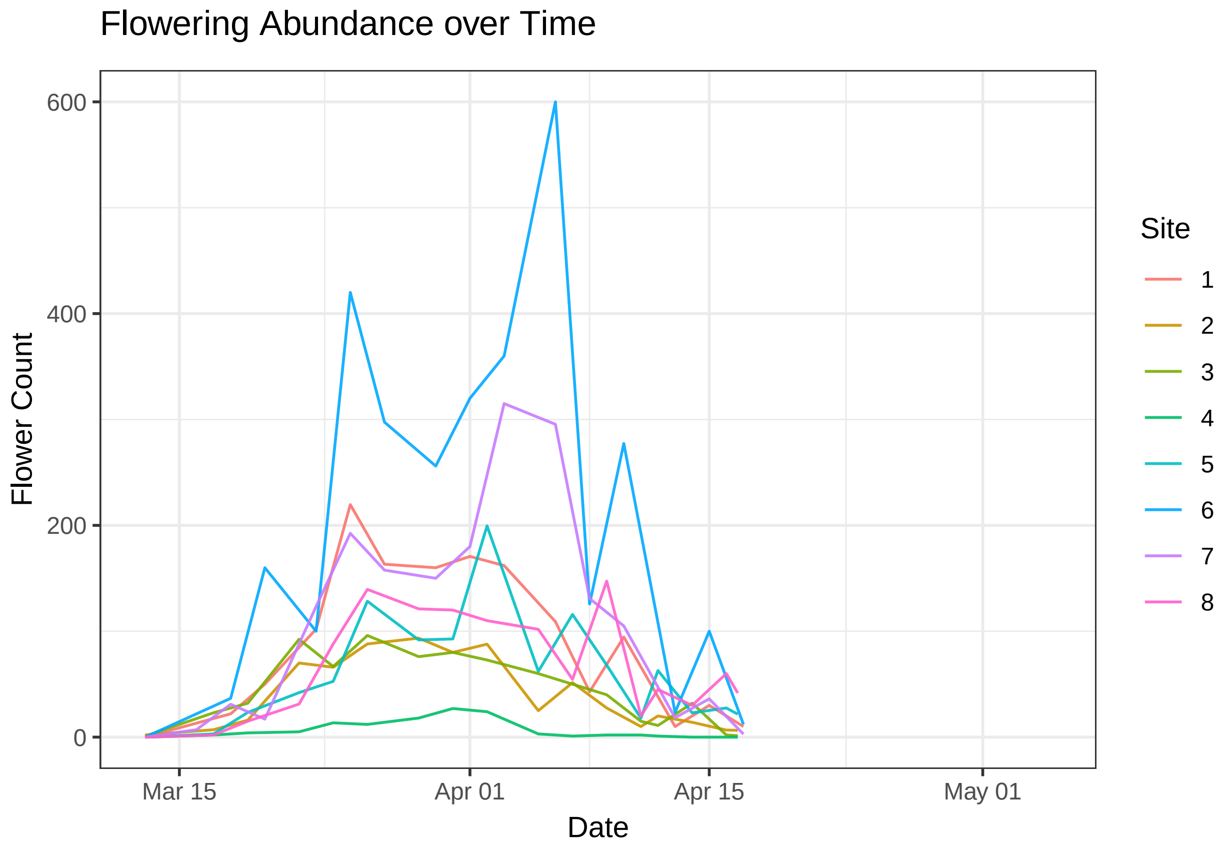
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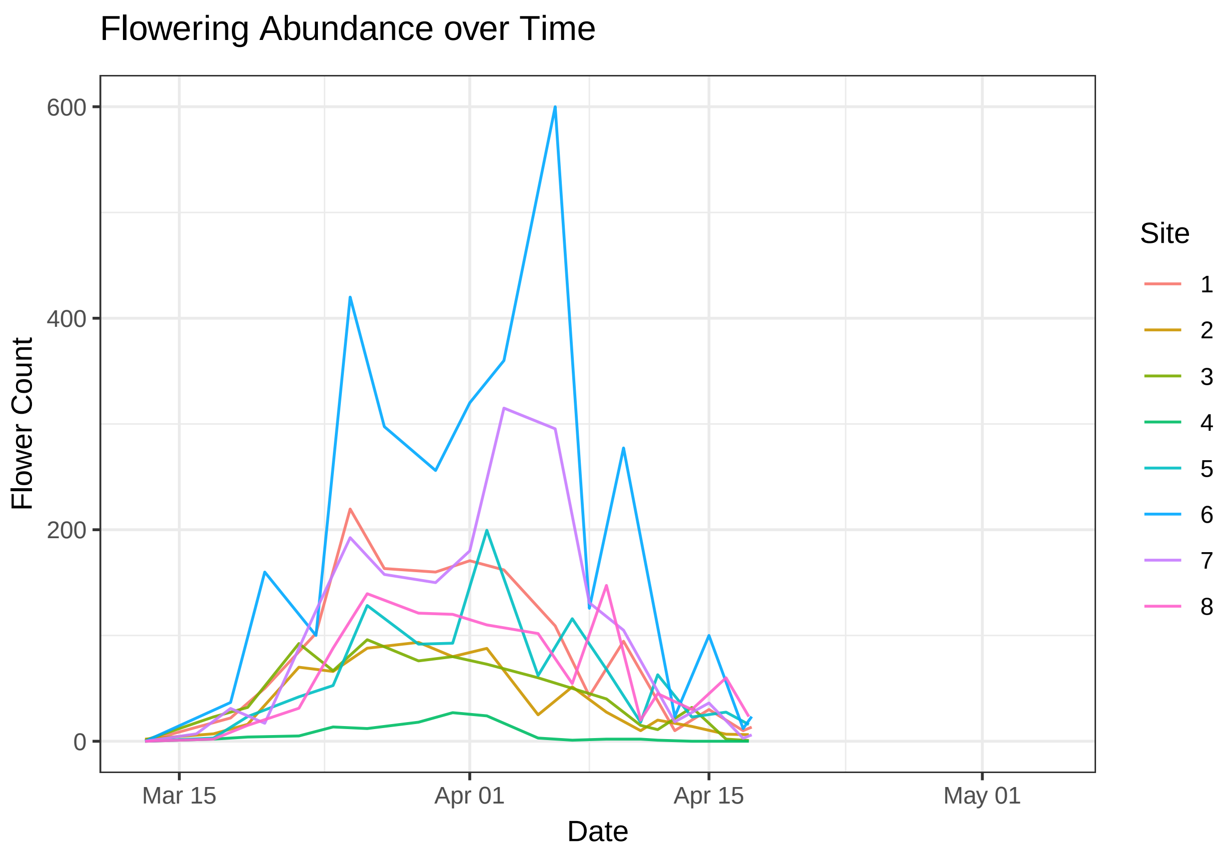
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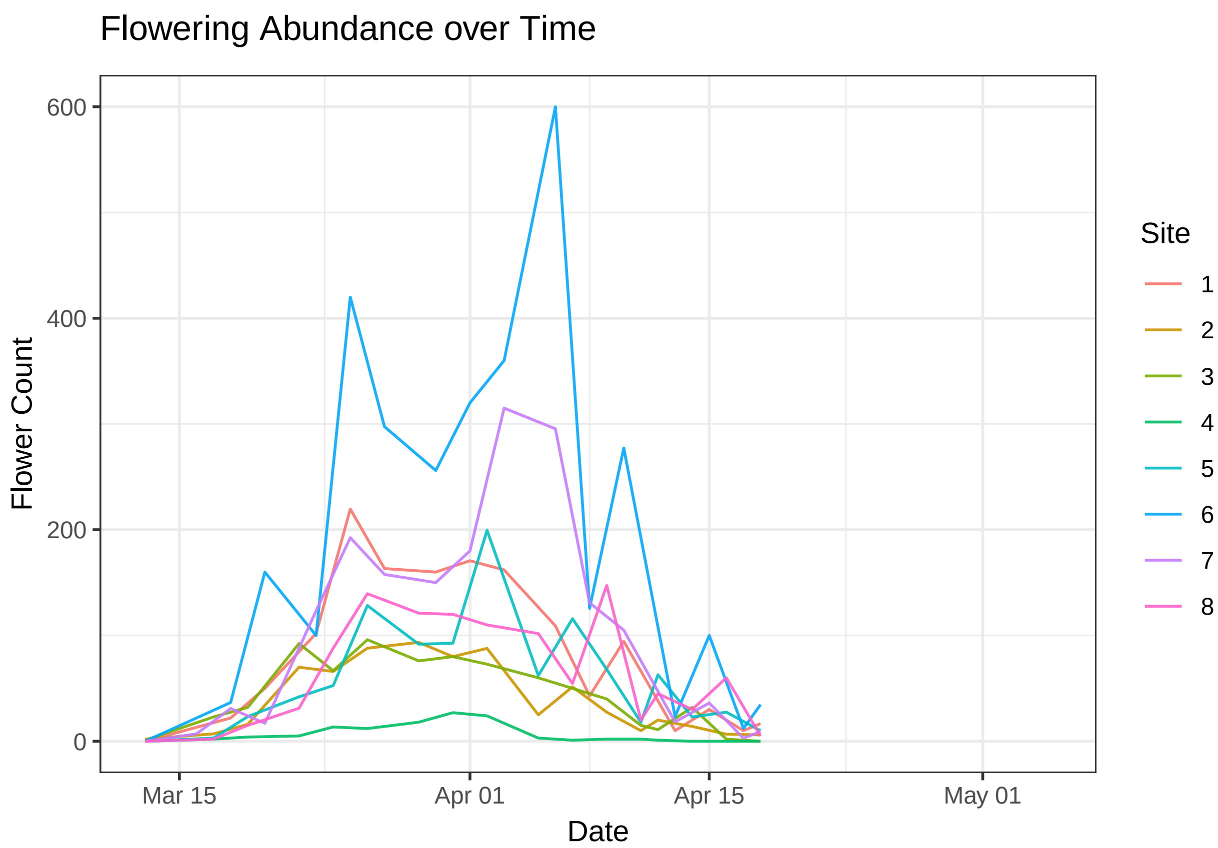

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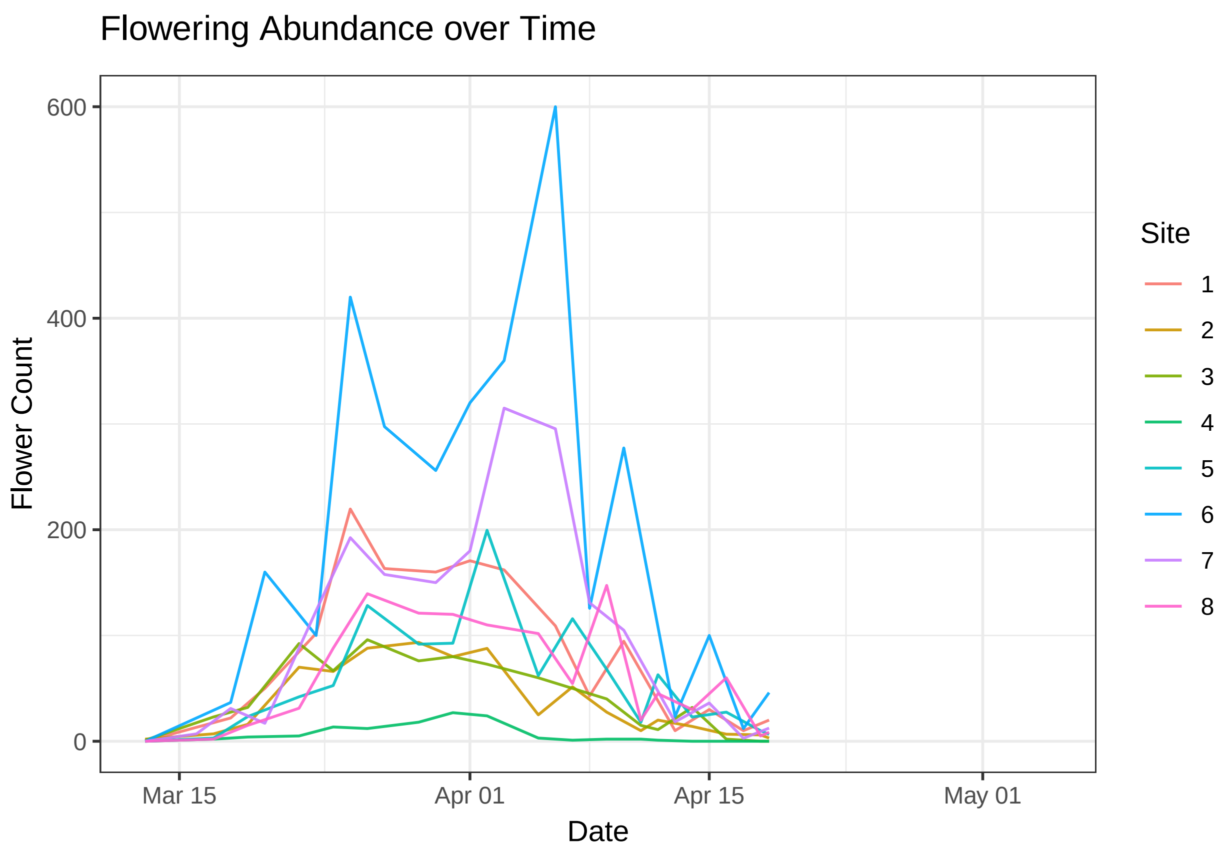
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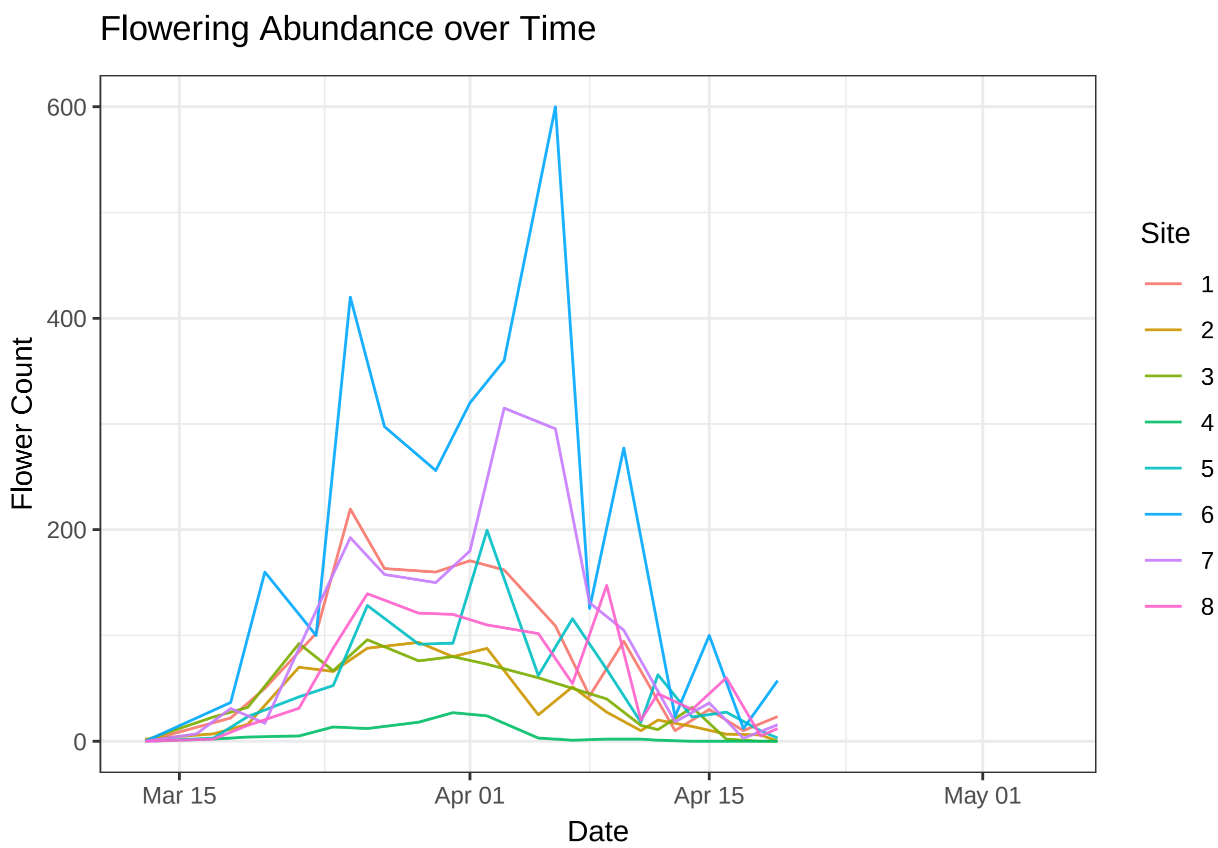
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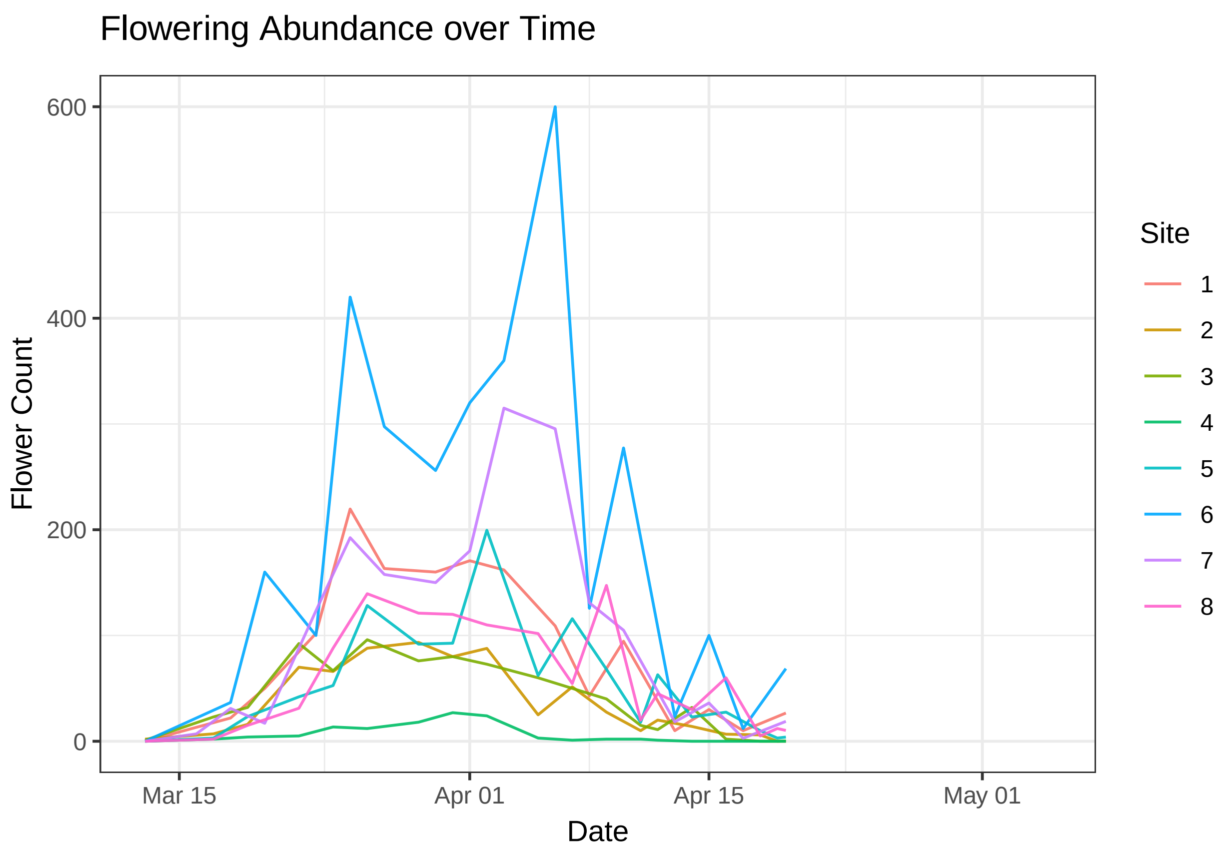

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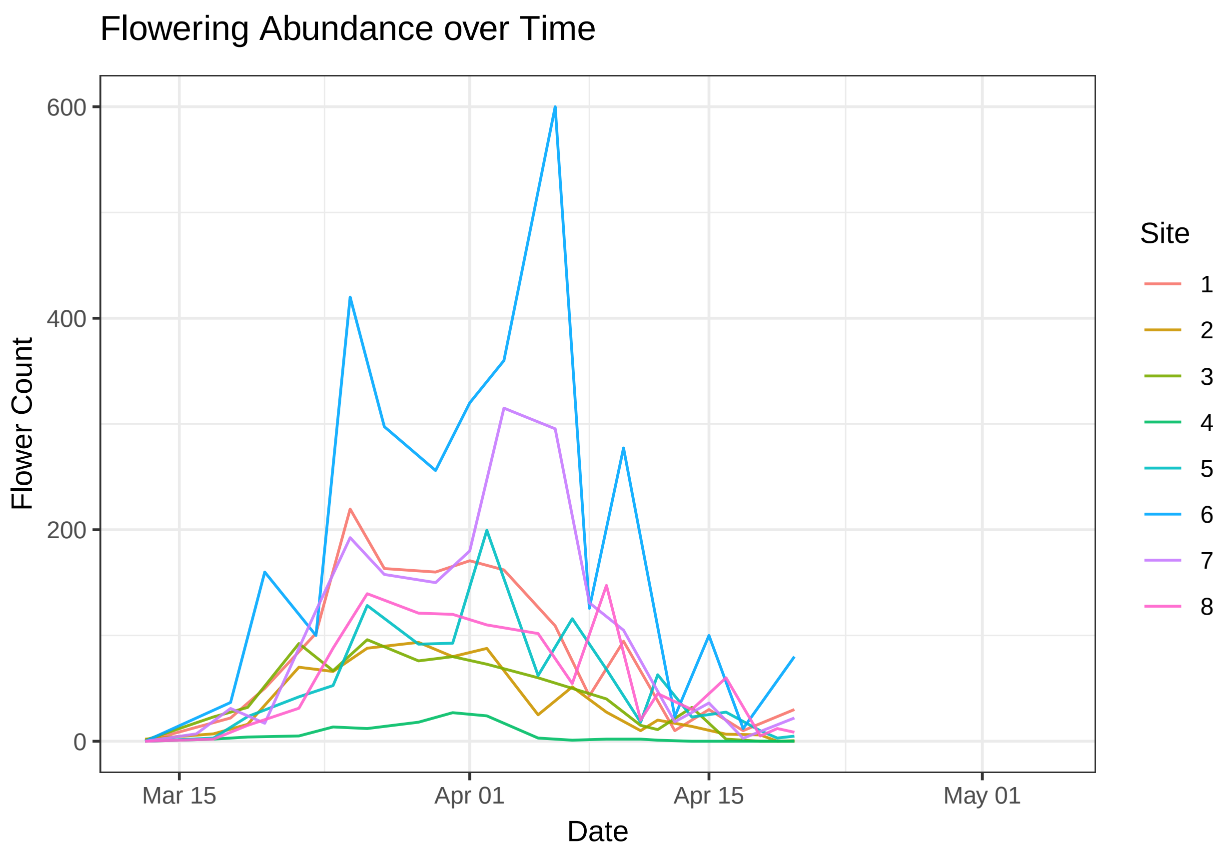
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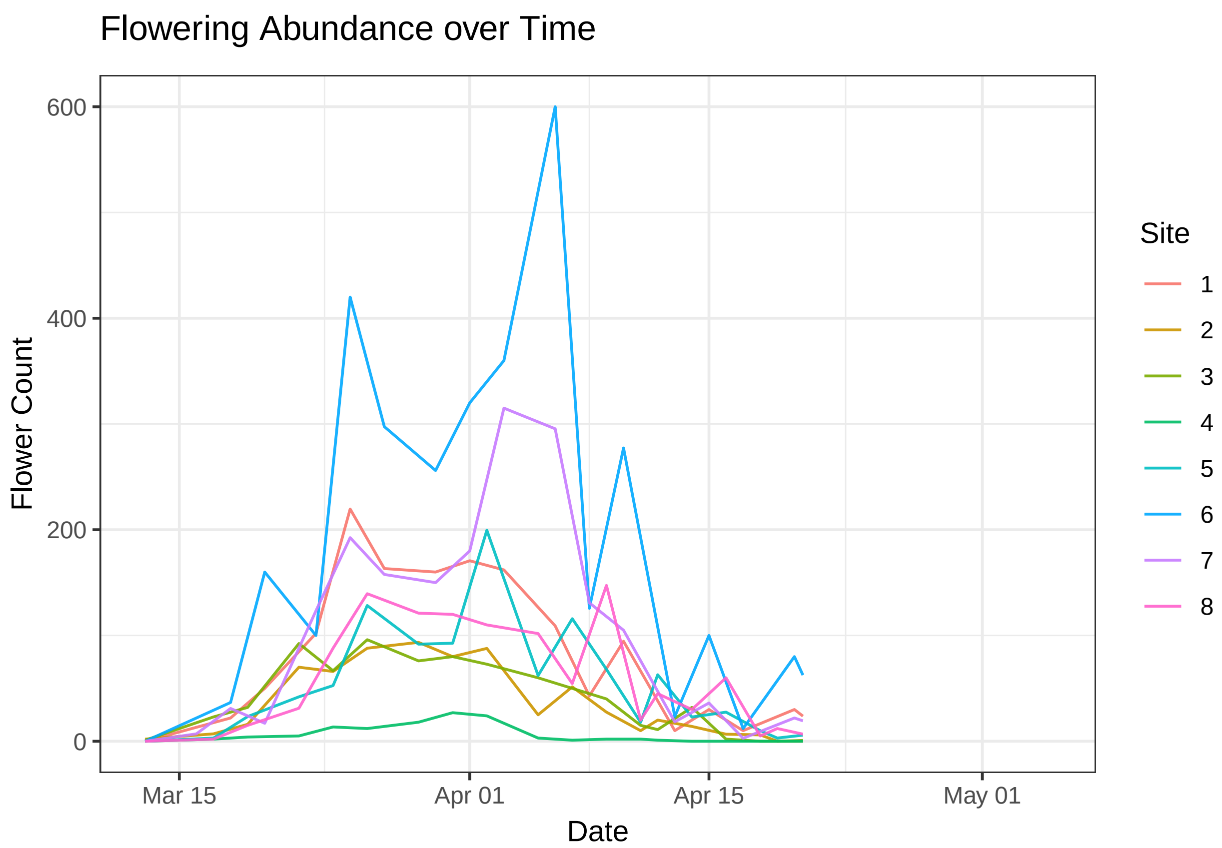
Flowering Abundance over Time
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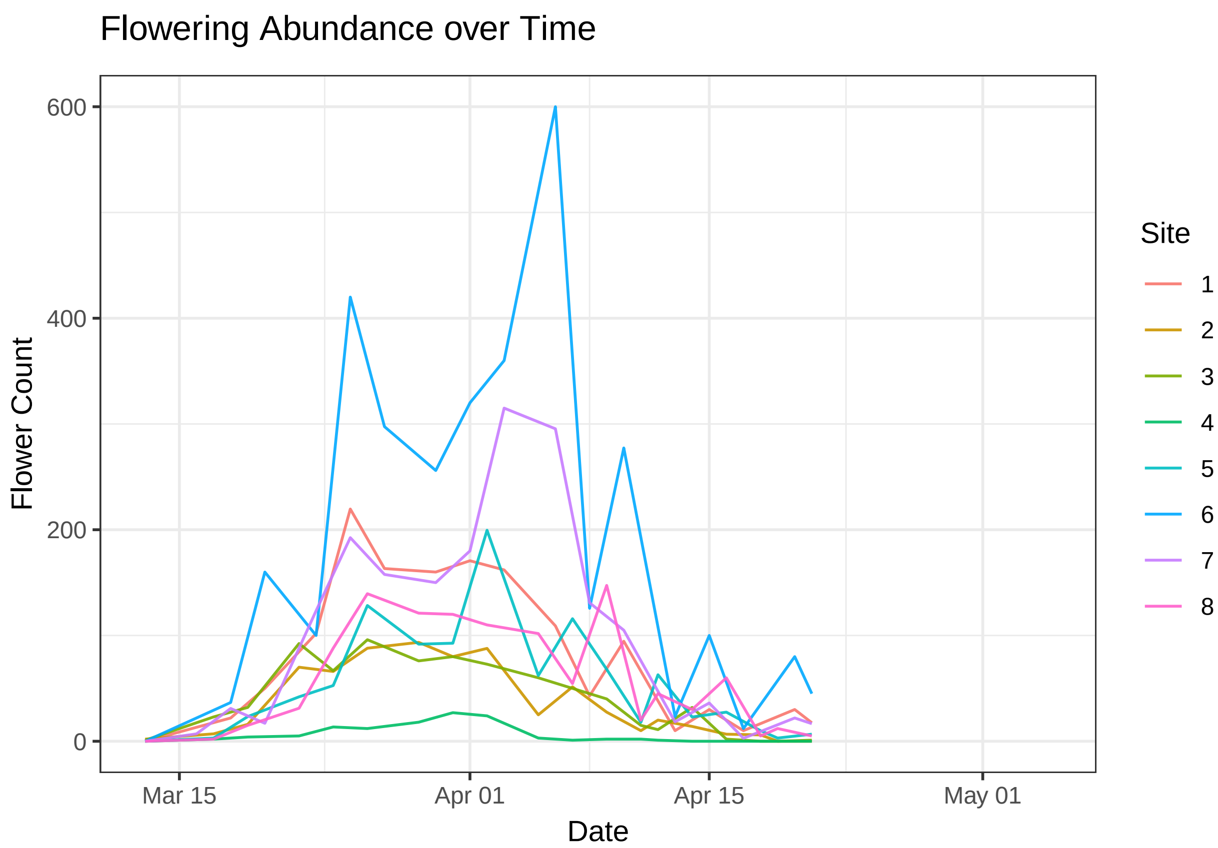

Flowering Abundance over Time
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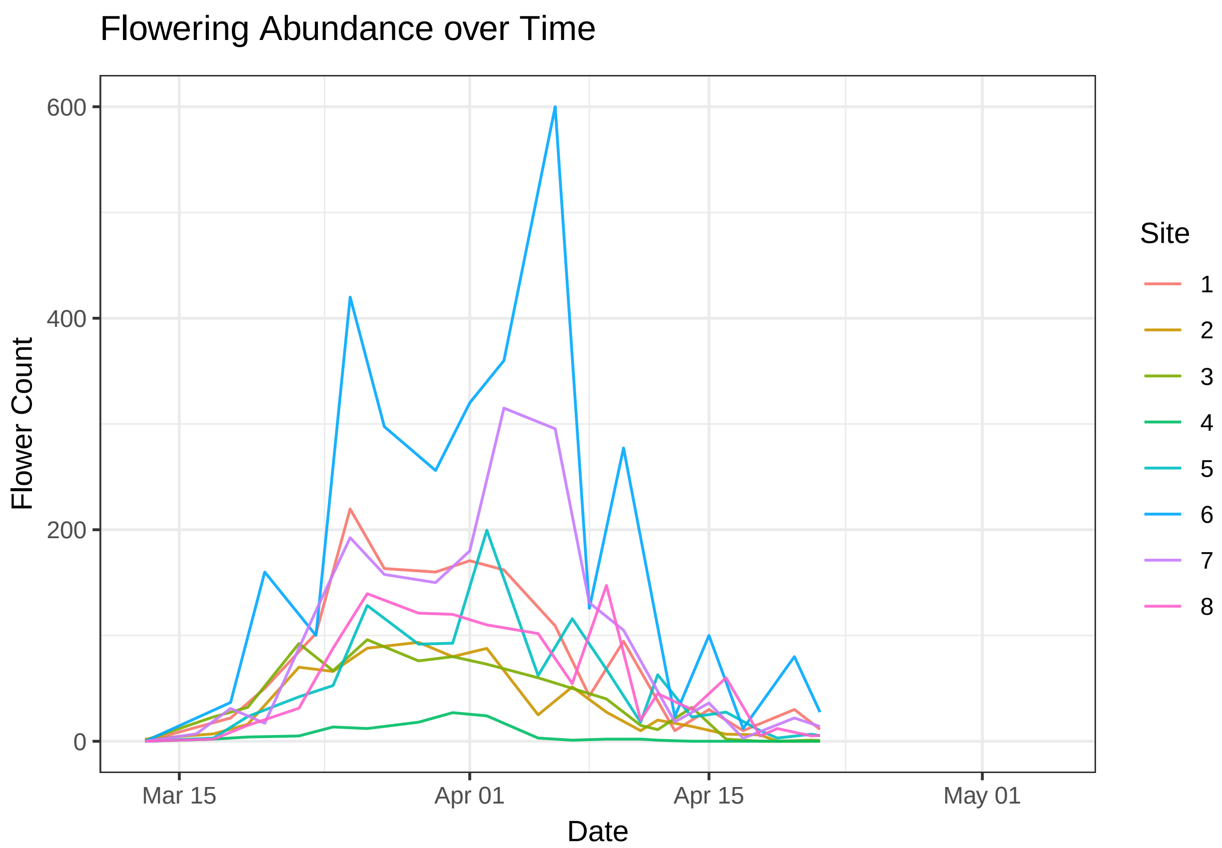
Flowering Abundance over Time
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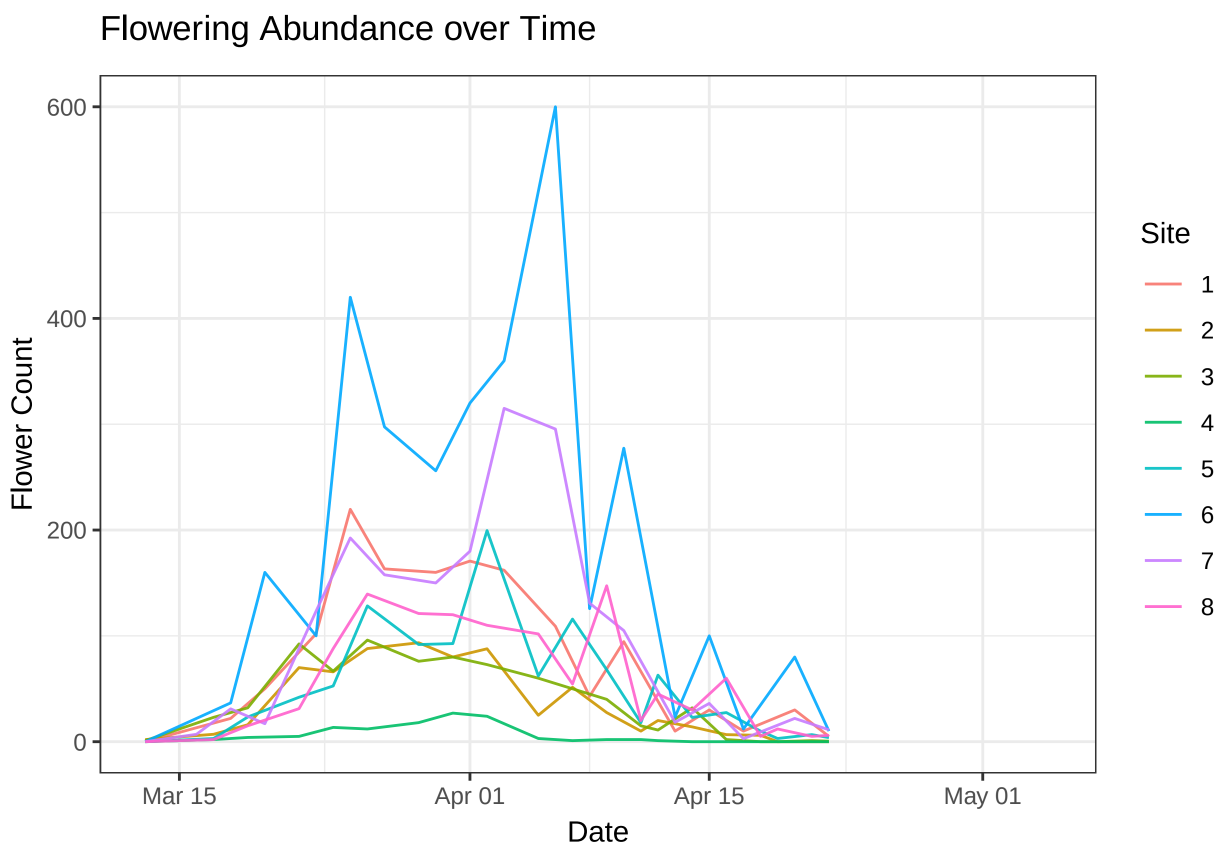
Flowering Abundance over Time
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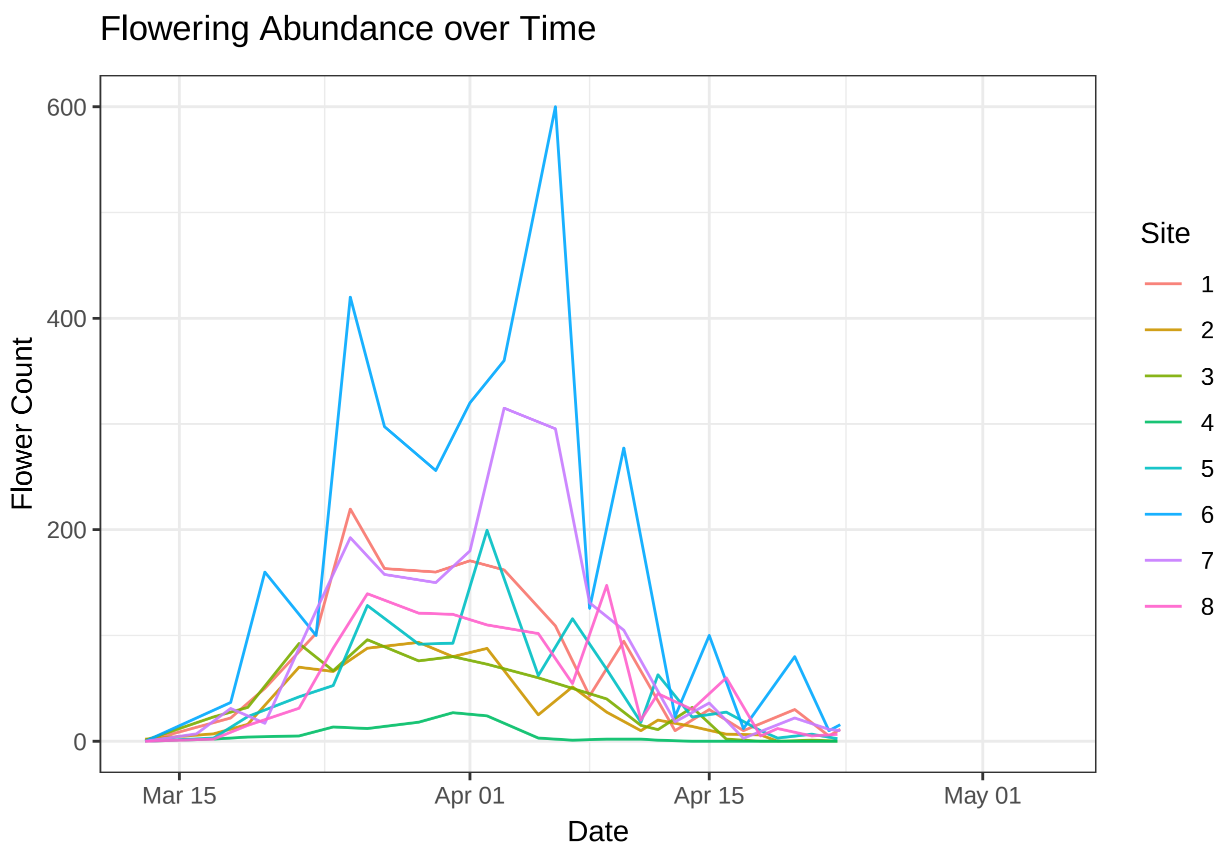
Flowering Abundance over Time
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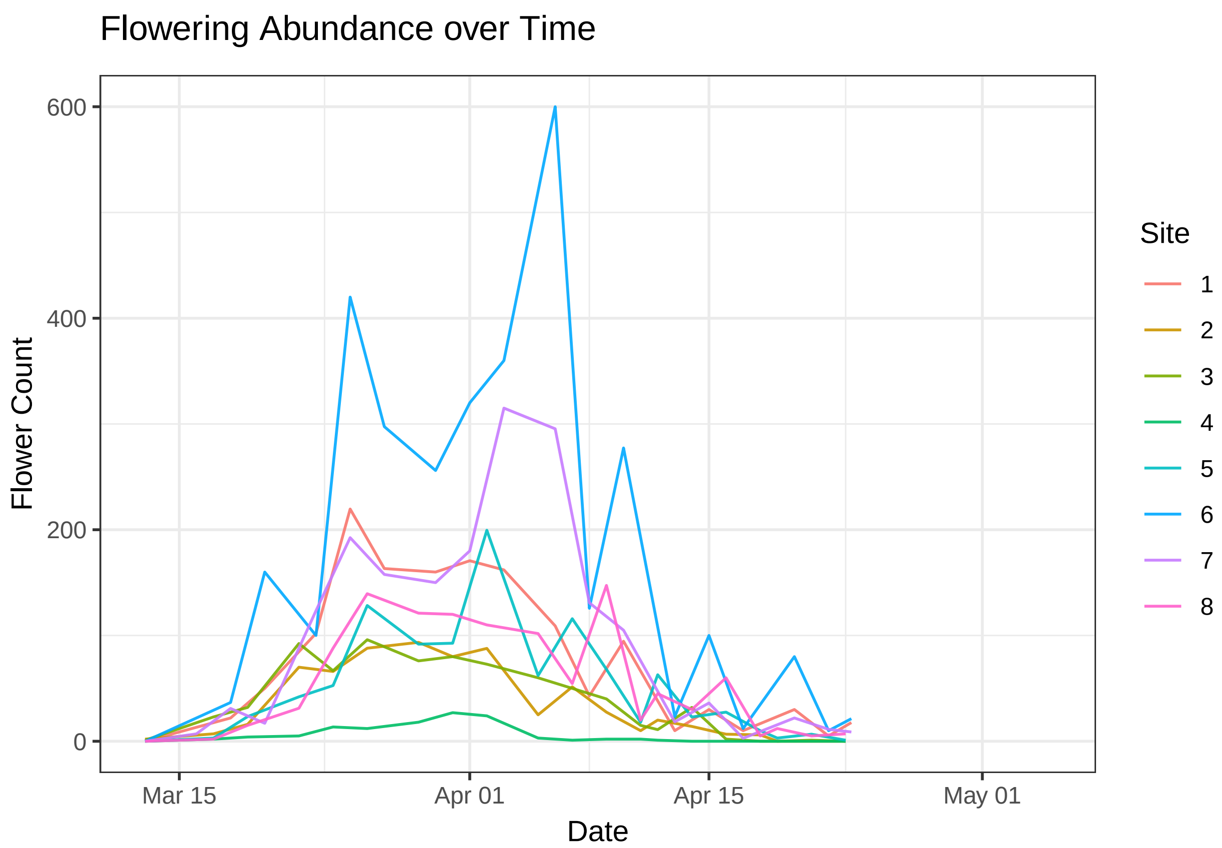

Flowering Abundance over Time
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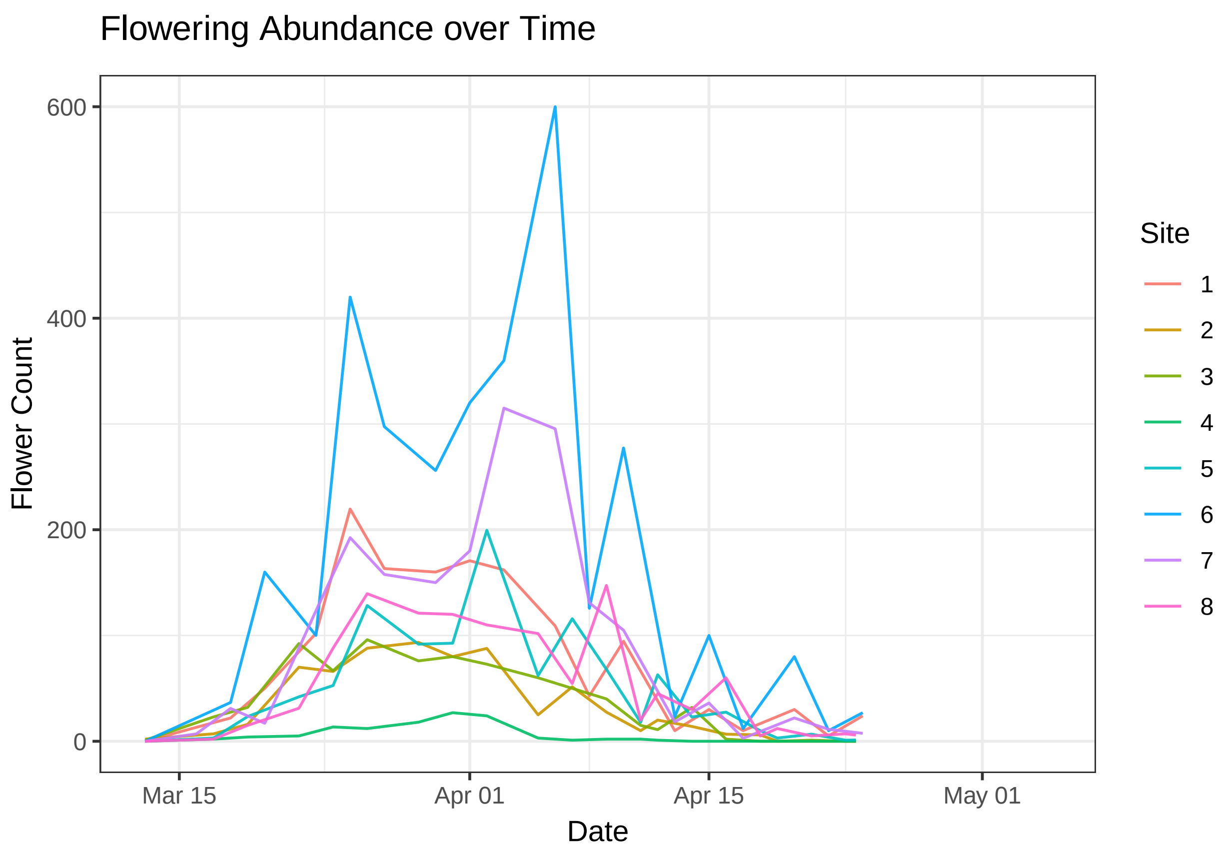

Flowering Abundance over Time
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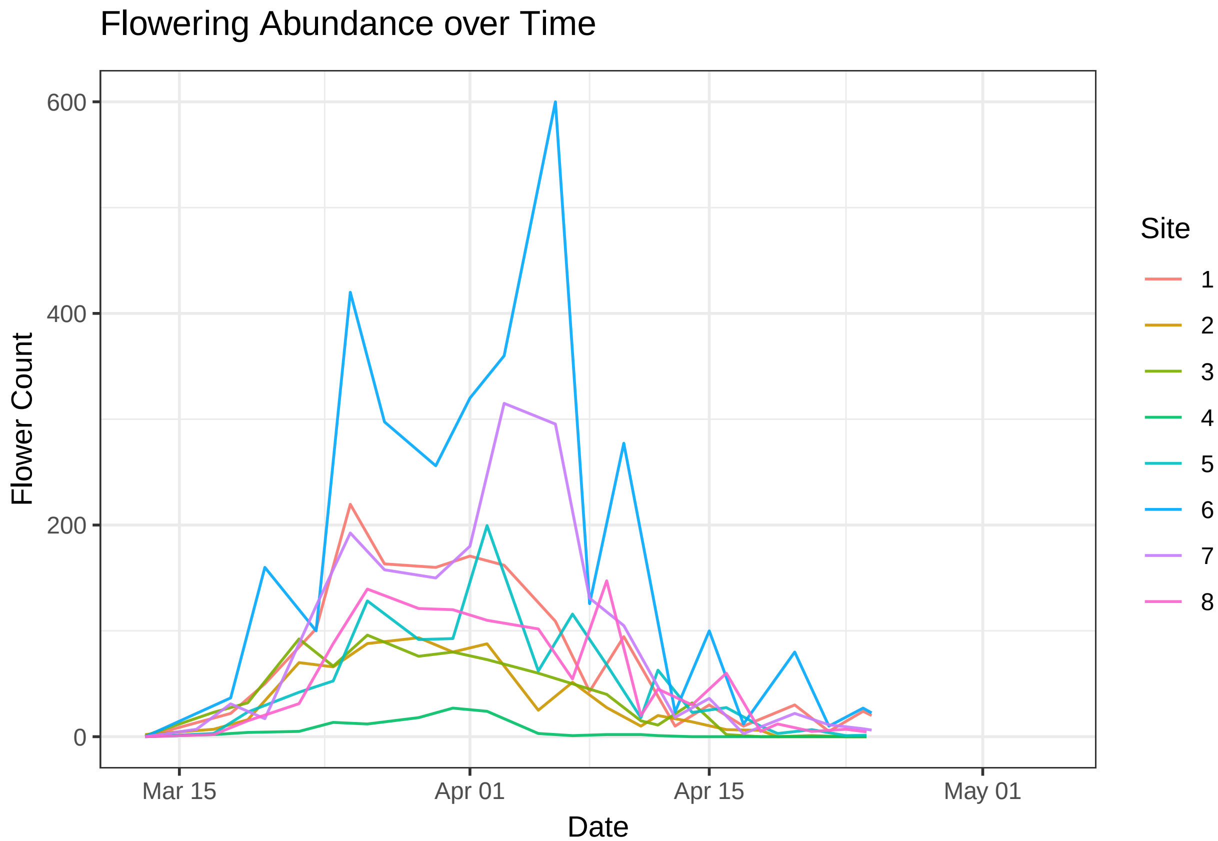

Flowering Abundance over Time
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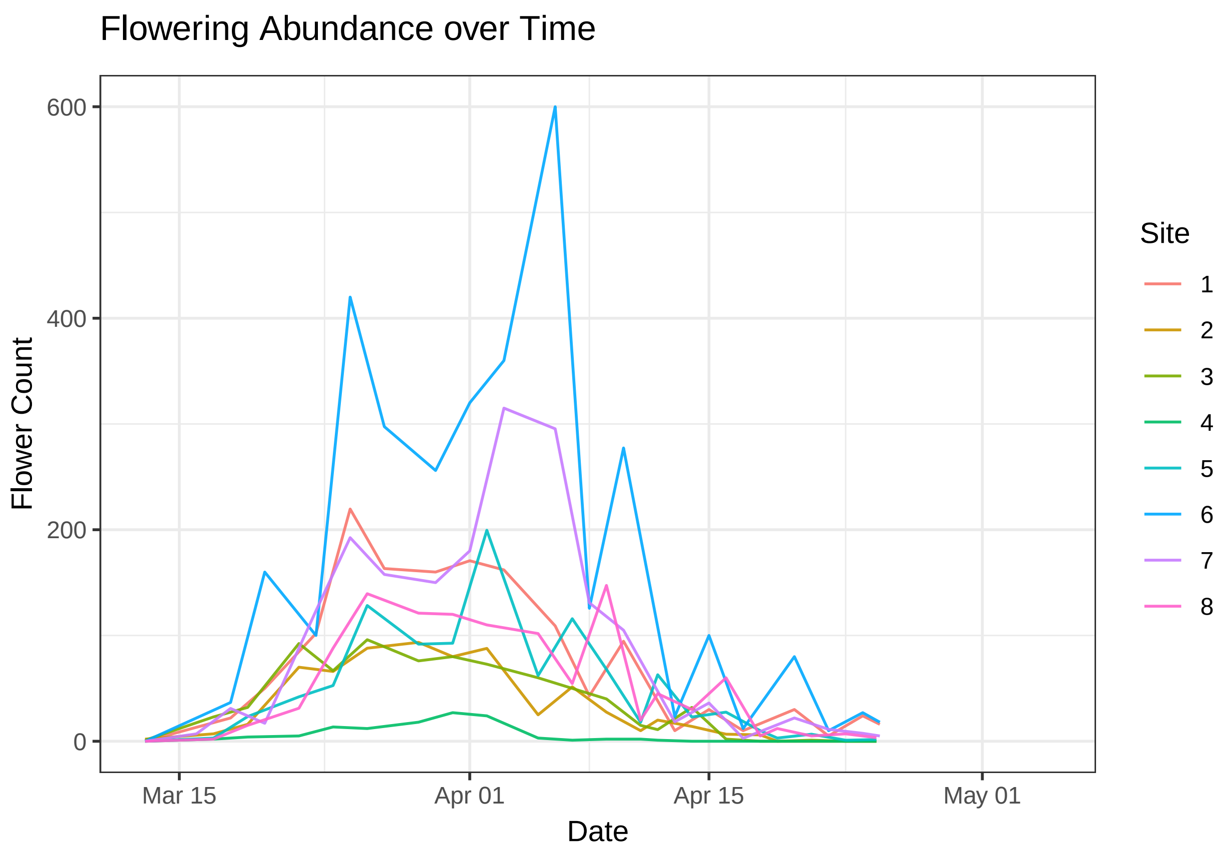
Flowering Abundance over Time
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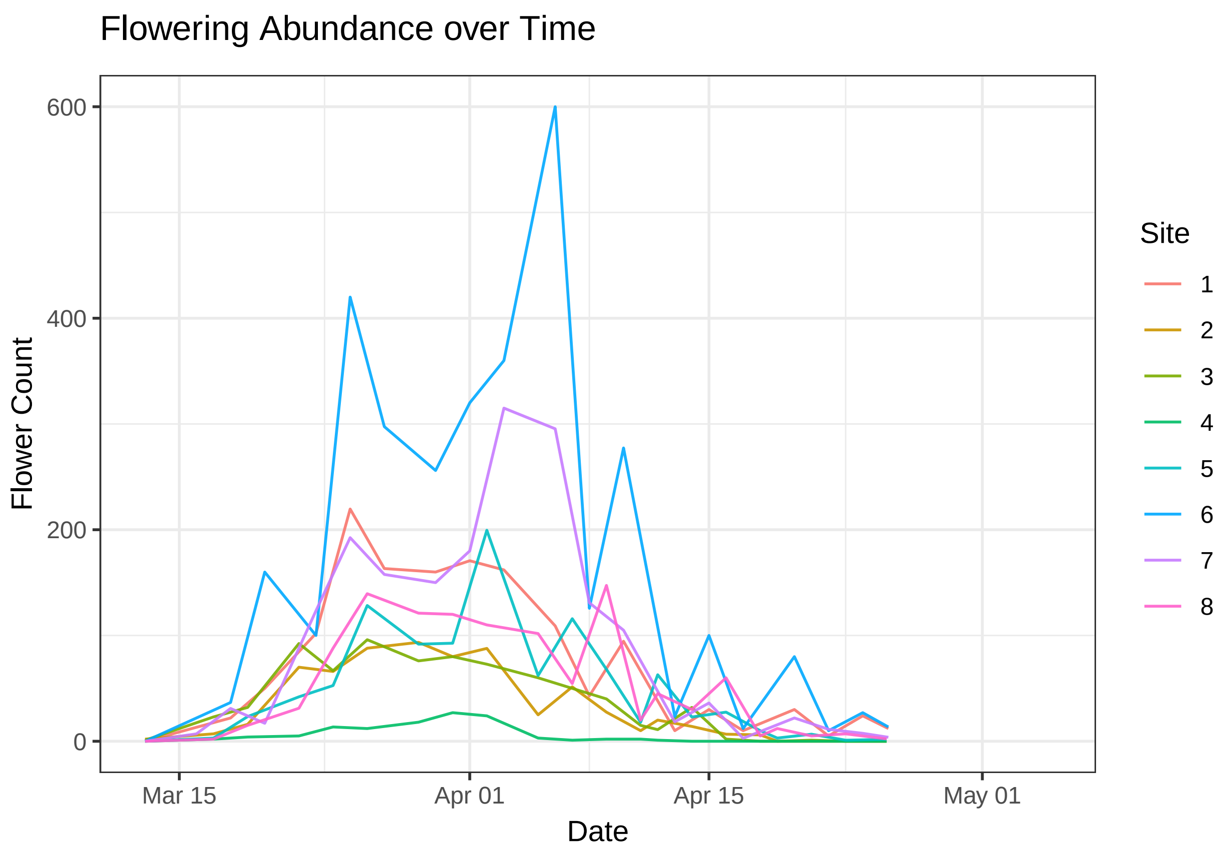
Flowering Abundance over Time
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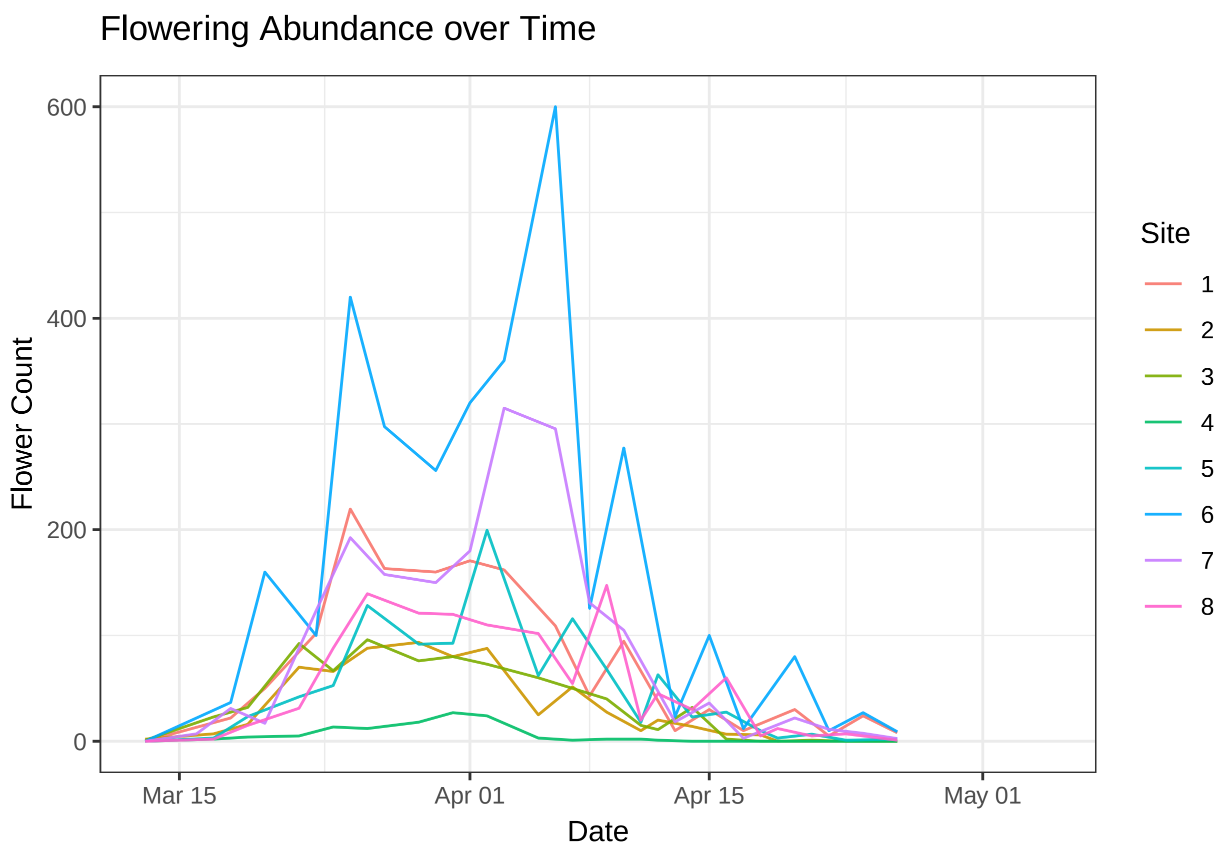

Flowering Abundance over Time
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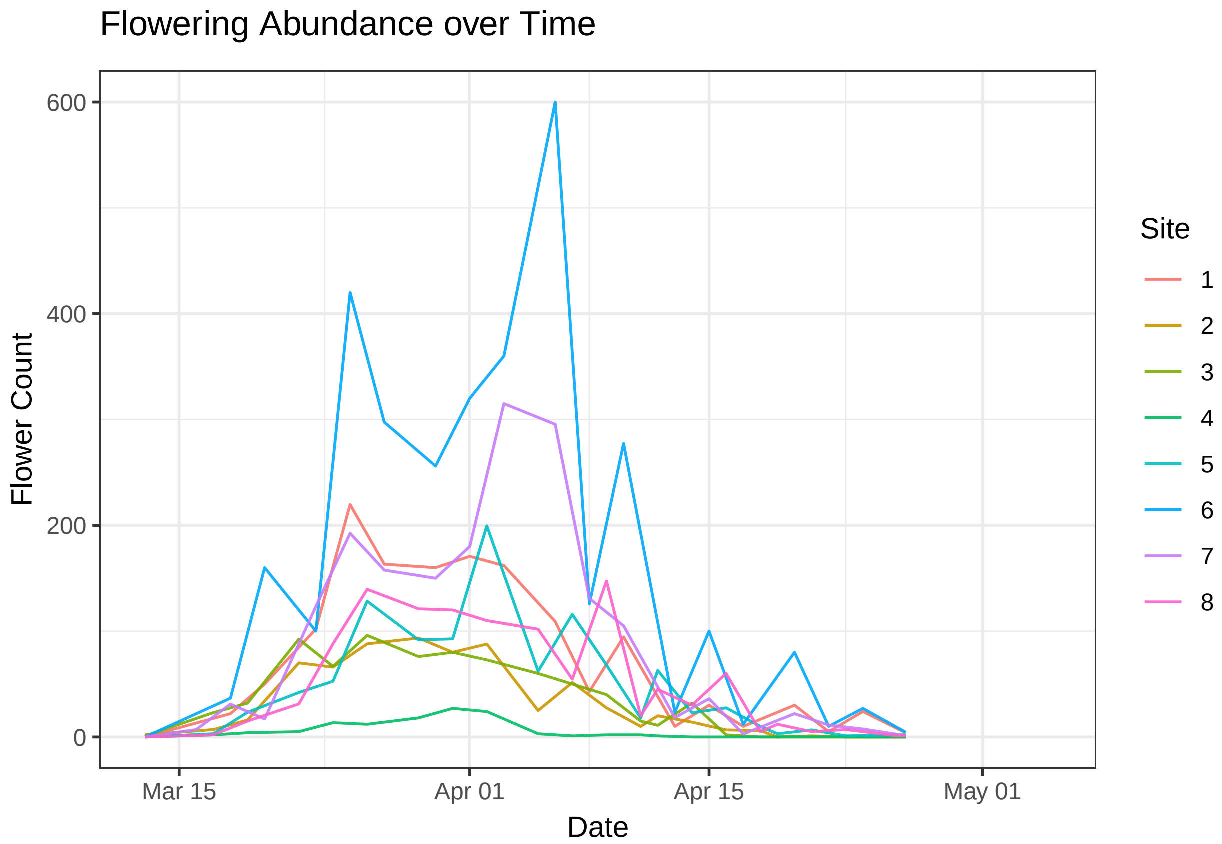

Flowering Abundance over Time
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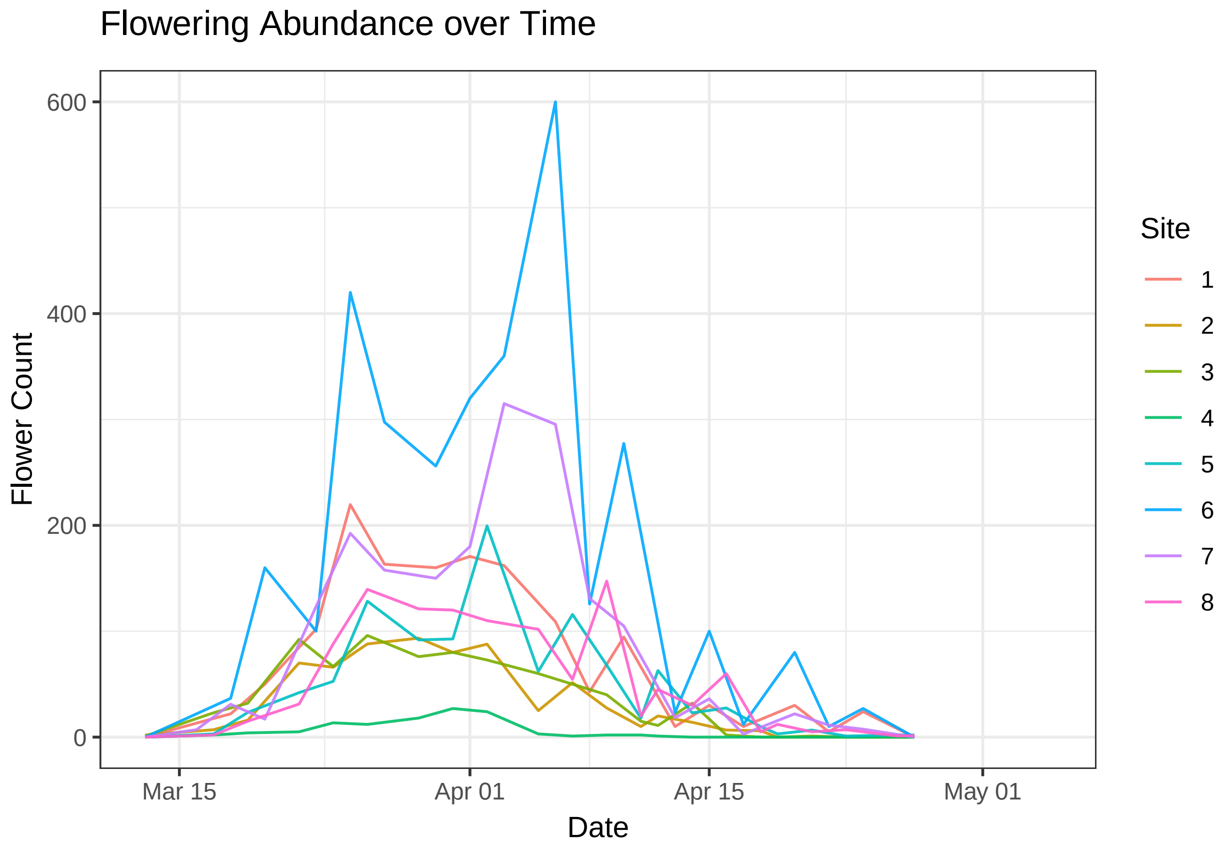
Flowering Abundance over Time
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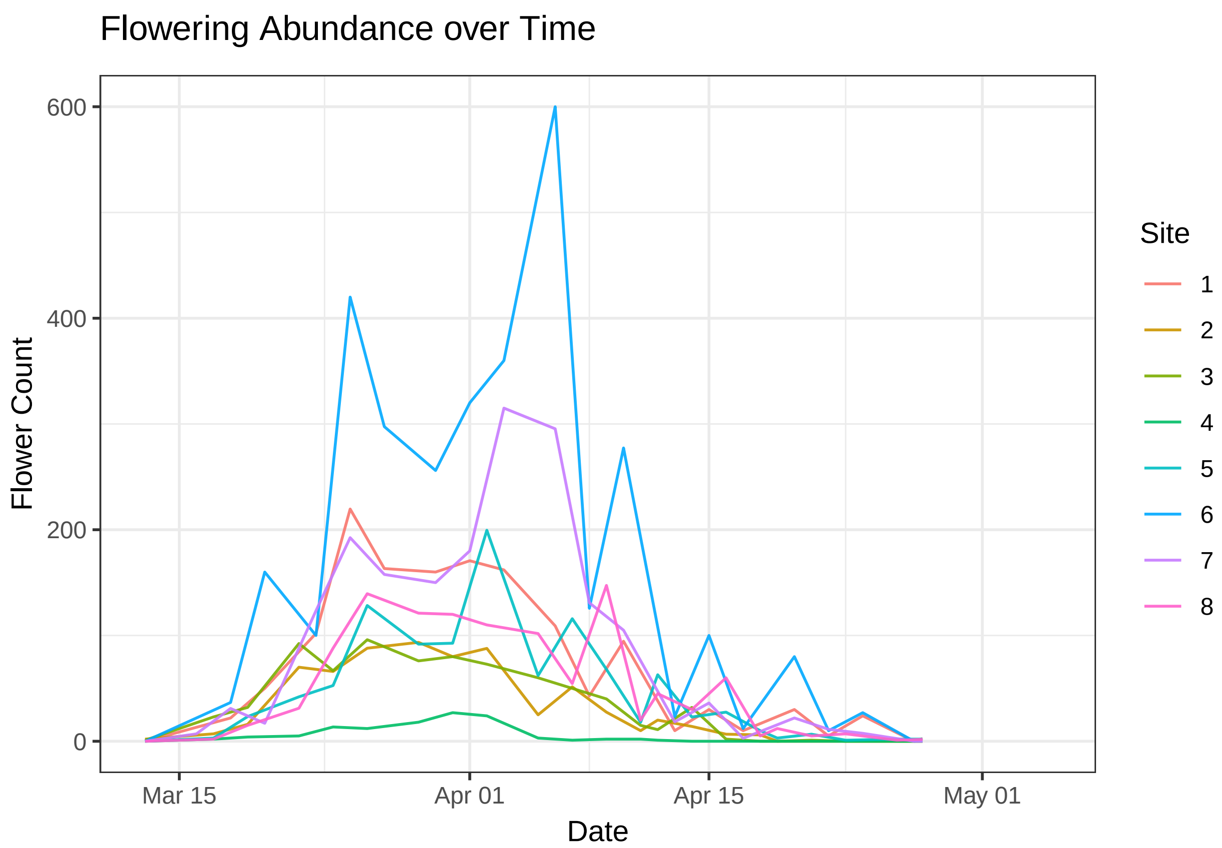
Flowering Abundance over Time
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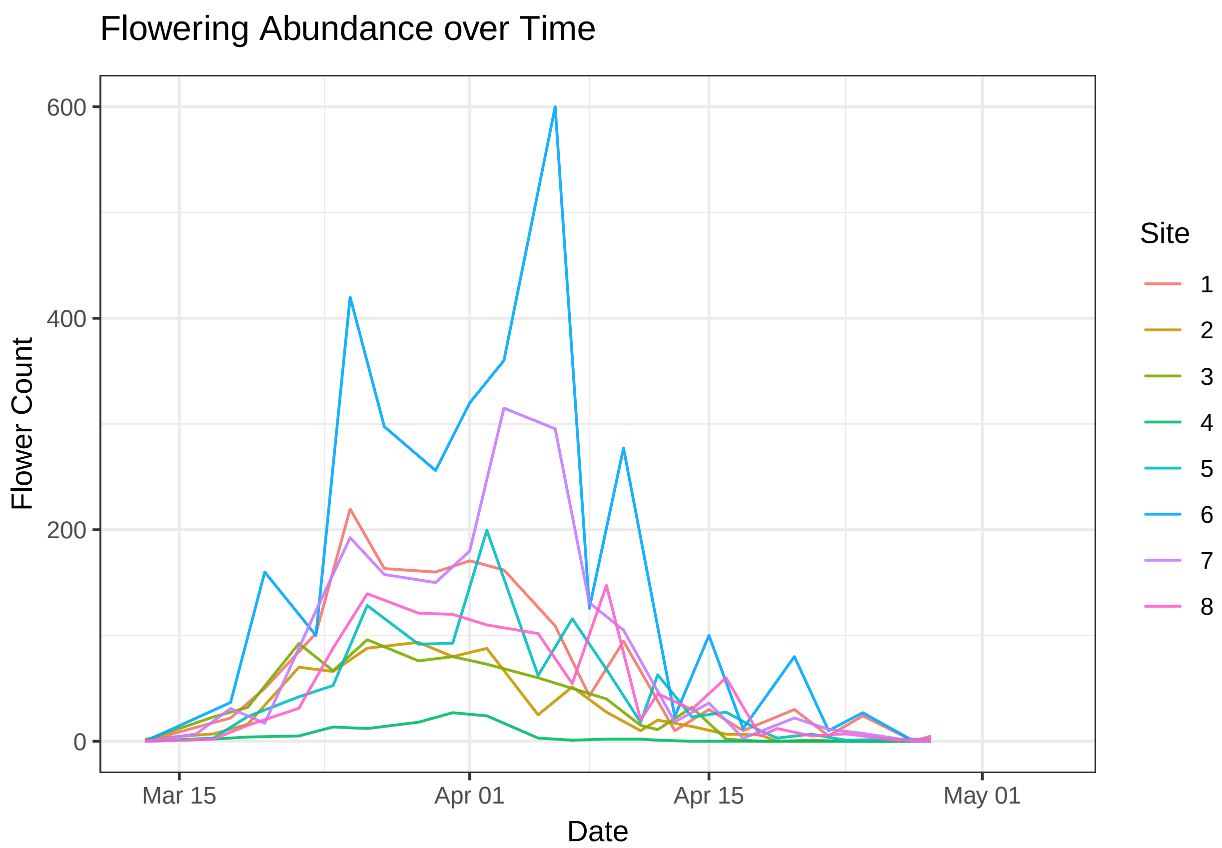
Flowering Abundance over Time
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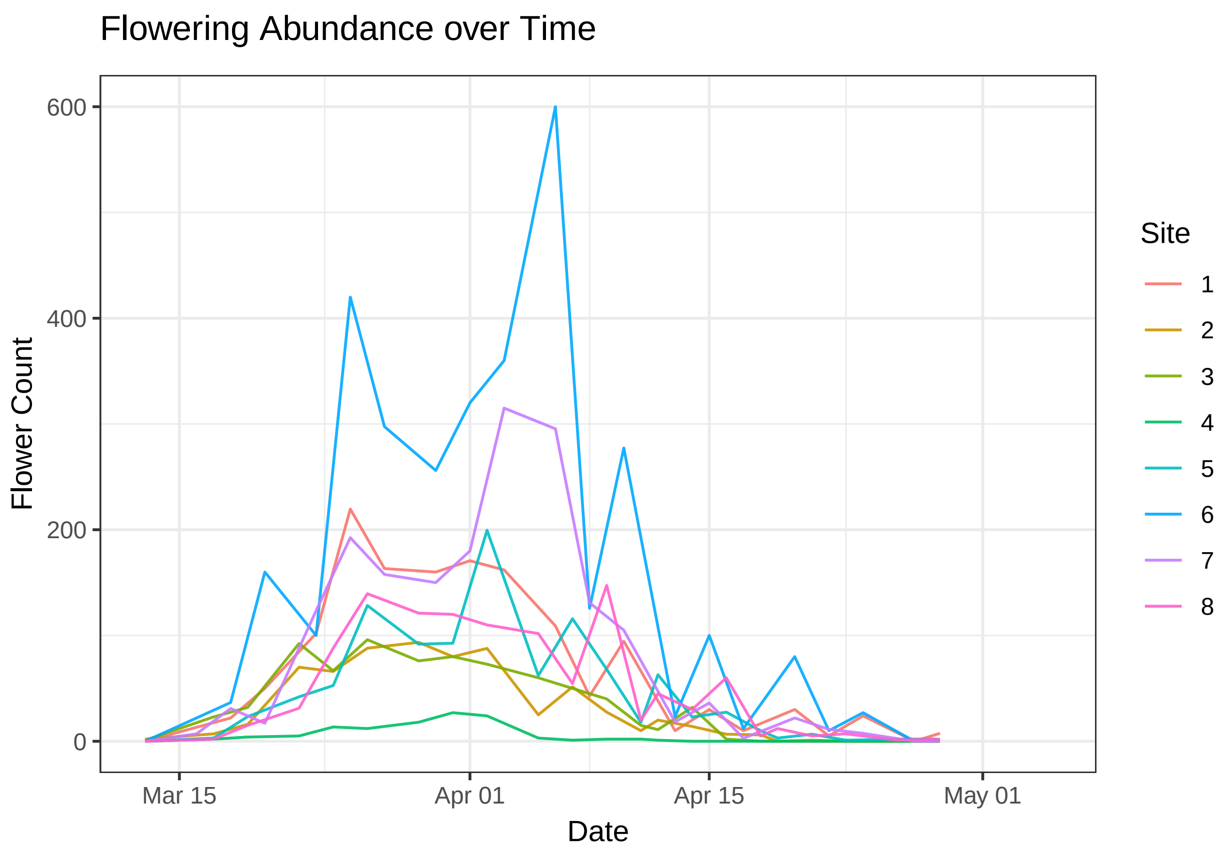
Flowering Abundance over Time
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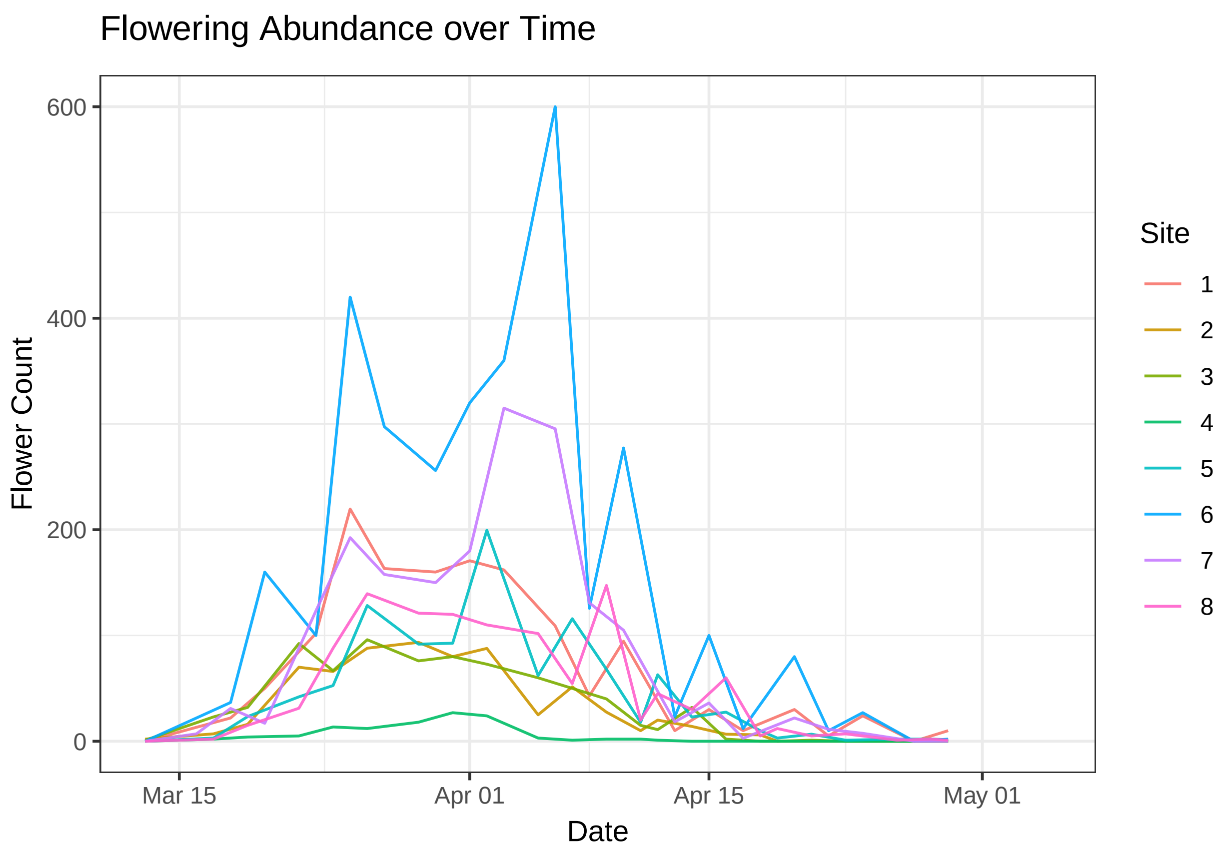

Flowering Abundance over Time
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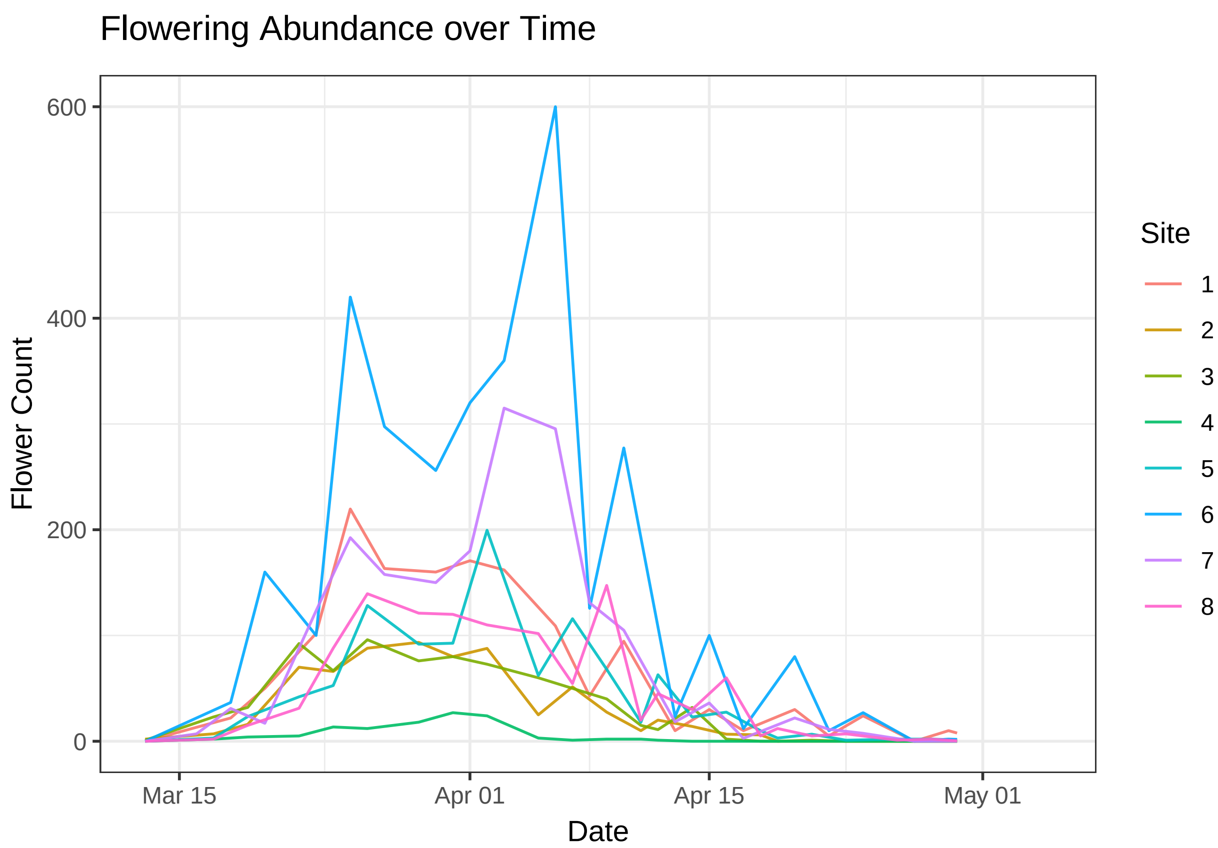
Flowering Abundance over Time
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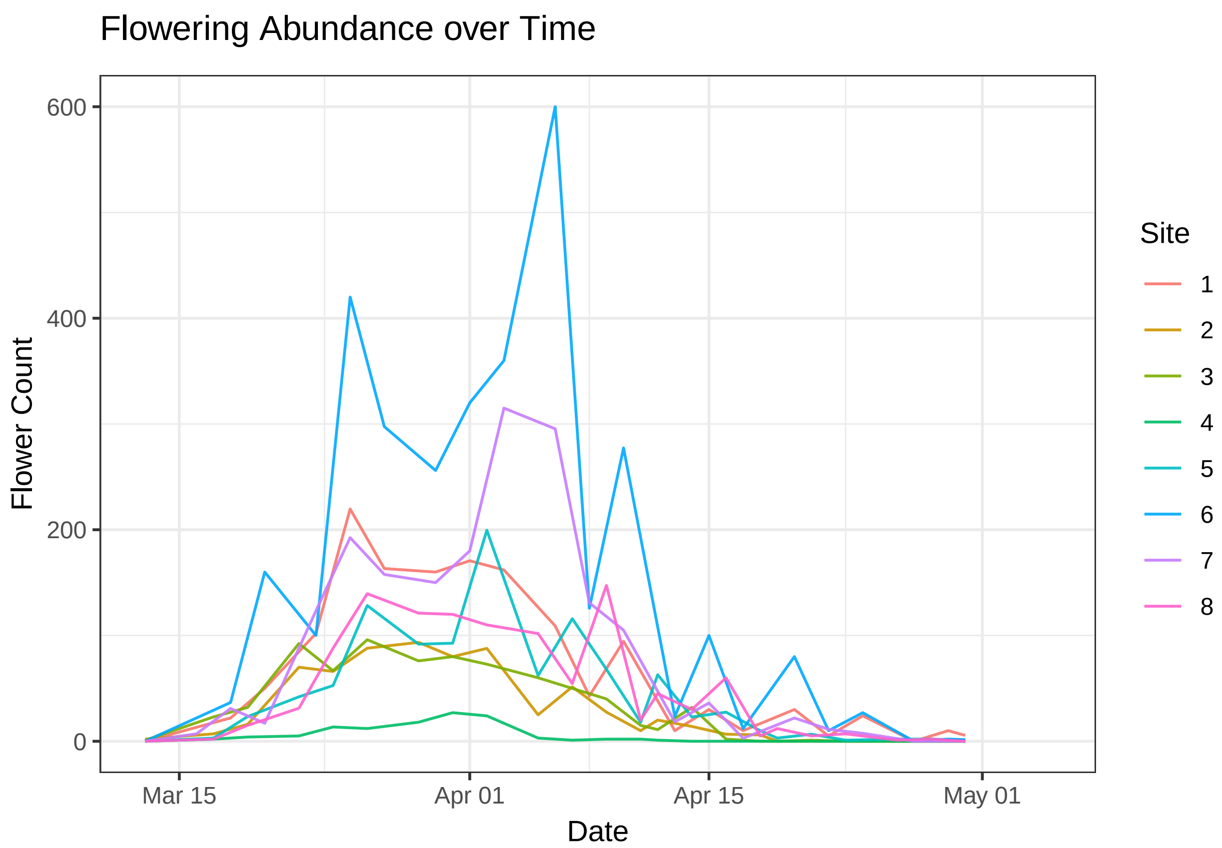

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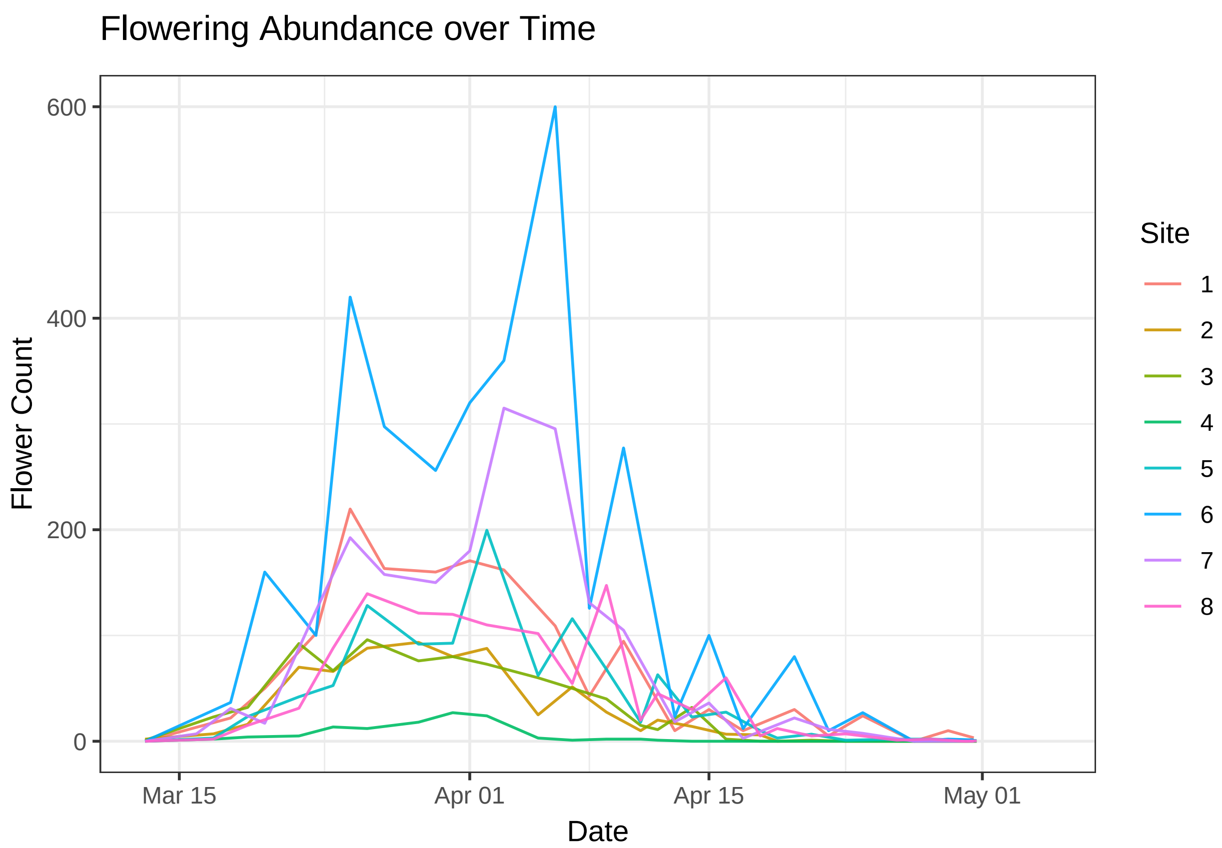
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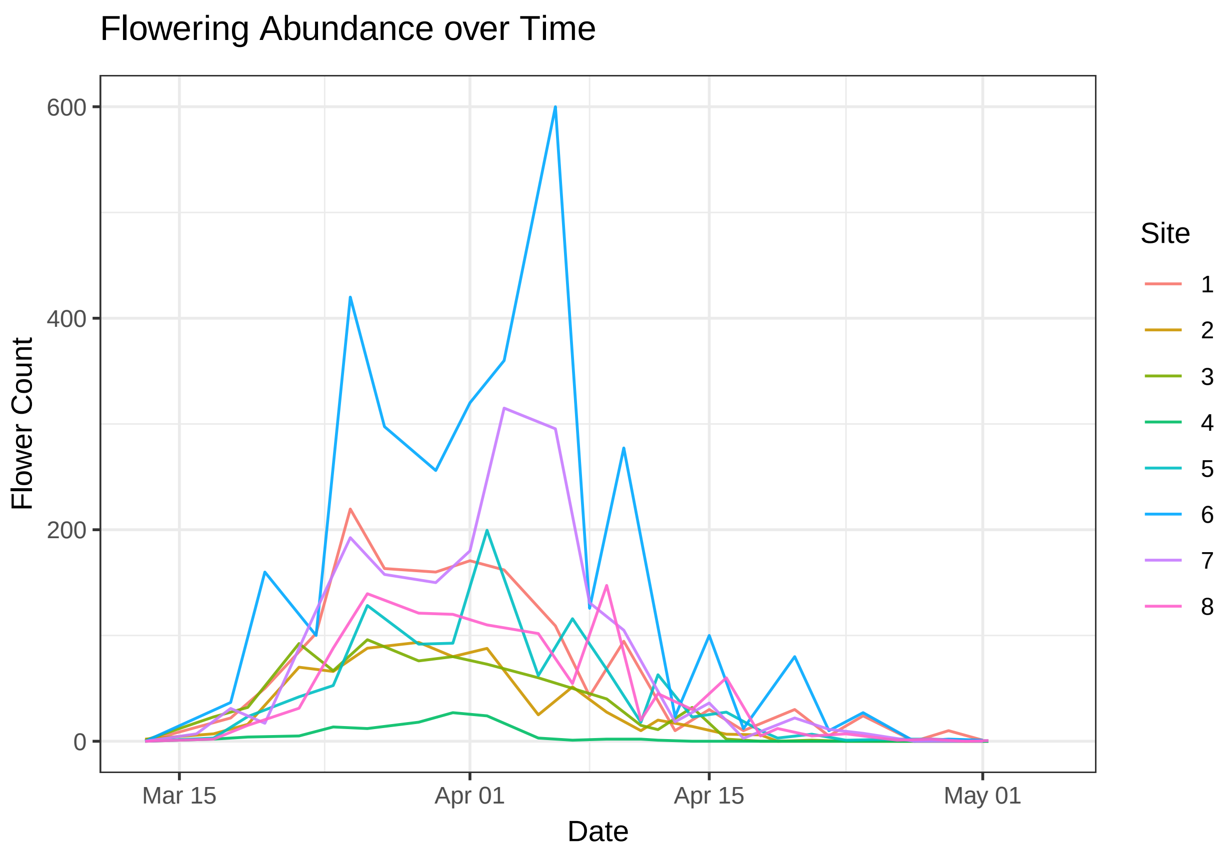
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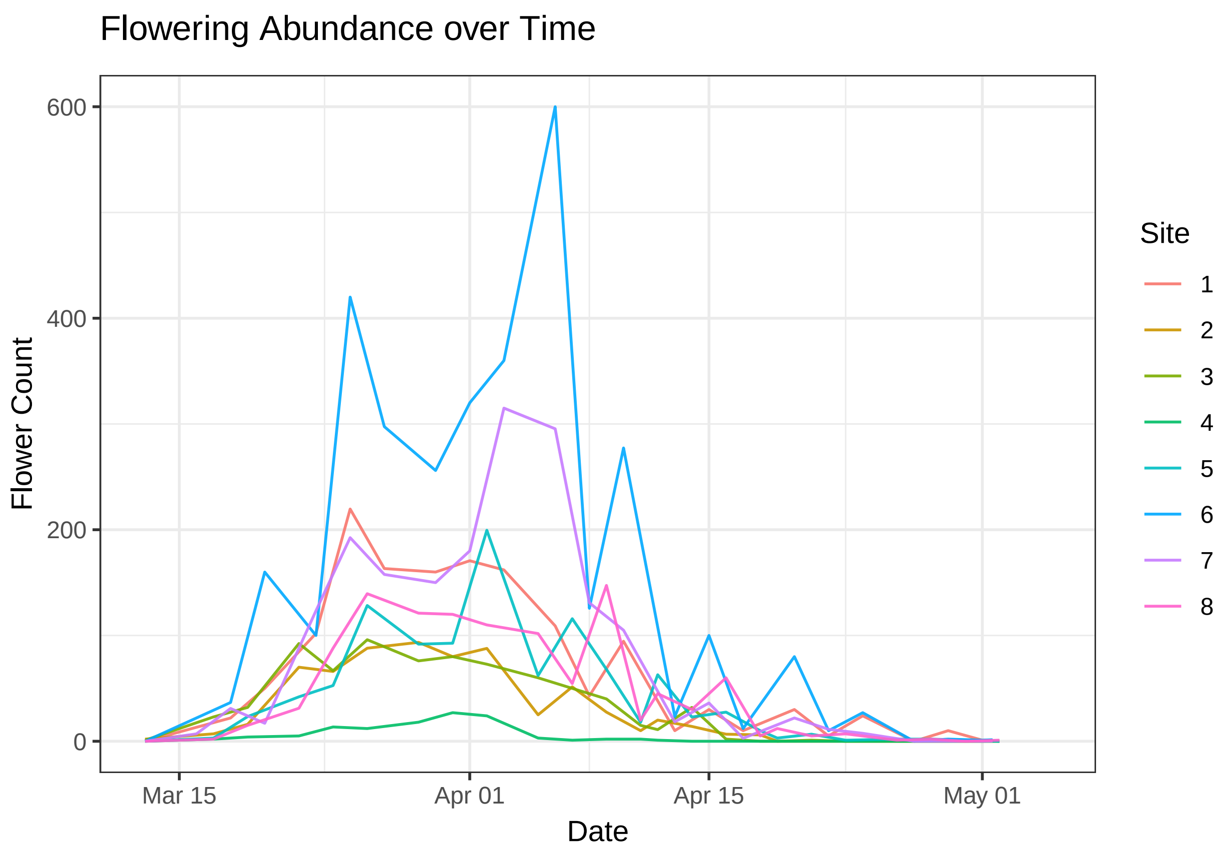
Flowering Abundance over Time
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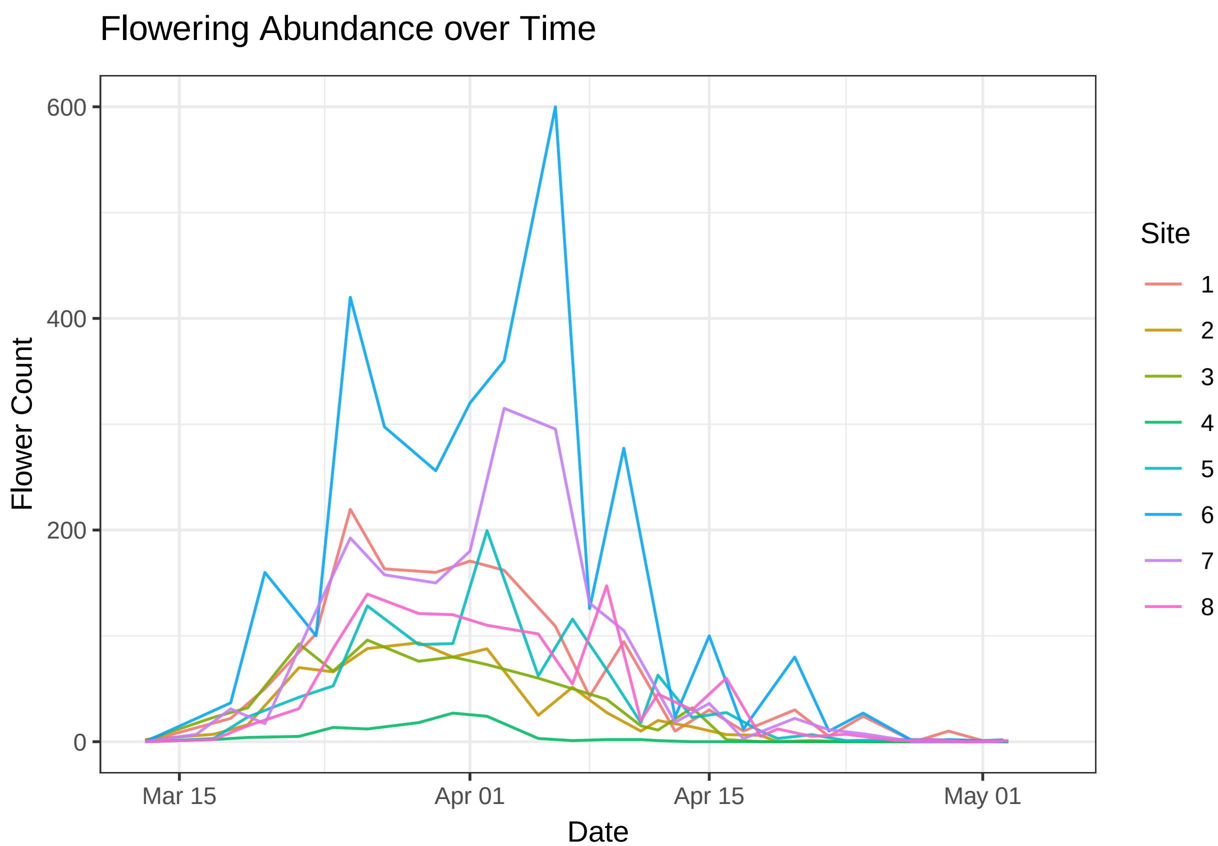

Flowering Abundance over Time
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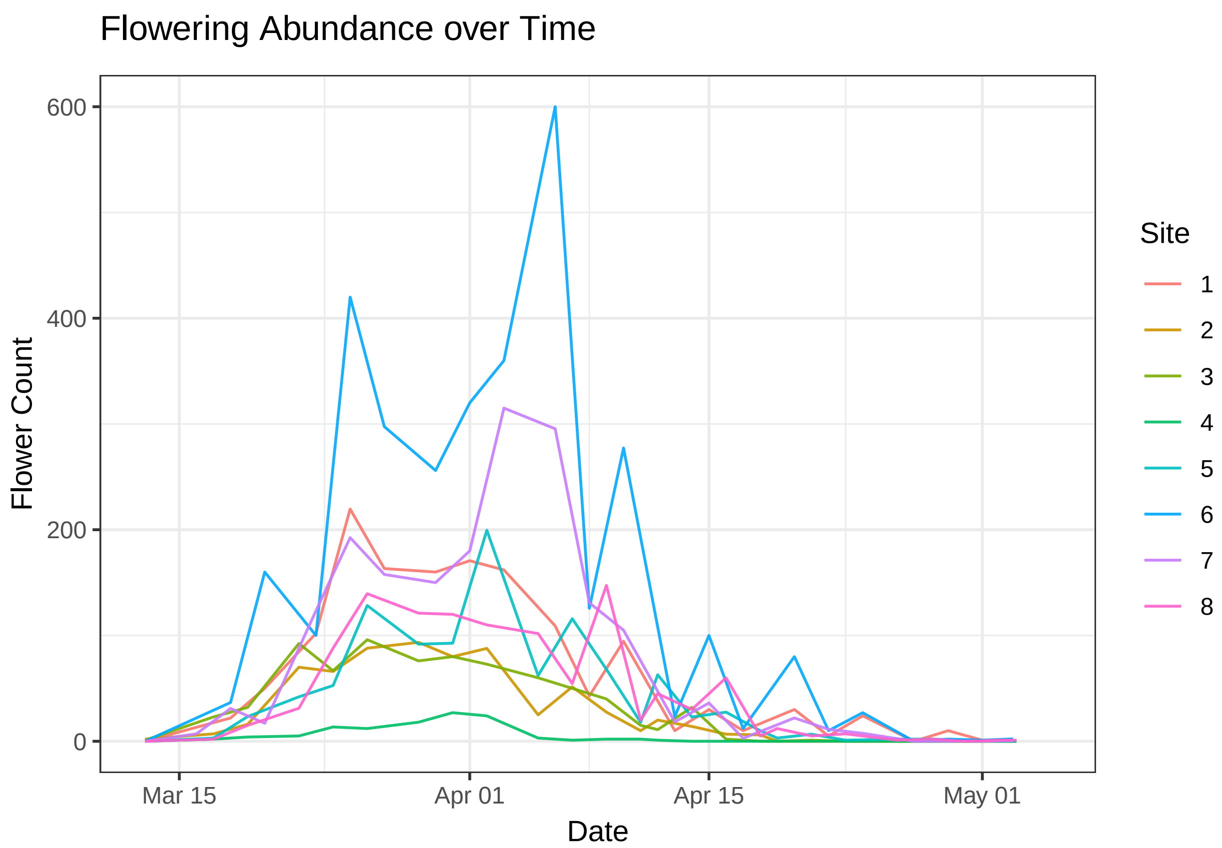
Flowering Abundance over Time
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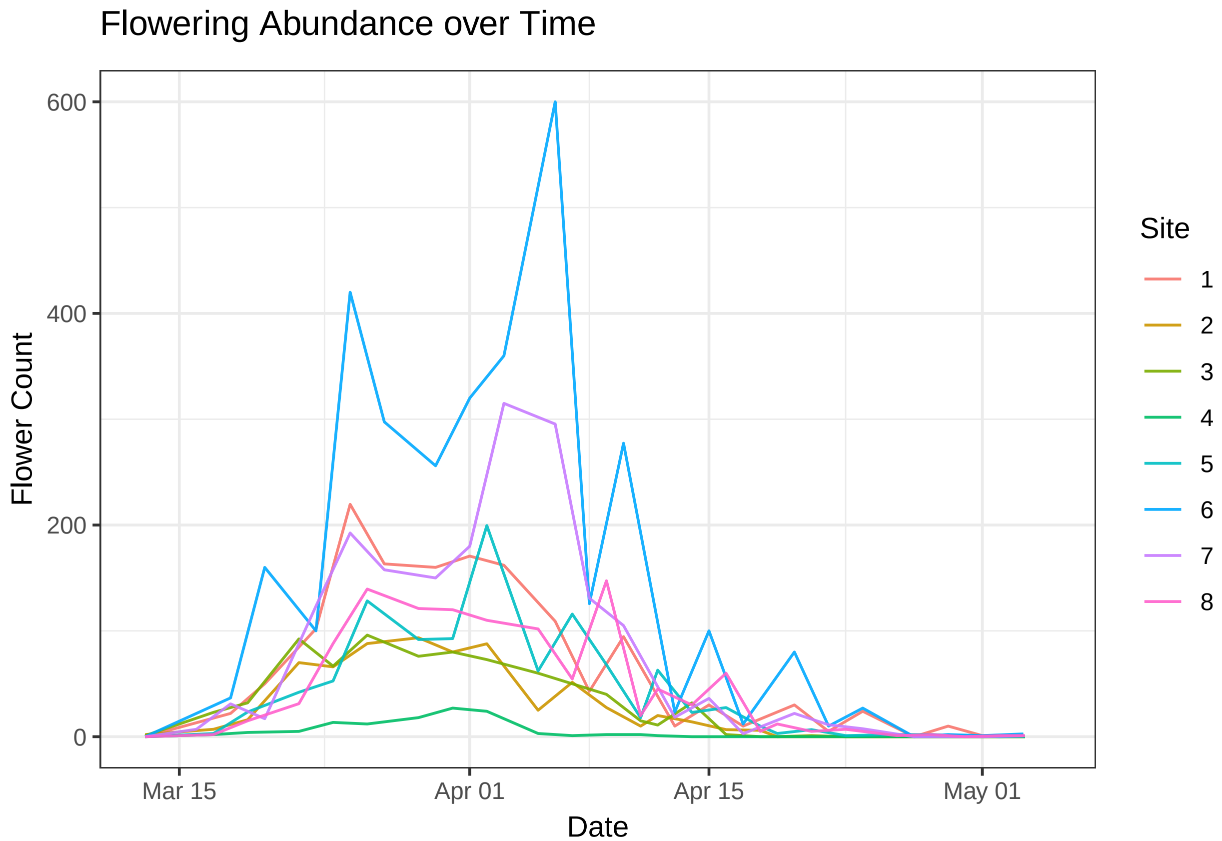

Flowering Abundance over Time
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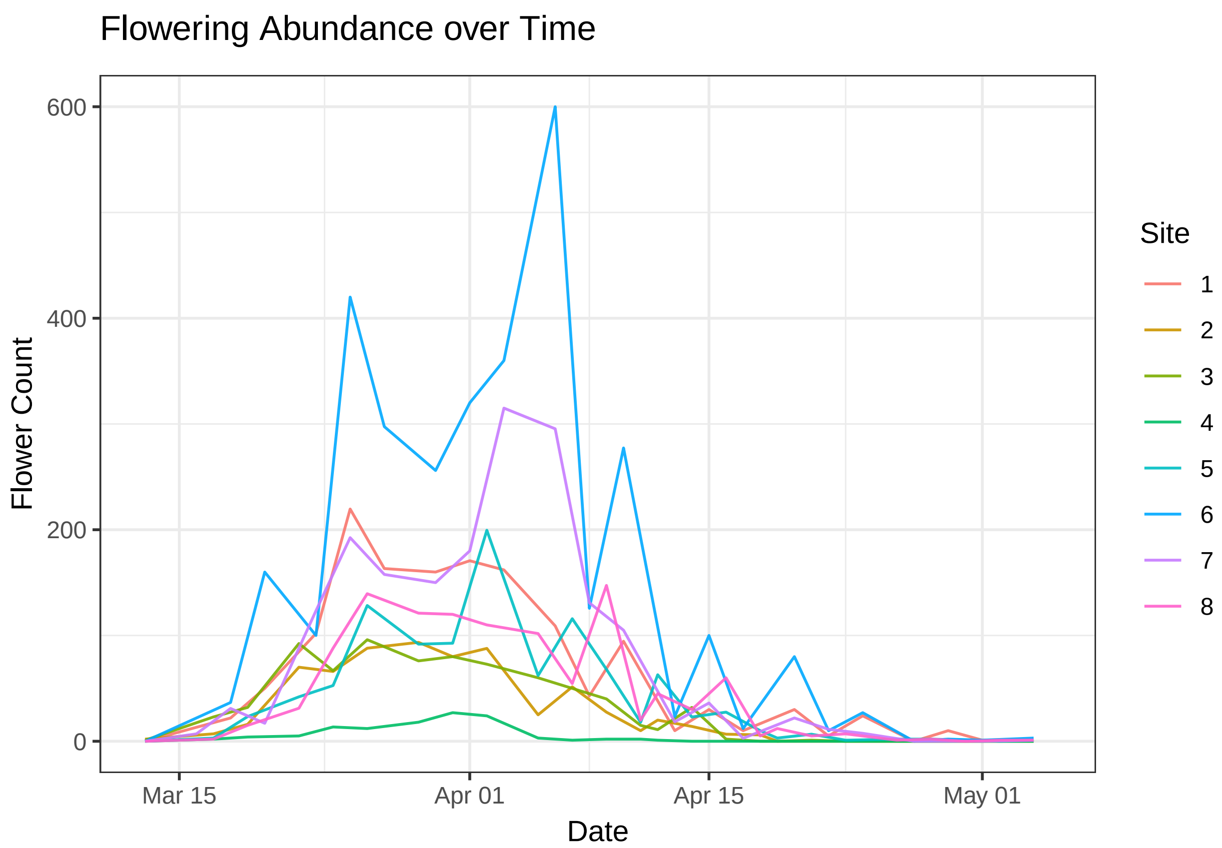

Flowering Abundance over Time
Flower Count
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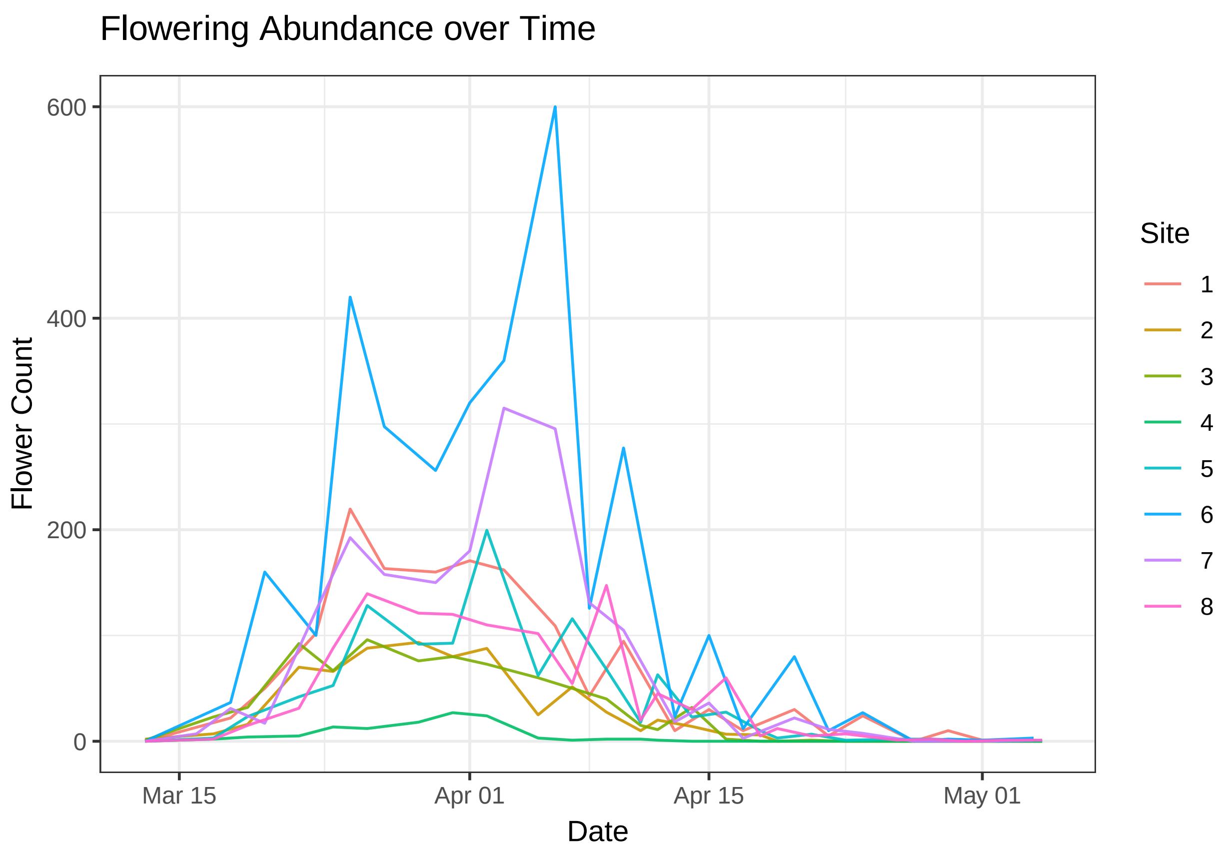
Flowering Abundance over Time
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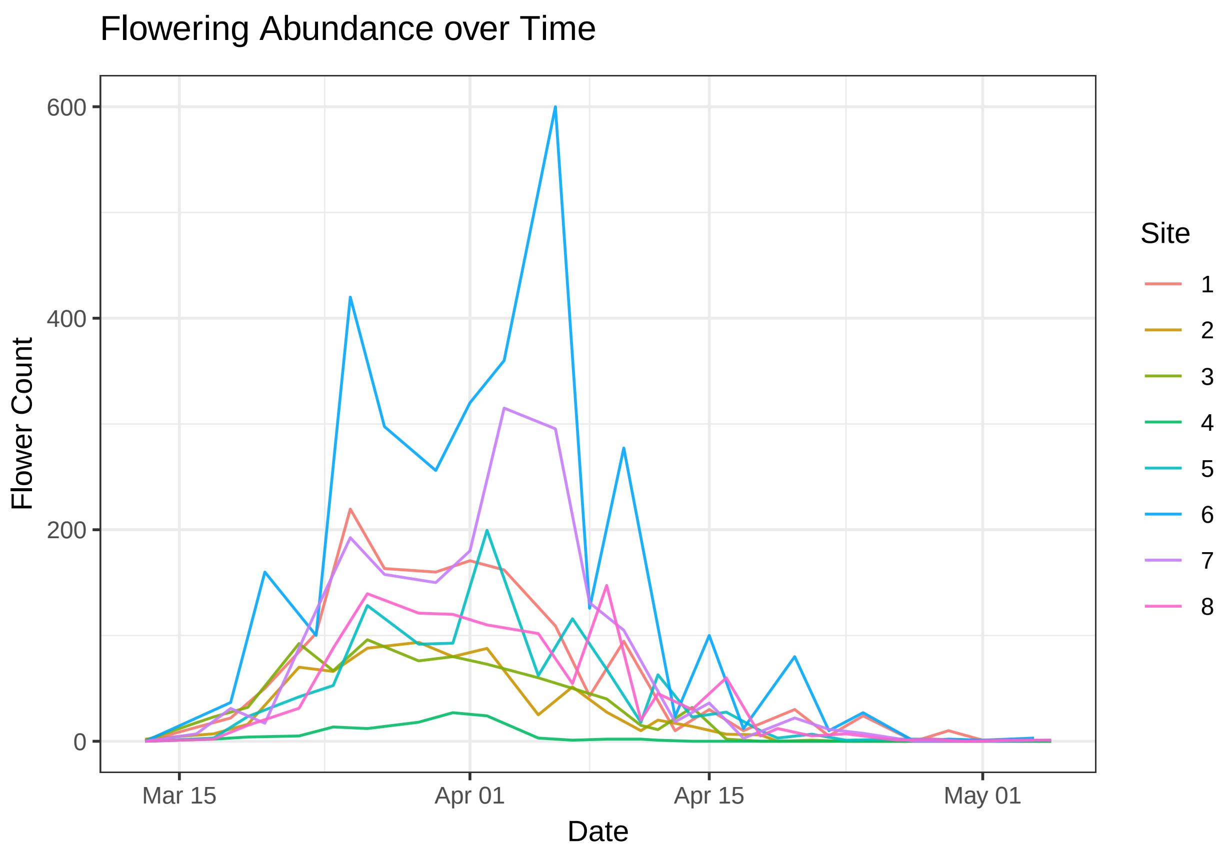
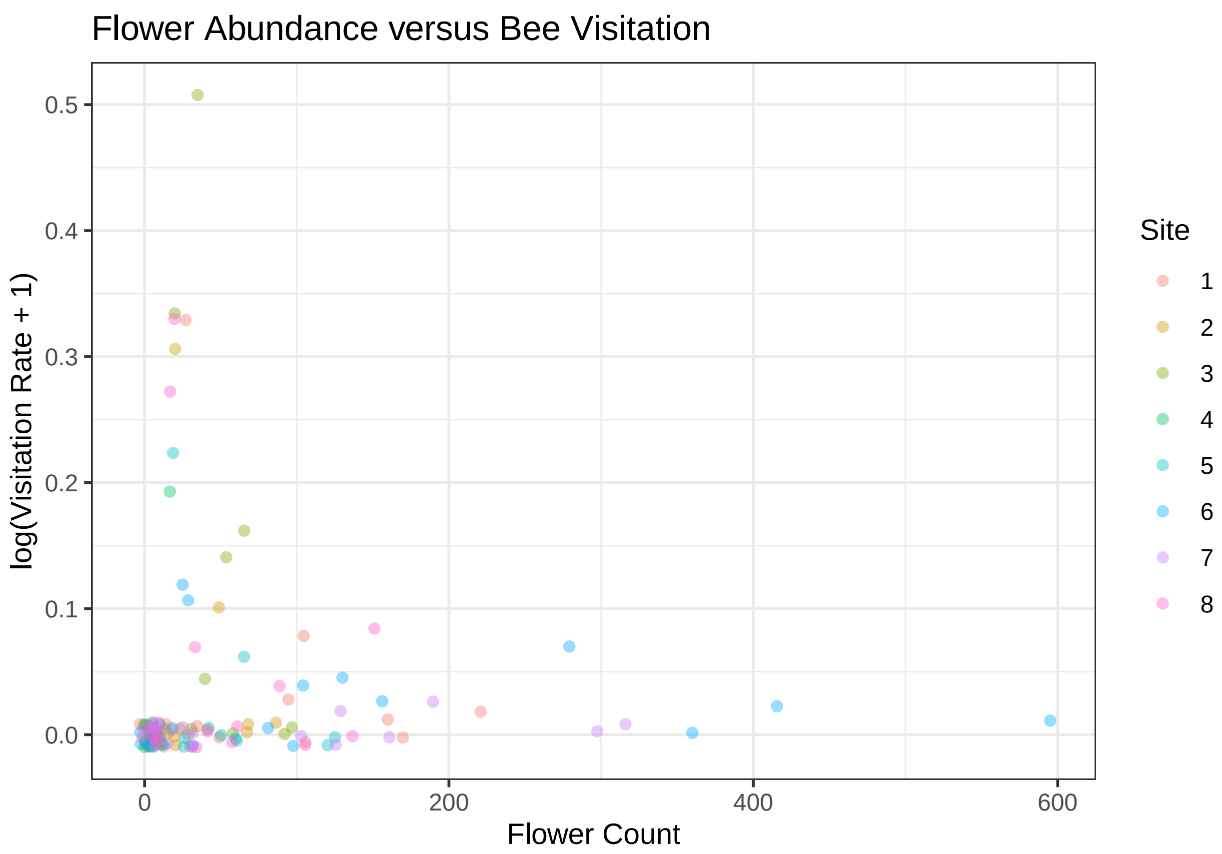


Visualize the relationship between flower abundance and bee visitation

```
(plot_T1_visitation <- T1 %>%
    ggplot(aes(x = flower_count,
               y = log1p(visitation_rate),
               color = site)) +
    geom_jitter(alpha = 0.4, height = 0.01, width = 5) +
    theme_bw() +
    labs(title = "Flower Abundance versus Bee Visitation",
         x = "Flower Count",
         y = "log(Visitation Rate + 1)",
         color = "Site")
  )
```

```
## Warning: Removed 65 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

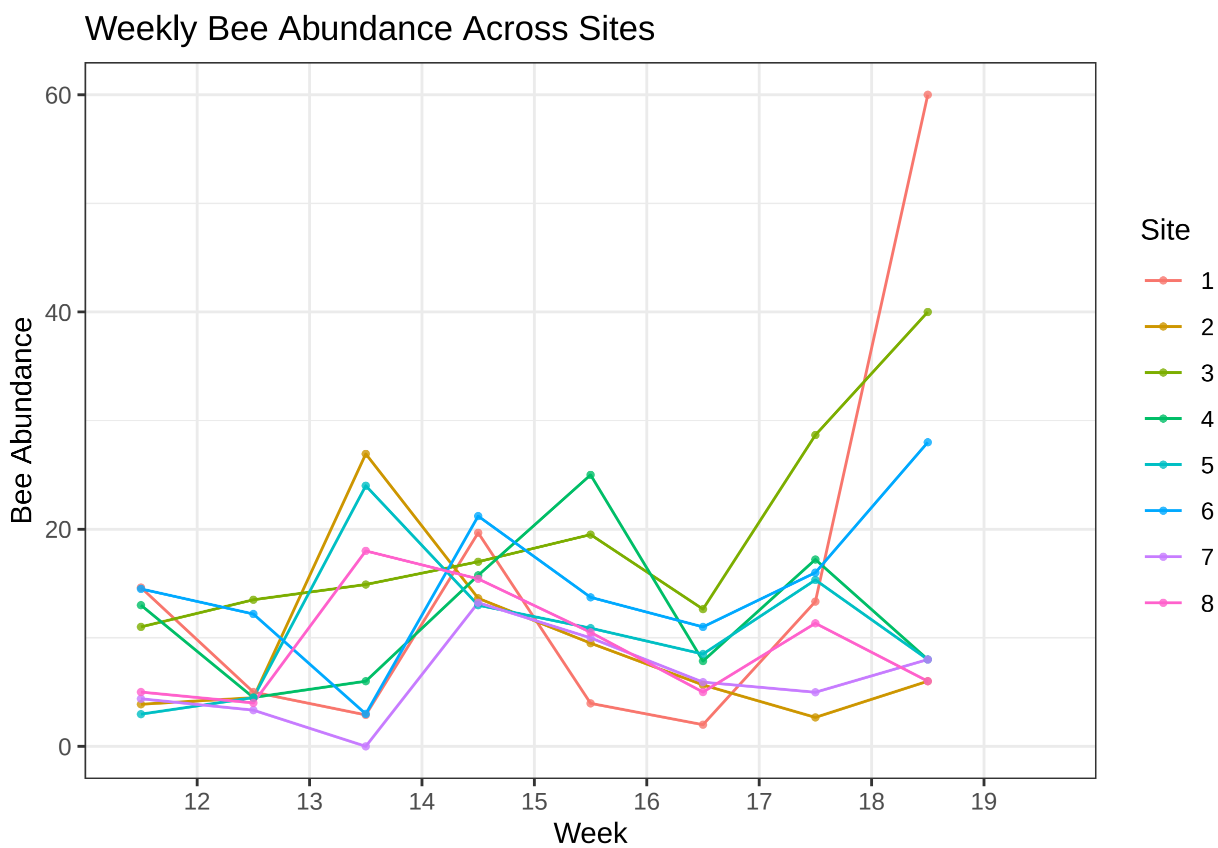


Visualize bee abundance over time across each site

```
(plot_T2 <- T2 %>%
    ggplot(aes(x = week,
               y = bee_abundance,
               color = site)) +
    geom_line() +
    geom_point(size = 0.8, alpha = 0.8) +
    theme_bw() +
    scale_x_binned(expand = c(0, 0)) +
    labs(title = "Weekly Bee Abundance Across Sites",
         x = "Week",
         y  = "Bee Abundance",
         color = "Site")
  )
```

```
## Warning: Removed 16 rows containing missing values or values outside the scale range
## (‘geom_line()‘).
```

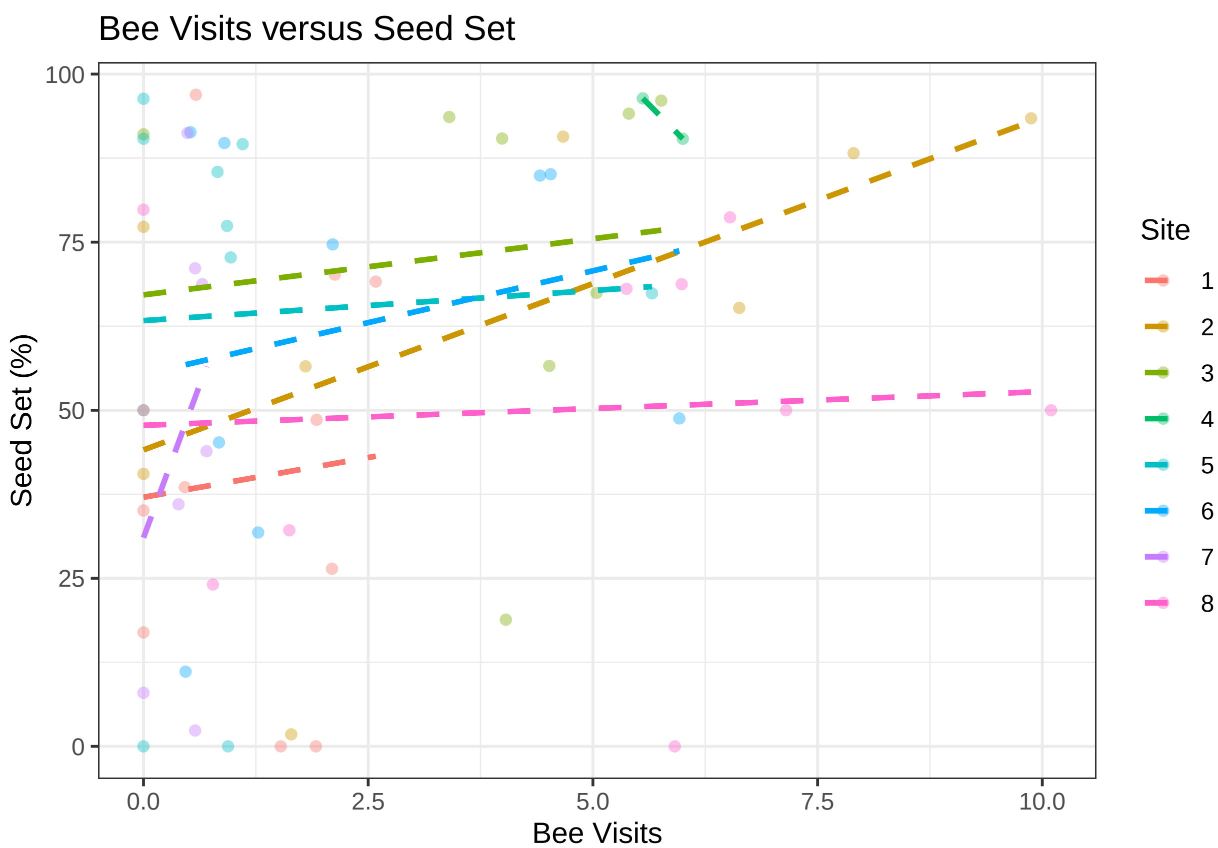
```
## Warning: Removed 16 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```



Visualize the relationship between Bee Visitation and Seed Set %

```
(plot_T3 <- T3 %>%
    ggplot(aes(x = bee_visits,
               y = seed_set,
               color = site)) +
    geom_point(alpha = 0.4) +
    geom_smooth(method = "lm", se = F, linetype = "dashed") +
    theme_bw() +
    labs(title = "Bee Visits versus Seed Set",
         x = "Bee Visits",
         y = "Seed Set (%)",
         color = "Site")
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

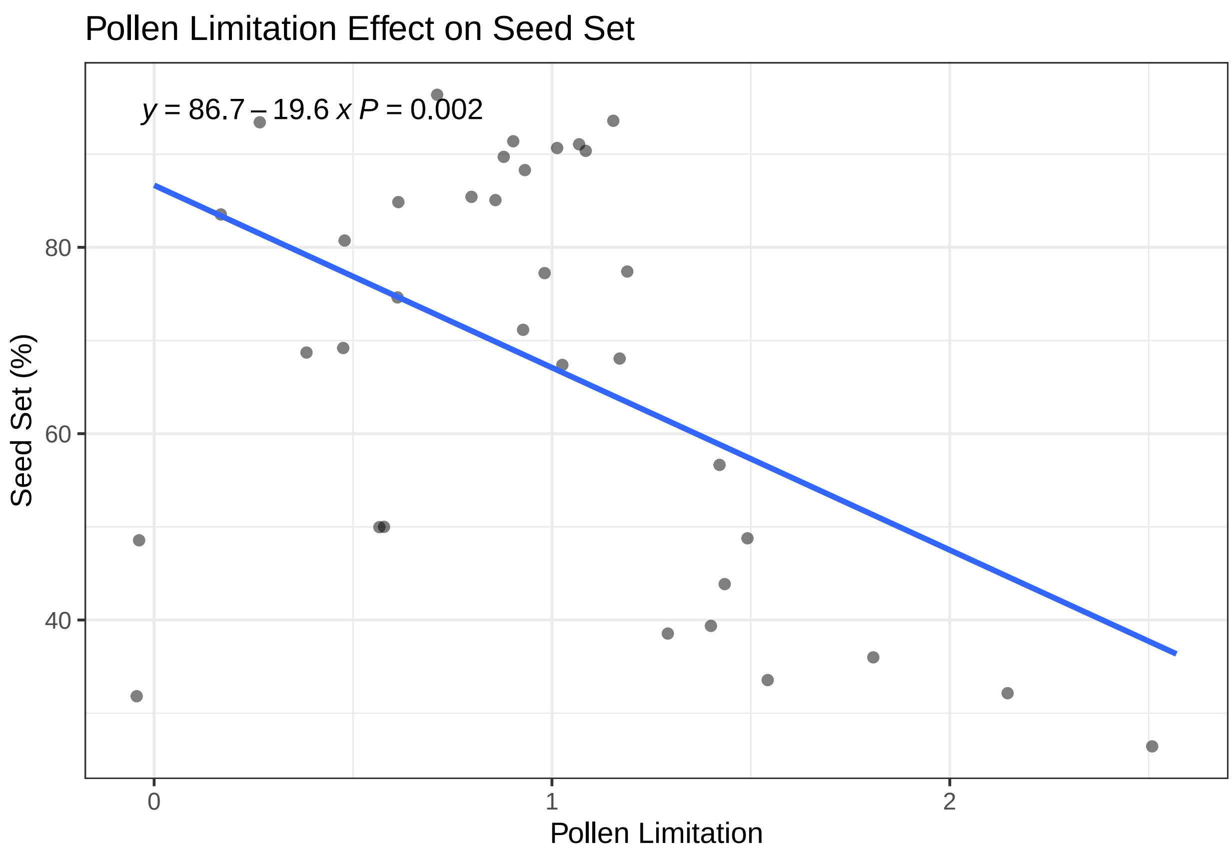


Visualize the relationship between Pollen Limitation and Seed Set %

```
(plot_T4 <- T4 %>%
    ggplot(aes(x = pollen_limitation,
               y = seed_set)) +
    geom_jitter(alpha = 0.5, width = 0.2) +
    geom_smooth(method = "lm", se = F) +
    theme_bw() +
    labs(title = "Pollen Limitation Effect on Seed Set",
         x = "Pollen Limitation",
         y = "Seed Set (%)") +
    stat_poly_eq(aes(label = paste(..eq.label..,
                                   ..p.value.label..,
                                   sep = "~")))
)
```

```
## Warning: The dot-dot notation (`..eq.label..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(eq.label)` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

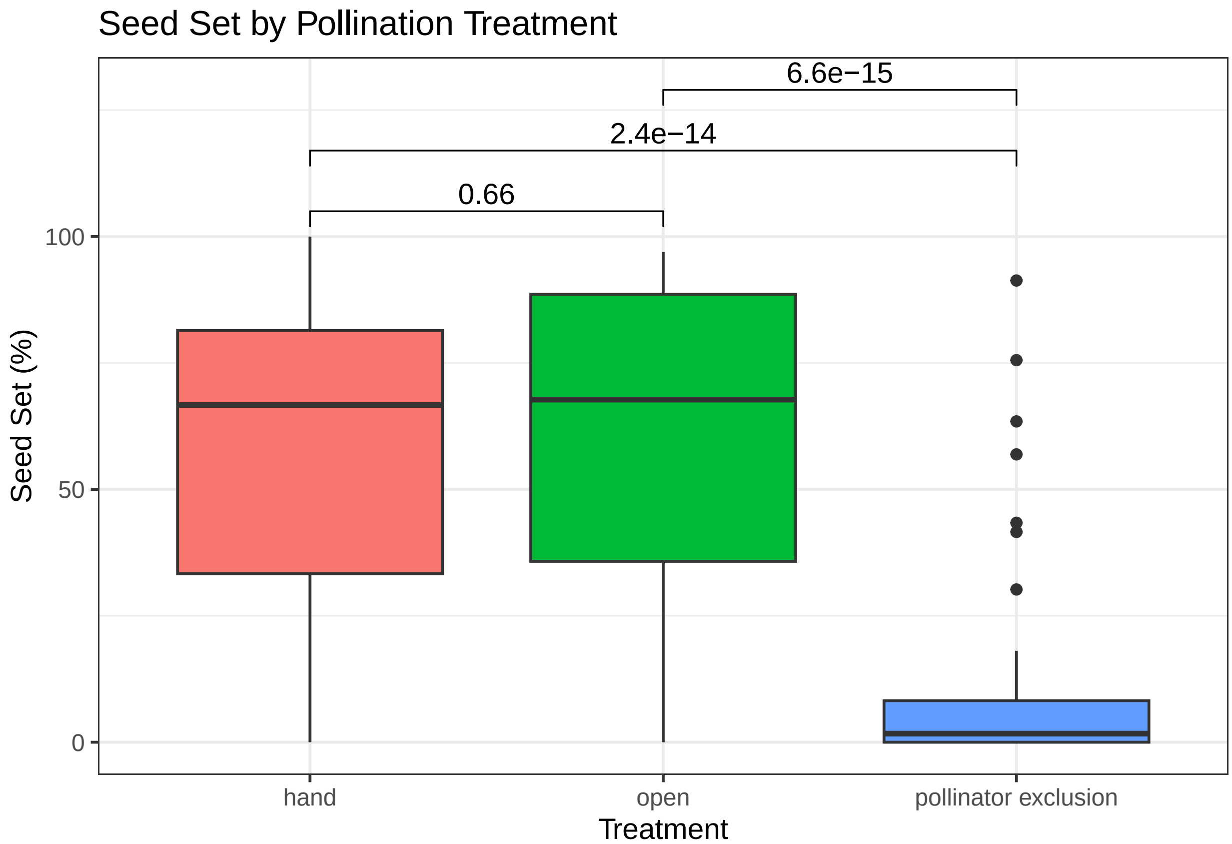
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
# ^ Fought with this for a wile, only works with the .. before and after each label,
# I'm not sure why! This is a new package/technique to me.
```

Visualize the results of each pollination treatment on Seed Set % -Add statistical significance w/ ggpubr

```
(plot_T5 <- T5 %>%
    ggplot(aes(x = treatment,
               y = seed_set,
               fill = treatment)) +
    geom_boxplot() +
    theme_bw() +
    labs(title = "Seed Set by Pollination Treatment",
         x = "Treatment",
         y = "Seed Set (%)") +
    theme(legend.position = "none") +
    stat_compare_means(method = "t.test",
                       label = "p.format",
                       comparisons = list(
                         c("hand", "open"),
                         c("hand", "pollinator exclusion"),
                         c("open", "pollinator exclusion")
                       ))
  )
```



Export Results

```
# Export summary statistics as csv files
#write_csv(T1_date_summary, "T1_date_summary.csv")
#write_csv(T1_site_summary, "T1_site_summary.csv")
#write_csv(T2_date_summary, "T2_date_summary.csv")
#write_csv(T2_site_summary, "T2_site_summary.csv")
#write_csv(T3_site_summary, "T3_site_summary.csv")
#write_csv(T4_site_summary, "T4_site_summary.csv")
#write_csv(T5_site_summary, "T5_site_summary.csv")

# Export figures as .png and .gif files
#anim_save("T1_FlowerAbundance.gif", plot_T1_time, width = 1600, height = 1000)
#ggsave("T1_VisitationRate.png", plot_T1_visitation, height = 5, width = 8)
#ggsave("T2_BeeAbundance.png", plot_T2, height = 5, width = 8)
#ggsave("T3_BeePollination.png", plot_T3, height = 5, width = 8)
#ggsave("T4_PollenLimitation.png", plot_T4, height = 5, width = 8)
#ggsave("T5_PollinationTreatment.png", plot_T5, height = 5, width = 8)
```